
Core Representation Theory

— *EYNTKA* Series —

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Part I

Representation Theory

There are many domains of mathematics which have made huge amounts of progress, having very-well understood objects and morphisms, theorems, results, and properties. One of the most well-understood area's of mathematics has become linear algebra, where vector-spaces and linear transformations. One of the goals of this chapter will be do be able to study the following 3 objects where the operations becomes matrix addition and multiplication

1. (Finite) groups as a subgroup of $GL(V)$, where the group operation becomes the matrix multiplication
2. Associative algebras
3. Lie Algebras

The further advantage of bringing these objects into linear algebra is that many field intersect with it: If the vector-spaces are infinite dimensional, we can make them Hilbert spaces and use tools from analysis. Vector-spaces are at the heart of differential geometry allowing us to use such theories to study groups!

For the reader who is really interested in number theory, we shall see in ref:HERE that matrix algebras have many parallels with some number theory results. The reader who goes onto learn class field theory¹ will to their delight find that these parallels continue. It was observed by Langland² that the study of simple k -algebras (algebras with trivial two-sided ideals) can be a higher-dimensional parallel to field extensions and the central results of class field mathematics known as the *Langland's Program*.

(The reader may want to go back to section ?? to review about direct sums of modules)

(Also, this entire link is really cool: [from wiki](#). Also has some cool examples of duality!!!)

¹See [NateCFT] for the continuation of this textbook into class field theory

²Or at least formalized by Langland, as these ideas were floating around during his time

Semisimple Rings and Modules

As mentioned in the introduction, representation theory is all about decomposing structures. In part ??, we showed in chapter ??, section ??, that all finite groups can uniquely be represented as a series of extensions with simple factors (i.e. G_i/G_{i+1} is simple). The simplest type of composition series would be when G is a direct product of simple groups, which would be in the terminology of exact sequences a series of trivial extensions with simple factors: such a group is sometimes *semi-simple* group¹. When it came to modules, we found that we may decompose it in different ways depending on what rings acts on it, in particular we saw in chapter ?? that if $R = k$ is a field k , then any k -module is free, and in chapter ?? we saw that if R is a Principal Ideal Domain, where we got that any R -module decomposed into a free part R -isomorphic to R^k , and a torsion part $R/(a_1) \times \cdots \times R/(a_n)$. It is evident that the ring that acts on the abelian group can have a large impact on its structure. In this section, we define a new type of ring, which will come to be called a *semi-simple* ring, that will force the module M to *always* be a direct product of *simple modules* (so that we are analysing the “building blocks” of modules).

To define this ring, we first shift focus and start by work with the module itself to find the “simplest” structure it may be², and explore in a manner which at the beginning parallels groups: we will look for *simple R -modules*, and direct product of simple modules (since they are the simplest extension of simple module).³. After defining such a structure, we will define which rings acting on modules will *always* make the module semi-simple

¹the term “semi-simple group” is not a completely standard terminology, as direct product of simple groups is not generally very heavily studied, see [this](#)

²“simple” here means no non-trivial sub-structures that can be used to build an extension, just like for simple groups or simple rings

³It might be tempting to say that this means we are exploring projective modules. However, we have yet to establish whether simple modules are free-modules. Are they?

1.1 Decomposing Modules

Definition 1.1.1: Decomposition Terminology

Let M be a nonzero R -module.

1. M is said to be *simple* (or *irreducible*) if the only submodules of M are 0 and M . Otherwise, M is *reducible*.
2. M is said to be *indecomposable* if there do not exist submodules $M_1, M_2 \subseteq M$ such that $M = M_1 \oplus M_2$. Otherwise, M is said to be *decomposable*.
3. The module M is said to be *semisimple* (or *completely reducible*) if it is the direct summand of simple modules.
4. The module M is said to be *semi-indecomposable* (or *completely indecomposable*) if it is a direct summand of indecomposable modules.
5. If M is semisimple, any direct summand of M is called a *constituent* of M (i.e. N is a constituent of M if there exists a submodule N' such that $M = N \oplus N'$).

A moment should be taken to find the relations between these definitions:

1. It is clear that simple modules are semi-simple and indecomposable. The prototypical example of a simple module is a field k over itself.
2. Semi-Simple $\not\Rightarrow$ Simple: take k^2 over k (the 2 dimensional vector-space over k). The canonical example of a semi-simple module would be vector-spaces, since they are all direct sums of the underlying field, which is a simple k -module.
3. Indecomposable $\not\Rightarrow$ Simple: this statement is equivalent to saying that not all short exact sequence split (i.e., not all modules are *projective*, see section ??). For a concrete example, let $\mathbb{Z}^2$ be a module over $\mathbb{Z}$ with the action:

$$r \cdot (a, b) = (a + rb, b)$$

Then it should be checked that it is not simple since $\mathbb{Z} \times \{0\}$ is a submodule, but it is indecomposable⁴.

4. Every semi-simple module is a semi-indecomposable module, however the converse is not true: $\mathbb{Z}$ is a semi-indecomposable $\mathbb{Z}$ -module (more generally any module over a PID is semi-indecomposable due to the Decomposition Theorem from chapter ??), however $\mathbb{Z}$ is not semi-simple a semi-simple $\mathbb{Z}$ -module as $p\mathbb{Z}$ is a submodule (the general case will be immediate after lemma 1.1.7)⁵

As stated in the introduction, we want to determine a semi-simple module or semi-simple modules in terms of the ring that is acting on it. The following definition was made for this purpose:

⁴Hint: Is it possible to write $\mathbb{Z} = n\mathbb{Z} \oplus m\mathbb{Z}$?

⁵This is a version of Krull-Schmidt's theorem for the decomposition of Noetherian or Artinian modules into the direct sum of indecomposable

Definition 1.1.2: Semi-Simple Ring

Let R be a ring. Then R is *semi-simple* if every R -module is semi-simple. Recall that R is *simple* if the only two-sided ideals are (0) and R .

Notice how a semi-simple ring was *not* defined as the direct product of simple rings. In the following proofs, we will show that the definition of semi-simple ring implies that R is the direct product of simple rings, however the converse is not true: there exists simple rings that are *not* semi-simple modules over themselves (and hence not simple R -modules). The problem is that of dimension; we will soon explore the relation between being semi-simple and Noetherian/Artinian. If we impose the condition that R be Noetherian and Artinian, then we are able to say that R is a semi-simple ring if and only if R is the direct product of simple rings.

We next establish many properties of simple and semi-simple modules, and show how closely related they are to the ring acting on them:

Lemma 1.1.3: Simple Module Characterization

Let M be an R -module. Then M is a simple R -module if and only if there exists a two-sided maximal ideal $\mathfrak{m} \subsetneq R$ such that

$$M \cong_R R/\mathfrak{m}$$

Proof:

($\Rightarrow$) Let M be a simple [non-trivial] R -module so that the only sub-modules of M are $\{0\}$ and M . Consider any $0 \neq x \in M$ and the R -module homomorphism:

$$\varphi : R \rightarrow M \quad r \mapsto r \cdot x$$

Since $\varphi(R) \subseteq M$ is a sub-module and M is a non-trivial R -module, it must be that $\varphi(R) = M$. Hence:

$$R/\ker(\varphi) \cong M$$

But then, $R/\ker(\varphi)$ cannot have any submodules, since M doesn't, and so $\ker(\varphi)$ must be maximal.

($\Leftarrow$) Let $R/\mathfrak{m} \cong M$ as R -module where $\mathfrak{m}$ is maximal. If $0 \subsetneq N \subsetneq M$, then there would exist an ideal $\subsetneq J \subsetneq R/\mathfrak{m}$, but that would make $\mathfrak{m} \subseteq J$, contradicting maximality of $\mathfrak{m}$.

If R is commutative, then $R/\mathfrak{m}$ is a field. For the non-commutative case, $R/\mathfrak{m}$ need not be a field, not even a division ring, the canonical example being $M_n(k)$ for $n \geq 2$ (see the remark ref:HERE or go over section ??). Do not confuse the isomorphism in the above statement as a ring isomorphism: the structure of M is isomorphic to the R -module structure of left-multiplication on $R/\mathfrak{m}$. Notice how this also shows that a simple R -module M is cyclic since $R \cdot \varphi(1) = \varphi(R \cdot 1) = \varphi(R/\mathfrak{m}) = M$ ⁶. If $R = k$ is a field, then M is isomorphic to a 1-dimensional vector-space over k . There do exist non-commutative rings R that are *not* of the form $M_n(k)$; we shall explore in ref:HERE that there is a family of quaternion algebras that are degree two extensions of a field k that is simple ring (more specifically, a simple k -algebra).

⁶In fact, any element must generate it, can you see why?

Using this result, we see that any semi-simple R -module is of the form:

$$M \cong_R \bigoplus_{i \in I} R/\mathfrak{m}_i$$

for maximal ideals $\mathfrak{m}_i \subseteq R$. In fact, we can further characterize $\mathfrak{m}_i$:

Corollary 1.1.4: Simple Module's And Annihilators

Let M be a simple left (or right) R -module. Then $M \cong R/\text{Ann}(m)$ for any nonzero m , and $\text{Ann}(m)$ is a maximal ideal.

Proof:

Since M is a simple left R -module, for any $0 \neq m \in M$, $mR \subseteq M$ is a submodule (by identifying R in M by the map $r \mapsto r \cdot 1$). As it is nonzero and M is simple, $M \cong_R Rm$. Then we have a surjective R -module homomorphism

$$R \rightarrow Rm \quad r \mapsto r \cdot m$$

Thus, $R/\mathfrak{m} \cong_R Rm$ for some ideal $\mathfrak{m}$ (note that the R -submodules of R are ideals). In particular, $\varphi(\mathfrak{m}) = \mathfrak{m}m = 0$, so $\mathfrak{m} = \text{Ann}(m)$ by definition. Thus, we have the isomorphism chain:

$$M \cong_R Rm \cong_R R/\text{Ann}(m)$$

Finally, since Rm is simple, $R/\text{Ann}(m)$ must have no other two-sided ideals, i.e. $\text{Ann}(m)$ is a two-sided maximal ideal, completing the proof.

Since we've seen the limitations of the structure of simple modules, it is also natural to ask if there are any limitations on the structure of the hom-set given two simple modules:

Lemma 1.1.5: Schur's Lemma

Let M and N be simple R -modules, $\varphi : M \rightarrow N$ an R -module homomorphism. Then either $\varphi = 0$ or φ is an isomorphism. As a consequence, $\text{End}_R(M)$ is a division ring.

Proof:

Since N is simple, either $\varphi(M) = N$ or $\varphi(M) = 0$. If $\varphi(M) = N$, it has to be that $\ker(\varphi) = 0$, since M is simple, and hence φ would be an isomorphism.

As a consequence, every element in $\text{End}_R(M)$ is either 0 or is a unit, i.e. has a multiplicative inverse, which by definition makes $\text{End}_R(M)$ a division ring under $(+, \circ)$.

Since $M \cong_R R/\mathfrak{m}$ for some maximal ideal $\mathfrak{m} \subsetneq R$, we have that $\text{End}_R(R/\mathfrak{m})$ is always a division ring, whether R is commutative or not. In a sense, this can be thought of as “generalization” to ??, where it was shown that $R/\mathfrak{m}$ is a field if R is commutative.

Notice that if $M = V$ was an 1-dimensional vector-space over k , then $\text{End}_R(V)$ is a known division ring, in particular it's ring-isomorphic to k . If V is a finite dimensional semi-simple module over k , i.e. it is n -dimensional, then $\text{End}_R(V)$ is ring-isomorphic to $M_n(k)$. We might hope for something

similar for the structure of $\text{End}_R(M)$ if M is semi-simple. Take a moment to ponder this before reading the next lemma (hint: recall ??)

Lemma 1.1.6: Hom's of Semisimple Modules

Let M be a semisimple R -module, so that:

$$M = S_1^{\oplus n_1} \oplus S_2^{\oplus n_2} \oplus \cdots \oplus S_k^{\oplus n_k}$$

where each S_i are simple R -modules, $S_i \not\cong S_j$. Then as a ring:

$$\text{End}_R(M) \cong_{\mathbf{Ring}} \text{End}_R(S_1^{\oplus n_1}) \oplus \cdots \oplus \text{End}_R(S_k^{\oplus n_k})$$

Furthermore, for each $S_i^{\oplus n_i}$, we have that direct sums “pull out” to become matrices:

$$\text{End}_R(S_i^{\oplus n_i}) \cong_{\mathbf{Ring}} M_{n_i}(\text{End}_R(S_i)) = M_{n_i}(D_i)$$

where $D = \text{End}_R(S_i)$ and $M_{n_i}(D_i)$ is the matrix ring over the division ring $\text{End}_R(S_i)$.

Proof:

We invoke ?? to get the result in terms of groups, and then use Schur's lemma and check that the ring operation is compatible:

$$\begin{aligned} \text{End}_R(M) &= \text{Hom}_R \left(\bigoplus_i S_i^{\oplus n_i}, \bigoplus_j S_j^{\oplus n_j} \right) \\ &\cong_{\mathbf{Grp}} \bigoplus_{i,j} \text{Hom}_R \left(S_i^{\oplus n_i}, S_j^{\oplus n_j} \right) \\ &\cong_{\mathbf{Grp}} \bigoplus_{i,j} \text{Hom}_R(S_i, S_j)^{\oplus n_i n_j} \end{aligned}$$

where when $S_i \neq S_j$, by Schur's lemma we must have:

$$\text{Hom}_R(S_i, S_j)^{\oplus n_i n_j} = 0$$

Hence we get that $\text{End}_R(M) \cong_{\mathbf{Grp}} \bigoplus_i \text{Hom}_R(S_i^{\oplus n_i}, S_i^{\oplus n_i}) = \bigoplus_i \text{End}_R(S_i^{\oplus n_i})$. Next, this map is compatible with the ring structure, since the map:

$$\varphi \rightarrow (\varphi|_{S_i^{\oplus n_i}} : S_i^{\oplus n_i} \rightarrow S_i^{\oplus n_i} \subseteq M)$$

is clearly compatible with composition, and hence

$$\text{End}_R(M) \cong_{\mathbf{Ring}} \bigoplus_i \text{End}_R(S_i^{\oplus n_i})$$

For part 2, we follow the same steps as for part (1): re-indexing so that $S^n = S_1 \oplus S_2 \oplus \cdots \oplus S_n$ so that $S_i = S_j = S$, we get

$$\text{Hom}_R \left(\bigoplus_i S_i, \bigoplus_j S_j \right) \cong_{\mathbf{Grp}} \bigoplus_{i,j} \text{Hom}_R(S_i, S_j) \stackrel{!}{=} \bigoplus_{i,j} D \cong_{\mathbf{Grp}} M_n(D)$$

where $\stackrel{!}{=}$ comes from Shur's lemma with $D = \text{End}_R(S)$ since $S_i = S_j = S$. The last equality comes from the fact that the matrix $M_n(D)$ when treated like a group is simply re-arranging where the addition is happening in $\bigoplus_{i,j} D$. To show that the ring structure is compatible, observe that

$$\varphi \mapsto \varphi_{ji}$$

is the desired map, where

$$\varphi \begin{pmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{pmatrix} = \begin{pmatrix} \varphi_{11} & \varphi_{12} & \cdots & \varphi_{1n} \\ \varphi_{21} & \varphi_{22} & \cdots & \varphi_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \varphi_{n1} & \varphi_{n2} & \cdots & \varphi_{nn} \end{pmatrix} \begin{pmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{pmatrix}$$

which is simply a formalization of what the group-isomorphism give us, showing us that composition is compatible, completing the proof. (see [this](#) for a more detailed explanation)

Hence, for any semi-simple R -module M , we get that:

$$\text{End}_R(M) \cong_{\mathbf{Ring}} \prod_i M_{n_i}(D_i) \cong_{\mathbf{Ring}} \prod_i \text{End}_R(S_i^{\oplus n_i})$$

and since each S_i is a simple R -module, $S \cong_R R/\mathfrak{m}_i$ for appropriate maximal ideal $\mathfrak{m}_i \subseteq R$, $\text{End}_R(S_i) \cong \text{End}_R(R/\mathfrak{m}_i)$. Thus:

$$\text{End}_R(M) \cong_{\mathbf{Ring}} \prod_i M_{n_i}(\text{End}_R(R/\mathfrak{m}_i)) = \prod_i M_{n_i}(D_i)$$

If $R = k$ is a field making $M = V$ a vector-space over k (say n -dimensional), then the only simple k -submodule up to isomorphism is k , and so this result reduces to:

$$\text{End}_R(V) \cong_{\mathbf{Ring}} M_n(\text{End}_k(k/(0))) = M_n(k)$$

where $\text{End}_k(k) \cong_{\mathbf{Ring}} k$, aligning with the results we found in section ???. This shows that when generalizing from linear algebra, due to the fact that a ring is non-commutative, we may have multiple non-isomorphic maximal ideals giving us multiple products of matrix rings. With a better understanding of simple modules and their endomorphisms, we continue by characterizing semi-simple modules:

Lemma 1.1.7: Semi-Simple Module Equivalence

Let M be a semi-simple module. Then the following are equivalent:

1. M is an external direct sum of simple modules:

$$M \cong \bigoplus_{i \in I} S_i$$

2. M is the internal direct sum of simple modules:

$$M = \bigoplus_{i \in \mathfrak{m}} S_i$$

3. If $N \subseteq M$, there exists a complimentary module, that is there exists another submodule $N' \subseteq M$ such that

$$M = N \oplus N'$$

i.e., every semi-simple R -module M is *projective* and *injective*.

Proof:

(1) $\Rightarrow$ (2) is ??.

For (2) $\Rightarrow$ (3), Let $N \subseteq M$. Then using Zorn's lemma, we take a maximal submodule $N' \subseteq M$ ordered by inclusion with the property that $N \cap N' = 0$ (upper-bounds naturally exist in this set, and 0 is in this set, and hence Zorn's lemma does apply). If $M = N + N'$, we're done, since we can use the isomorphism from ???. For the sake of contradiction, let's say $N + N' \subsetneq M$. Then since $M = \sum_i S_i$, there must exist some $i \in I$ and $s_i \in S_i$ such that $s_i \notin N + N'$, since $0 \in S_j$ for all $j \in I$. Hence $S_i \not\subseteq N + N'$, so $S_i \cap (N + N') \subsetneq S_i$, however S_i is simple, and hence:

$$S_i \cap (N + N') = 0$$

But then:

$$(S_i + N') \cap N \subseteq N' \cap N = 0$$

Thus, by maximality of N' , $S_i + N' = N'$, so $S_i \subseteq N' \subseteq N + N'$, contradicting the assumption that $s_i \notin N + N'$, as we sought to show.

For (3) $\Rightarrow$ (1), Let $\mathcal{S}$ be the collection of simple submodule of M where their direct sum maps injectively into M . Since $0 \in \mathcal{S}$ and taking the sum's of all the modules is an upper bound, Let N be the maximal submodule with this property. By ??,

$$N = \sum_{i \in I} S_i \cong \bigoplus_{i \in I} S_i$$

If the map is surjective, we are done. For the sake of contradiction, suppose it is not. Then by part (3), there exists an N' such that $M = N + N'$. If $N' = 0$, we found a contradiction. So, let's say $N' \neq 0$. Then it is enough to show that N' contains a simple module, say S , since then we can make the collection $\{(S_i)_{i \in I}, S\}$ which will map injectively into M , contradicting maximality.

(here is what the professor posted Suppose every submodule N of M has a complement (i.e. a submodule N' of M such that $M = N \oplus N'$ (internal direct sum)). Then every nonzero submodule N' of M contains a simple submodule.

Proof: first we show that every submodule K of M has the same property (every submodule has a complement): suppose L is a submodule of K . by assumption, $M = L \oplus L'$ for some submodule L' of M . then note that $K = L \oplus (L' \cap K)$.

Now, suppose N' is nonzero submodule of M . Take a nonzero $x \in N'$, so Rx is a nonzero submodule and by 1st isomorphism theorem we have $Rx \cong_R R/I$ for some proper left ideal I . By the previous paragraph, every submodule of Rx has a complement, i.e. every submodule of R/I has a complement. Take any maximal left ideal $\mathfrak{m}$ containing I . Then M/I is a submodule of $R/\mathfrak{m}$ with complement L' in R/I , i.e. $R/I = \mathfrak{m}/I \oplus L'$, so $L' \cong (R/I)/(\mathfrak{m}/I) \cong R/\mathfrak{m}$ is simple, i.e. L' is a simple submodule of N' (via the isomorphism $Rx \cong R/I$)

This theorems shows that every vector-space over k is a semisimple k -module, however by remark internalSumVsSumRmk, $\mathbb{Z}$ is *not* a semi-simple $\mathbb{Z}$ -module, since $2\mathbb{Z}$ is a submodule with no compliment (showing that not all free modules are semi-simple). We can characterize semi-simple rings by saying that R is semi-simple if and only if every R -module is projective. This also shows us that $\mathbb{Z}$ is not a semi-simple ring; though $\mathbb{Z}$ is projective over itself, any finite abelian group is a non-projective $\mathbb{Z}$ -module. Thus, a module of a semi-simple ring is stronger then being a projective module. Hence being semi-simple is stronger (in a sense much stronger) than being projective, but may not be free: even though many example of semi-simple modules are free-modules, we would require the underlying ring to be semi-simple over itself. In fact, it is enough that the underlying ring be a semi-simple module over itself, giving us an easier condition for defining a semi-simple ring! We first introduce an important result we will use:

Corollary 1.1.8: Extension Closure Of Semisimple Modules

Let M an R -module. Then submodule and quotient modules of M are semisimple if and only if M is semi-simple. Furthermore, if $(M_i)_{i \in I}$ is a collection of semi-simple modules, then $\bigoplus_{i \in I} M_i$ is a semi-simple module.

Proof:

The direct sum of semi-simple modules is semi-simple by definition, and so we check the other two statements.

Let M be a semi-simple R -module so that $M = \bigoplus_{i \in I} S_i$ is by lemma 1.1.7. First, consider a surjection $M \twoheadrightarrow N$.

$$\varphi(M) = \varphi \left(\bigoplus_{i \in I} S_i \right) = \bigoplus_{i \in I} \varphi(S_i)$$

then by Schur's lemma, either $\varphi(S_i) = 0$ or $\varphi(S_i) \cong S_i$, and hence N is semisimple.

If $N \subseteq M$, then by lemma 1.1.7, we have that $M = N \oplus N'$, and so by ??

$$M/N' \cong N$$

which we just showed makes N semi-simple.

Using these two facts, it is not difficult to show that if $N \subseteq M$ and $M/N \cong N'$ are semi-simple, then so is M , where if $x \in M$, $x = \sum_i n_i + \sum_j n'_j$.

These properties of semi-simple modules allow us to characterize semi-simple rings much more easily:

Corollary 1.1.9: Semi-Simple Rings Condition

Let R be a ring. Then R is a semi-simple ring if and only if R is a semi-simple R -module, that is we only need to check that R is a semi-simple R -module for one R -module structure (namely R itself)

Proof:

By definition, R if R is a semi-simple ring then any R -module is a semi-simple R -module, including R over itself.

Now R is given some R -action becomes a semi-simple R -module, then by corollary 1.1.8 so is $\bigoplus_{i \in I} R$. We will use this to show that any R -module M is semi-simple, which will make R semi-simple by definition.

Let M is any other R -module, by the universal property of the direct sum and free-modules:

$$M = \bigoplus_i Rm_i$$

for appropriate m_i (for ex. we can pick all nonzero elements of M). Then we get a surjective map:

$$\bigoplus_{i \in I} R \rightarrow M \quad (r_i)_{i \in I} \mapsto \sum r_i m_i$$

hence, M is the quotient of a semi-simple module, and hence semi-simple. Thus, R is semi-simple, as we sought to show.

Due to this, we get a nice description of semi-simple rings which is even stronger than semi-simple modules:

Lemma 1.1.10: Decomposing Semi-Simple Rings

Let R be a semi-simple ring. Then:

$$R = I_1 \oplus \cdots \oplus I_n$$

for a finite collection of I_k , where I_k are simple left-submodules of R , i.e. *minimal* left ideals^a of R

^aThese are *left ideals* as we are working with *left* submodules

Proof:

Since R is a semi-simple ring, it is a semi-simple R -module and so by definition:

$$R = \bigoplus_{i \in \mathcal{C}} I_i$$

for simple submodules I_i (i.e. minimal left ideals I_i). To make this a finite sum, since $1 \in R$, we

get for an appropriate finite subset $A \subseteq \mathcal{C}$

$$1 \in I_{\alpha_1} + \cdots + I_{\alpha_n}$$

But then, since these are all ideals,

$$R = R \cdot 1 \subseteq I_{\alpha_1} \oplus \cdots \oplus I_{\alpha_n} \subseteq R$$

and hence $R = I_{\alpha_1} \oplus \cdots \oplus I_{\alpha_n}$, completing the proof.

1.1.11 Remark As a consequence of this, any semi-simple ring R is a left-Noetherian/Artinian ring. Interestingly, we will soon show that this also implies that R is a *right*-Noetherian/Artinian ring (see rightNoethArtRemark)

Furthermore, this shows how it is necessary for the condition that R be left-Noetherian/Artinian so that the notion of being semi-simple coincide between the two definitions.

Since we characterized simple R -modules for any ring R to be R -isomorphic to $R/\mathfrak{m}$, and we've given a structure of R if R is a semisimple ring, we can make lemma 1.1.3 more precise:

Corollary 1.1.12: Structure Of Simple Modules Of Semi-Simple Ring

Let R be a semi-simple ring so that $R = I_1 \oplus I_2 \oplus \cdots \oplus I_n$ for minimal left ideal I_k (equivalently, simple left R -modules). Then if M is a simple left R -module,

$$M \cong_{R\text{-Mod}} I_i$$

for some $1 \leq i \leq n$.

Proof:

Let M be a simple nonzero left R -module, and consider the R -module homomorphism:

$$\varphi : R \rightarrow M \quad r \mapsto r \cdot x$$

where $x \neq 0$. Since M is simple, it is nonzero, so φ is surjective. Since $R = \bigoplus_i^n I_i$, for some i 's

$$\varphi|_{I_i} \neq 0$$

and since $\varphi|_{I_i} = \pi_i \circ \varphi$ we have the R -module homomorphism:

$$\varphi|_{I_i} I_i \rightarrow M$$

and since both modules are simple and this is a nonzero homomorphism, by Schur's lemma this map must be an isomorphism, as we sought to show.

Note that there can be multiple I_i 's that are isomorphic to M , which we will show in a following theorem.

This shows that any simple R -module over a semi-simple ring R is in fact isomorphic to a simple

R -module over R (i.e. R -isomorphic to a minimal ideal of R). Thus, studying simple R -modules over semi-simple rings comes down to studying the ring R itself, and that there are only *finitely many* simple R -modules. Furthermore, we have that $R/(\mathfrak{m}_1 \oplus \cdots \mathfrak{m}_{n-1}) \cong_R \mathfrak{m}_n$. Here is a good example of where the ring structure of the left hand side cannot exist in the right hand side. For example, if R is commutative, then $R/\mathfrak{m}$ is a field that is R -module isomorphic to a minimal ideal, but ideals cannot be fields (over the same identity⁷). Thus, in the case where R is semi-simple, for a maximal ideal $\mathfrak{m}$, $R/\mathfrak{m}$ is R -isomorphic to a minimal ideal of R .

Exercise 1.1.1

1. Let S be a simple module. Show that any nonzero submodule of the injective envelope $E(S)$ is indecomposable, but not simple.
2. Show that if M is a simple R -module, then it necessarily has to be cyclic (the converse, however, is not true).
3. Show that $A = \mathbb{Q}[x, y]/(xy - yx - 1)$ is a simple ring, but not a semi-simple ring (this is a special example of an algebra called a *Weyl Algebra*)
4. Let R be a ring, not necessarily commutative, and let $[R, R] = \{rs - sr \mid r, s \in R\}$. Show that the trace map induces an isomorphism

$$\frac{M_n(R)}{[M_n(R), M_n(R)]} \cong \frac{R}{[R, R]}$$

From this perspective, matrix rings are the natural “nonabelian” extension of a ring!

1.2 Artin-Wederburn

We just showed how studying simple and semi-simple R -modules comes down to studying the ring R , especially if the ring R is semi-simple. Thus, it is fruitful to classify semi-simple rings, which leads us to the main classification result for semi-simple rings, namely we build on lemma 1.1.10 and show that we can find more precisely how to represent each minimal ideal $\mathfrak{m}_i$, and furthermore show that this classification is exhaustive, and that any semi-simple ring uniquely breaks down into the result we’ll show. We will then finish by showing that if we upgrade our ring to k -algebras (that is, we can put a vector-space structure on our ring), we get some stronger results due to the fact we can count dimensions.

First, we try to relate what we’ve shown for the hom-sets of simple modules with the structure of simple modules. You may perhaps recall that we commented in exercise ref:HERE (in 9.1.1) and example ?? that R is not necessarily isomorphic to R^{op} . There is in fact a very nice way of linking R^{op} and R , which will be demonstrated after formally define R^{op} for reference

⁷ $k \times \{0\} \subseteq k \times k$ is an ideal and a field with identity $(1, 0)$

Definition 1.2.1: Opposite Ring

Let R be a ring. Then we define a new ring labelled R^{op} that has the same underlying set and addition operation, with the multiplication operation $*$ being:

$$x * y = yx$$

where the multiplication yx happens in R

If R is commutative, then $R = R^{\text{op}}$, and so we only get non-trivial results when R is not commutative. An example of $R \not\cong R^{\text{op}}$ would be a free-algebra over two variables (say $k\langle x, y \rangle$).

Proposition 1.2.2: Properties Of Opposite Ring

Let R be a ring and R^{op} its opposite ring. Then:

1. $(R^{\text{op}})^{\text{op}} = R$
2. $(R \times S)^{\text{op}} = R^{\text{op}} \times S^{\text{op}}$
3. $M_n(R)^{\text{op}} = M_n(R^{\text{op}})$ where the isomorphism is $A \mapsto A^t$ (essentially since $(AB)^t = B^t A^t$)

Proof:

exercise

With this, we can make a link between R and R^{op} :

Lemma 1.2.3: Yoneda's Lemma for opposite Rings

Let R be a ring and consider the ring $\text{End}_R(R)$ to be the set of all R -module homomorphisms from R (as an R -module) to itself. Then $\varphi : R^{\text{op}} \rightarrow \text{End}_R(R)$, $r \mapsto (x \mapsto xr)$ is a ring isomorphism:

$$R^{\text{op}} \cong_{\text{Ring}} \text{End}_R(R)$$

Proof:

Let φ_r be the given map. Let's first check that $\varphi_r \in \text{End}_R(R)$. Then we see that:

$$\begin{aligned} \varphi_r(\lambda x + \lambda y) &= (\lambda x + \lambda y)r \\ &= \lambda(xr) + \lambda(yr) \\ &= \lambda\varphi_r(x) + \lambda\varphi_r(y) \end{aligned}$$

showing that $\varphi_r \in \text{End}_R(R)$. Next, we check that φ is indeed a ring-homomorphism. By the properties of modules, it is easy to see that:

1. $\varphi_1 = \text{id}$, and hence φ sends the identity to the identity.
2. $\varphi_{r+r'} = \varphi_r + \varphi_{r'}$
3. $\varphi_{rr'} = \varphi_{r'} \circ \varphi_r$ (don't forget we are working with R^{op})

Finally, showing bijectivity, for injectivity we see that if $\varphi_r = 0$, then $0 = \varphi_r(1) = r$, and hence $\varphi_r = \varphi_0$. For surjectivity, let $\varphi \in \text{End}_R(R)$. Let's say $\varphi(1) = r$. Then $\varphi(x) = x\varphi(1) = xr$, so $\varphi = \varphi_r$, completing the proof.

In the special case where R is commutative, then $R \cong R^{\text{op}}$, showing that in this case $R \cong \text{End}_R(R)$. Using this, we will use lemma 1.2.3 along with lemma 1.1.6 to upgrade lemma 1.1.10:

Theorem 1.2.4: Artin-Wedderburn, one Direction

Let R be a semisimple ring. Then:

$$R \cong_{\text{Ring}} M_{n_1}(D_1) \times \cdots \times M_{n_k}(D_k)$$

where each D_i is a division ring and $M_{n_i}(D_i)$ is the matrix ring over D_i

The key realization about this theorem is that while the multiplication on the left hand side can “send you” all over the place, the multiplication on the right hand side only sends you back into your particular component. To see this more clearly, let's say R was a k -algebra for a field k so that it has a basis. If R was a semi-simple ring, then that means that when multiplying elements of R (so figuring out where the basis elements go while multiplying), we can isolate the problem down to the components on the right hand side, and figure out where they go through matrix multiplication!

Proof:

First, by lemma 1.1.10, for we get simple R -submodule of R , i.e. minimal ideals of R ,

$$R = I_1 \oplus \cdots \oplus I_n$$

Now, we can always re-order the sums (i.e. $I_1 \oplus I_2 \cong I_2 \oplus I_1$, since $R \cong I_1 \oplus \cdots \oplus I_n$ by ??). Hence, we will group all the I_i 's that belong to the same isomorphism class (as R -modules), letting us re-write R as:

$$R \cong_{\text{Ring}} I_1^{n_1} \oplus \cdots \oplus I_k^{n_k}$$

where $I_i \not\cong I_j$ for $i \neq j$. Hence by lemma 1.2.3 and lemma 1.1.6:

$$\begin{aligned} R^{\text{op}} &\cong_{\text{Ring}} \text{End}_R(R) \\ &\cong_{\text{Ring}} M_{n_1}(D_1) \times \cdots \times M_{n_k}(D_k) \end{aligned}$$

where $D_i = \text{End}_R(I_i)$ is a division ring. Finally, Taking the opposite ring on both sides, we get:

$$\begin{aligned} R &\cong_{\text{Ring}} (R^{\text{op}})^{\text{op}} \\ &\cong_{\text{Ring}} M_{n_1}(D_1)^{\text{op}} \times \cdots \times M_{n_k}(D_k)^{\text{op}} \\ &\cong_{\text{Ring}} M_{n_1}(D_1^{\text{op}}) \times \cdots \times M_{n_k}(D_k^{\text{op}}) \end{aligned}$$

and it is clear the D_i^{op} is still a division ring, hence completing the proof.

Thus, if we have a simple-ring then the ring breaks down into the product of matrix rings over division rings. In the special case where $R = k$ is a field, then k has only one minimal ideal, namely itself, and $\text{End}_k(k) \cong M_1(k) \cong k$, and hence we reduce to $k \cong k$. If $R = k^n$ is a product of the same field, then the minimal ideals of R are $R = k \oplus \cdots \oplus k$, so that we get back $k^n \cong_{\text{Ring}} k^n$.

We will now show that the converse of this theorem is true, that the product of matrix rings over division rings is semi simple. We start by showing that matrix ring are simple modules⁸.

Proposition 1.2.5: Matrix Ring over Division Rings are semi-simple

Let $R = M_n(D)$ be a matrix ring over a division ring. Let $C = D^{\oplus n}$ be the left $M_n(D)$ -module where the action is left matrix multiplication (imagine C as a column). Then C is a simple $M_n(D)$ -module, and

$$M_n(D) \cong_{M_n(D)\text{-Mod}} C^{\oplus n} = (D^{\oplus n})^{\oplus n}$$

showing that $M_n(D)$ is semisimple ring and $M_n(D) \cong_{\text{Ring}} C^{\oplus n}$

Proof:

Let $R = M_n(D)$ and $C = D^{\oplus n}$ (C for *column*). For notational simplicity, let $E_{ij} \in R$ be the elementary matrix which is 1 in position ij and zero everywhere else. Let $e_i \in C$ be the column where it is 1 at position i and zero everywhere else:

$$E_{ij} = \begin{pmatrix} 0 & & & 0 \\ & \ddots & & \\ & & 1 & \\ & \ddots & & \ddots \\ 0 & & & & 0 \end{pmatrix} \quad e_i = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}$$

Then $E_{ij}e_k = \delta_{jk}e_i$ where δ_{jk} is the Kronecker Delta:

$$\delta_{jk} = \begin{cases} 1 & j = k \\ 0 & j \neq k \end{cases}$$

Now, we will start by showing that C is a simple $M_n(D)$ -module. Since $C = D^{\oplus n}$,

$$C = \bigoplus_i De_i$$

for some basis $e_1, e_2, \dots, e_n$. To show that C is simple, it is enough to show that for any $0 \neq x \in C$, $C = Rx$ since then any nonzero submodule would in fact be C . Given our above notation, we have that:

$$x = \sum_k d_k e_k$$

for appropriate $d_k \in D$, where at least one of these is nonzero, let's say $d_j \neq 0$. Then consider $E_{ij}x \in Rx$:

$$E_{ij}x = \sum_k \delta_{jk} d_k e_k = d_j e_i$$

since $d_j \neq 0$. Since D is a division ring, we get that $d_j^{-1}e_i \in Rx$, for all $1 \leq i \leq n$. But then, we get:

$$C = \sum De_i \subseteq Rx \subseteq C$$

⁸This will be an example of R being both a simple R -module and a simple ring

showing that $C = Rx$ for any $0 \neq x \in C$, showing that C is indeed simple.

Now, we can re-write this ring as the internal direct sum of matrices I_i which are zero outside the i th column:

$$R = M_n(D) = I_1 \oplus \cdots \oplus I_n$$

$$I_k = \begin{pmatrix} 0 & \cdots & a_{1,k} & \cdots & 0 \\ 0 & & a_{2,k} & & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & & a_{n-1,k} & & 0 \\ 0 & \cdots & a_{n,k} & \cdots & 0 \end{pmatrix}$$

This decomposition works since each I_i is stable under *left* multiplication by elements of $M_n(D)$ (since left multiplication is a *column* transformation, right multiplication would be *row* transformation). Finally, it is clear that $C \cong_{R\text{-mod}} I_i$ by simply moving the column to the right place, and so each I_i is simple. Hence, we have decomposed $M_n(D)$ as a direct product of simple left $M_n(D)$ -modules, and hence $M_n(D)$ is a semisimple $M_n(D)$ -module, which by corollary 1.1.9 makes $M_n(D)$ a semi-simple ring. Finally, since $C^{\oplus n}$ represents the n rows of n -length columns, we see that the action of $M_n(D)$ on $C^{\oplus n}$ is identical to that of matrix multiplying elements of $C^{\oplus n}$, which gives us that $M_n(D) \cong_{\text{Ring}} C^{\oplus n}$, as we sought to show.

1.2.6 Remark In fact, $M_n(D)$ is a simple ring, meaning the only *two-sided* ideals of R are 0 and R ; the above proof can be modified to show this. This gives us a nice class of non-commutative simple rings (since all simple commutative rings are fields).

Notice that it is quite special that the ring operation and the module operation coincided, this because of the special way in which we decomposed the $M_n(D)$ -module $M_n(D)$ and by the fact that it is a module over itself. In the previous proposition, we showed that $D^{\oplus n}$ is a simple $M_n(D)$ -module. In fact, all simple modules of $M_n(D)$ are isomorphic to $D^{\oplus n}$:

Corollary 1.2.7: Structure of Simple $M_n(D)$ -Module

Let $R = M_n(D)$ be a matrix ring over a division ring. Then any simple $M_n(D)$ -module is isomorphic to $D^{\oplus n}$.

Proof:

Let M be a simple $M_n(D)$ -module. By proposition 1.2.5

$$M_n(D) \cong_{\text{Ring}} D^{\oplus n} \oplus \cdots \oplus_{\text{Ring}} D^{\oplus n}$$

And by corollary 1.1.12,

$$M \cong D^{\oplus n}$$

Corollary 1.2.8: Endomorphism Of Simple $M_n(D)$ -Module

Let S be a simple $M_n(D)$ -module. Then

$$\text{End}_{M_n(D)}(S) \cong_{\mathbf{Ring}} D^{\text{op}}$$

Proof:

Since S is simple, $S \cong_R D^{\oplus n}$ for some n . Let $\varphi_d : S \rightarrow S$, $\varphi_d(x) = x \cdot d$ where $x \cdot d$ is x times dI . Then:

$$\varphi_d(Ax) = (Ax) \cdot d = A(x \cdot d) = A\varphi_d(x)$$

and

$$\varphi_d(x + y) = (x + y) \cdot d = (x + y)dI = xdI + ydI = x \cdot d + y \cdot d = \varphi_d(x) + \varphi_d(y)$$

hence $\varphi_d \in \text{End}_R(S)$. Conversely, if $\varphi \in \text{End}_R(S)$, Since $S \cong D^{\oplus n}$, and

$$D^{\oplus n} \cong (D^{\oplus n} \ 0^{\oplus n} \ \dots \ 0^{\oplus n})$$

we will treat S as a $M_n(D)$ submodule of $M_n(D)$. Now, since φ is a group homomorphism, we know that:

$$\varphi \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} \varphi_{11} & \cdots & \varphi_{n1} \\ \vdots & \ddots & \vdots \\ \varphi_{n1} & \cdots & \varphi_{nn} \end{pmatrix} \begin{pmatrix} 1 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} \varphi_{11}(1) + \cdots + \varphi_{n1}(1) \\ \vdots \\ 0 \end{pmatrix}$$

hence let $d = \sum_i \varphi_{i1}(1)$, and let the right-action on $\text{End}_{M_n(D)}(S)$ be right multiplication by the identity matrix dI . Then: Then for all $x \in S$, we have:

$$\begin{aligned} \varphi \begin{pmatrix} x_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ x_n & \cdots & 0 \end{pmatrix} &= \varphi \left(\begin{pmatrix} x_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ x_n & \cdots & 0 \end{pmatrix} \begin{pmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{pmatrix} \right) \\ &\stackrel{!}{=} \begin{pmatrix} x_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ x_n & \cdots & 0 \end{pmatrix} \varphi \begin{pmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{pmatrix} \\ &= \begin{pmatrix} x_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ x_n & \cdots & 0 \end{pmatrix} \cdot d \end{aligned}$$

where $\stackrel{!}{=}$ comes from the fact that we can treat that matrix as an element of $M_n(D)$, and since φ is an $M_n(D)$ -module homomorphism, we can pull it out. But then, the action is completely dependent on d , and so we have a bijection between right multiplication by $d \in D$ and $M_n(D)$ -endomorphisms. It is an easy task to check that:

$$\varphi_{dd'} = \varphi_{d'} \circ \varphi_d$$

showing that

$$D^{\text{op}} \cong_{\mathbf{Ring}} \text{End}_{M_n(D)}(S)$$

as we sought to show.

The next thing we will need is to show that if R, S are semisimple rings, then so is $R \times S$. Since R and S are semi-simple R -modules and S -modules, if we can show that $R \times S$ is a semi-simple $(R \times S)$ -module, we are done. For this to be the case, we must be able to decompose $R \times S$ into simple $(R \times S)$ -submodules. Fortunately, many universal properties play in our favour to be able to easily identify simple $(R \times S)$ -modules of $R \times S$ given we know the simple R -modules and S -modules. To see the key idea of the following proof, we define some terms

Definition 1.2.9: Idempotent And Orthogonal Elements

Let R be a ring. Then $e \in R$ is said to be *idempotent* if

$$e^2 = e$$

Furthermore, if e_1, e_2 are idempotent elements in R , if

$$e_1 e_2 = e_2 e_1 = 0$$

then they are said to be *orthogonal*. If e is idempotent and cannot be written as the sum of two (commuting) idempotent elements, then e is said to be a *primitive* idempotent element.

Example 1.2.10: Idempotent Elements

Naturally, for any ring R , $1 \in R$ is an idempotent element.

For a more interesting example the product ring $R \times S$, and take $e = (1, 0) \in R \times S$, $f = (0, 1) \in R \times S$. Then:

1. $e^2 = e$ and $f^2 = f$ (i.e. they are both *idempotent*)
2. $ef = fe = 0$ (implying also that $ef \in Z(R \times S)$)
3. $e + f = 1$ (i.e. e and f are *not* primitive)

What happens if the number of terms in the product increases?

Next, if M is an R -module and N is an S -module, we can M and N to be $(R \times S)$ -module by ignoring what happens to S (resp. R), i.e. think of the action being defined by the following function

$$R \times S \rightarrow R \rightarrow \text{End}_{\mathbf{ab}}(M)$$

and similarly for a S -module N . Hence, we can define $M \oplus N$ as an $(R \times S)$ -module. Conversely, if M is an $(R \times S)$ -module, then using what we've shown in example 1.2.10, given $e = (1, 0)$ and $f = (0, 1)$:

$$M = eM + fM$$

Notice that eM is killed by Rf , so it's a $A \cong A \times B/(\{0\} \times B)$ -module, and similarly for fM . In other words, idempotent elements characterize the splitting of modules! This gives us the following:

Proposition 1.2.11: Direct Product Decomposition

If $A\text{-Mod}$ and $B\text{-Mod}$ are the categories of A -modules and B -modules, and $A\text{-Mod} \times B\text{-Mod}$ is the category where every object and morphism is a product of objects or morphisms. Then there is an equivalence:

$$A\text{-Mod} \times B\text{-Mod} \xrightleftharpoons[\psi]{\varphi} (A \times B)\text{-Mod}$$

sending:

$$\begin{aligned} (M, N) &\mapsto M \oplus N \\ (eM, fM) &\mapsto M \end{aligned}$$

where we further get: $\varphi(\psi(M \oplus N)) \cong M \oplus N$ and $\psi(\varphi(M)) \cong M$

Proof:

easy to check

Proposition 1.2.12: Classifying Simple $(A \times B)$ -Modules

Let P be a simple $(A \times B)$ -module. Then

$$P \cong_{A \times B\text{-Mod}} M \oplus (0) \quad \text{or} \quad P \cong_{A \times B\text{-Mod}} (0) \oplus N$$

where M is a simple A -module and N is a simple B -module.

Proof:

By proposition 1.2.11, $P \cong_{A \times B\text{-Mod}} M \oplus N$ where M is an A -module and N is a B -module. But then:

$$P \cong_{A \times B\text{-Mod}} (M \oplus (0)) \oplus ((0) \oplus N)$$

since P is simple, one of these has to be zero, and the other one has to be simple.

Corollary 1.2.13: Product Of Semi-Simple Rings

Let R and S be semi-simple rings. Then $R \times S$ is a semi-simple ring. Furthermore, if $R_1, R_2, \dots, R_n$ are the simple R -module up to isomorphism and $S_1, S_2, \dots, S_m$ are the simple S -modules, then the simple $R \times S$ -modules are of the form:

$$R_i \oplus (0) \quad (0) \oplus (S_j)$$

Proof:

By proposition 1.2.11, since R is a semi-simple R -module and S is a semi-simple S -module, $R \times S$ is a semi-simple $(R \times S)$ -module, and so $R \times S$ is a semi-simple ring. Then by proposition 1.2.12,

if $P \subseteq R \times S$ is a simple $(R \times S)$ -module, then they it must be of the form:

$$R_i \oplus (0) \quad (0) \oplus (S_j)$$

Corollary 1.2.14: Structure Of $\prod_i M_{n_i}(D_i)$: Artin Wedderburn part II

Let $R = M_{n_1}(D_1) \times \cdots \times M_{n_k}(D_k)$. Then

1. R is semi-simple
2. There exist exactly k simple R -modules up to isomorphism, each of the form:

$$0 \oplus \cdots \oplus D_i^{\oplus n_i} \oplus \cdots \oplus 0$$

Proof:

The ring R is semi-simple by corollary 1.2.13 using induction. Furthermore, each $M_{n_i}(D_i)$ has $D_i^{\oplus n_i}$ is it's unique simple $M_{n_i}(D_i)$ -module, since there are k matrix rings in the product, there must be exactly k simple R -modules, as we sought to show.

1.2.15 Remark We can as before get that

$$R^{\text{op}} \cong M_{n_1}(D_1^{\text{op}}) \times \cdots \times M_{n_k}(D_k^{\text{op}})$$

and hence R^{op} is left Noetherian/Artinian, which we know now is true if and only if R is right Noetherian/Artinian (Since a left R^{op} -module is a right R -module). By leftNoethArtRemark, we have that any semisimple ring is left-Noetherian/Artinian, and hence any semisimple ring is both left and right Noetherian/Artinian, even though R is not necessarily commutative!

Thus, we have the following statemnt, were the last thing we need to prove is uniqueness

Theorem 1.2.16: Artin-Wedderburn

Let R be a ring. Then R is a semi-simple ring if and only if

$$R \cong_{\text{Ring}} M_{n_1}(D_1) \times \cdots \times M_{n_k}(D_k)$$

where each D_k is a division ring and $M_{n_i}(D_i)$ is the matrix ring over D_i , and each D_i n_i are unique up to isomorphism

Proof:

All that is left to prove is the uniqueness. Let:

$$M_{n_1}(D_1) \oplus \cdots \oplus M_{n_s}(D_s) \cong_{\text{Ring}} R \cong_{\text{Ring}} M_{n'_1}(D'_1) \oplus \cdots \oplus M_{n'_t}(D'_t)$$

By corollary 1.2.14, R has exactly s and t simple module, and so $s = t$. Hence, for any $D_i^{\oplus n_i}$, there must exist a $(D'_i)^{\oplus n'_i}$ such that

$$D_i^{\oplus n_i} \cong_R (D'_i)^{\oplus n'_i}$$

Since $D_i^{\oplus n_i} \cong_R (D'_i)^{\oplus n'_i}$, $\text{End}_{M_{n_i}(D_i)}(D_i^{\oplus n_i}) \cong_{\mathbf{Ring}} \text{End}_{M_{n'_i}(D'_i)}((D'_i)^{\oplus n'_i})$. By corollary 1.2.8,

$$D_i^{\text{op}} \cong_{\mathbf{Ring}} \text{End}_{M_{n_i}(D_i)}(D_i^{\oplus n_i}) \cong_{\mathbf{Ring}} \text{End}_{M_{n'_i}(D'_i)}((D'_i)^{\oplus n'_i}) \cong_{\mathbf{Ring}} (D'_i)^{\text{op}}$$

which, taking op on both sides, we see that:

$$D_i \cong_{\mathbf{Ring}} D'_i$$

Finally, we need to show that $D^{\oplus n} \cong_R D^{\oplus m}$ implies $n = m$. Treating $D^{\oplus n}$ and $D^{\oplus m}$ as D -modules, then we get that each have a basis of dimension n and m . Importantly, the results of the Replacement Theorem (??) applies to division rings (notice that commutativity was not necessary), and so ?? and ?? give us that $n = m$, completing the proof.

This theorem also shows us that if R is a [finitely-generated] semi-simple ring, then R is indeed the direct product of simple rings (since $M_{n_i}(D_i)$ is a simple ring). Notice too that this theorem just gave us the answer to the classification of all *finite simple rings*: namely they would have to be of the form $M_n(\mathbb{F}_q)$ ⁹ for some finite field of order q (which is a power of some prime p). Contrast this to how hard it is to find all finite simple groups! Furthermore, the finite dimensional assumption was key: the *Weyl Algebra*:

$$W = k\langle x, y \rangle / (xy - yx - 1)$$

is a simple algebra that is not the matrix ring of a division algebra.

Example 1.2.17: Artin Wedderburn

$$\text{End}_{M_2(k)}(M_2(k)) = \text{End}_{M_2(k)}((k^2)^{\oplus 2}) = M_2(\text{End}_{M_2(k)}(k^2)) = M_2(k)$$

Corollary 1.2.18: Artin-Wedderburn For k -Algebras

Let R be a semi-simple k -algebra for some field k . Then

$$R \cong_{k\text{-Alg}} M_{n_1}(D_1) \times \cdots \times M_{n_l}(D_l)$$

where each D_i is a division k -algebra, finite dimensional as a k -vector space:

$$\dim_k R = \sum n_i^2 \dim_k(D_i)$$

and has exactly l simple R -modules as we've classified before.

Proof:

By the Artin-Wedderburn theorem, we have that

$$R \cong_{\mathbf{Ring}} M_{n_1}(D_1) \times \cdots \times M_{n_l}(D_l)$$

Let's now further assume R is a k -algebra for some field k , so that there is a map $k \hookrightarrow Z(R)$.

⁹Why is the division ring a field? The answer lies in ??

Then by the above isomorphism:

$$\begin{aligned} k \mapsto Z(R) &\cong Z(M_{n_1}(D_1) \times \cdots \times M_{n_l}(D_l)) \\ &\cong Z(M_{n_1}(D_1)) \times \cdots \times Z(M_{n_l}(D_l)) \\ &\cong Z(D_1) \times \cdots \times Z(D_l) \end{aligned}$$

where the last one comes from the fact that $Z(M_{n_i}(D_i))$ are the diagonal elements, and $z \mapsto zI$. This shows that D_i each have a natural k -algebra structure induced by R , and so the above isomorphism was already a k -algebra isomorphism:

$$R \cong_{k\text{-Alg}} M_{n_1}(D_1) \times \cdots \times M_{n_l}(D_l)$$

Finally, treating $M_{n_i}(D_i)$ as k -vector spaces we have:

$$M_{n_i}(D_i) \cong_{k\text{-Vect}} D^{\oplus n_i^2}$$

which gives the desired result, completing the proof.

Corollary 1.2.19: Special Case: k Algebraically Closed

Let $k = \bar{k}$ be an algebraically closed field and D a k -algebra that is finite-dimensional as a k -vector-space. Then $D = k$

Proof:

Let's say $k \subsetneq D$. Then pick any $\delta \in D \setminus k$. Then

$$k \subseteq k[\delta] \subseteq D$$

the key thing to observe is that $k[\delta]$ is still commutative!! This is essentially the same reasoning for why any cyclic group is commutative! Since D is a finite-dimensional vector-space over k , some power of δ^n must be a k -linear combination of every smaller term. Since D is a division ring, every δ has an inverse, and so every element of $k[\delta]$ is in fact a unit. Hence $k[\delta]/k$ is an algebraic field extension. However, k is an *algebraically closed field*, and hence $k[\delta] = k$. Since δ was arbitrary, we must have that $D = k$, as we sought to show.

Corollary 1.2.20: Artin-Wedderburn For $\bar{k}$ -Algebras

Let R be a semi-simple k -algebra for some algebraically closed field $k = \bar{k}$. Then

$$R \cong_{k\text{-Alg}} M_{n_1}(k) \times \cdots \times M_{n_l}(k)$$

showing that R is a finite dimensional vector-space over R with dimension:

$$\dim_k(R) = \sum_i^l n_i^2$$

and has exactly l simple R -modules as we've classified before.

As a consequence of this, if R is a semi-simple $\mathbb{C}$ -algebra, the simple R -modules are all R -isomorphic to $\mathbb{C}^{n_i}$ for appropriate power n_i . As a final word of warning, it is natural to wonder if all simple rings are of the form $M_n(D_i)$

Exercise 1.2.1

1. Let k be a field, k^n be a direct product of fields a k^n -module over itself. Let $0 \oplus \cdots \oplus k \oplus \cdots \oplus 0 = k_i \subseteq k^n$. Then $\text{End}_{k^n}(k_i) \cong_{\mathbf{Ring}} k$, $\text{End}_{k^n}(k^n) \cong_{\mathbf{Ring}} k^n$, and $\text{End}_{k^n}(k^n) \cong_{\mathbf{Ring}} M_n(\text{End}_{k^n}(k_i)) \cong_{\mathbf{Ring}} M_n(k)$. But that would imply $k^n \cong_{\mathbf{Ring}} M_n(k)$ – what's the contradiction?
2. Let k be an algebraically closed field, and let R be a finite-dimensional k -algebra. Let M be a simple R -module. Prove that if r is an element of the centre $Z(R)$ of R , there exists $c \in k$ such that $rm = cm$ for all $m \in M$

1.3 Central Simple Algebras**Definition 1.3.1: Algebra Representation**

Let A be a finite dimensional (associative) algebra over k . Then a *representation* of A is a k -homomorphism:

$$\rho : A \rightarrow \text{End}_k(V)$$

for some vector space V .

Recall that one definition of an R -module is given by a ring homomorphism $\rho : R \rightarrow \text{End}_{\text{Ab}}(M)$ for an abelian group M . If we restrict to k -homomorphisms and k -endomorphisms, we can define the A -module $\rho : A \rightarrow \text{End}_k(V)$, hence we can think of representation of an algebra A as a left A -modules. This makes sense, as representation theory can be thought of the study of associative algebras (the study of group representation is the study of group algebras, the study of the representation of lie algebras is universal enveloping algebras, the study of the representation of quiver algebras is the study of path algebras, and so forth). Because of this, we shall see that many of the results for the representation of associative algebras outlined below to be familiar to anyone who has taken a basic course in representation theory.

Next, we define the object that shall represent the generalization of number rings:

Definition 1.3.2: Central Simple Algebra

Let A be an algebra over a field k . Then A is called *simple* if it is a simple ring. It is called a *central simple algebra* (CSA) if it is simple, is finite dimensional over k , and its center is k .

The idea behind this definition is that it generalizes field extensions F/k , for we have a distinguished field k , and we are interested in the “algebraic case”, hence the extension ought to be finite. Contrasting to fields, a CSA need not be commutative and need not be a division ring.

Example 1.3.3: Simple and Central Simple Algebras

1. Naturally, $M_n(k)$ is a central simple algebra of dimension n^2
2. It is possible to be simple and have center k but not be finite: the Weyl algebra $k[x, \delta_x]$ is infinite dimensional over k .
3. $\mathbb{C}$ is a central simple algebra over $\mathbb{C}$, but not over $\mathbb{R}$ (as its center contains more than $\mathbb{R}$
4. Let $A = \mathbb{H}$ be the quaternions (or Hamiltonians). Then this is a CSA over $\mathbb{R}$ with dimension 4. We shall soon define a quaternion algebra, and see that all 4-dimensional central simple algebras over a field k is a quaternion algebra, and it shall be limited to 2-by-2 matrices or division algebras.

Let us now generalize some of basic representation theory results. Let E be an irreducible representation of A (take for example a minimal left ideal of A).

Lemma 1.3.4: Schur’s Lemma - updated

Let E be an irreducible representation of a k -algebra A , and let $K \subseteq \text{End}_k(E)$ be the endomorphisms that commute with the operations of A : it is the centralizer of A in $\text{End}_k(E)$, $K = C_{\text{End}_k(E)}(A)$. Then K is a division ring.

Note that this makes K an A -algebra

Proof:

Take $u \in K$. We want to show that $u^{-1} \in K$. Consider $u^{-1}(0) = V \subseteq E$. For $x \in V$, we have:

$$0 = a \cdot u(x) = u(a \cdot x)$$

hence, V is stable under A , and as $u \neq 0$ we must have $V = 0$ as u is invertible in $\text{End}_k(E)$. It follows that $u^{-1} \in K$.

Notice from this that we may say that the representation E must be faithful (the map must be injective). Indeed, the kernel of this representation would be an ideal of A but

Lemma 1.3.5: Minimal Centralizer of Representations

Let E be a semi-simple and faithful representation of A , $K = C_{\text{End}_k(E)}(A)$, $B = C_{\text{End}_k(E)}(K)$ (i.e. the centralizer of A and K in $\text{End}_k(E)$). Then

$$A = B$$

Proof:

Let $x \in E$ and $b \in B$. We shall show that there exists an $a \in A$ such that $a(x) = b(x)$.

Let $Ax = V \subseteq E$ be a stable subspace. Then V admits a (stable) complement: $E = V \oplus V^\perp$. Let $\pi : E \rightarrow V$ be the natural projection map. Then certainly π is A -linear and commutes with K , namely $\pi \in K$. Then:

$$\pi(b(x)) = b(\pi(x)) = b(x)$$

namely, $b(x) \in V = aX$, implying $a(x) = b(x)$, as we sought to show.

Lemma 1.3.6: Generator lemma

Let E be a semi-simple and faithful representation of A , $K = C_{\text{End}_k(E)}(A)$, $B = C_{\text{End}_k(E)}(K)$ (i.e. the centralizer of A and K in $\text{End}_k(E)$). Let $x_1, \dots, x_n \in E$ and $b \in B$. Then there exists values $a_i \in A$ such that:

$$ax_1 = bx_1, \quad ax_2 = bx_2 \quad \dots \quad ax_n = bx_n$$

Proof:

Let F be the direct sum space of n copies of representation E . Then F is formed of elements $(y_1, \dots, y_n)$, $y_i \in E$, on which A acts by

$$a \cdot (y_1, \dots, y_n) = (ay_1, \dots, ay_n)$$

The representation F is still semi-simple. The centralizer of A in $\text{End}_k(F)$ is formed of matrices (u_{ij}) , $u_{ij} \in K$, and it follows that, if we let B operate on F by the formula:

$$b \cdot (y_1, \dots, y_n) = (by_1, \dots, by_n)$$

the elements of $\text{End}_k(F)$ thus defined will still be in the centralizer of the centralizer of A (in the theory of semi-groups, this is sometimes called the bicommutant). By lemma 1.3.5 applying to the point $x = (x_1, \dots, x_n) \in F$, and to $b \in B$ giving us the desired result.

Serre writes K_n for $M_n(K)$ for a division ring K , but we shall stick to $M_n(D)$ where our division ring is D instead of K . If D contains k in its center, we have:

$$M_n(D) = M_n(k) \otimes D$$

as can be easily seen.

We present one more important result that has two useful corollaries. Let A be a k -algebra. Then A acting on itself by left multiplication gives something called the *regular left representation*. If

$A = M_n(D)$, it has basis e_{ij} , $1 \leq i, j \leq n$. The elements e_{i,j_0} for a fixed j_0 gives a subspace of $M_n(D)$, say $N_{j_0} \subseteq M_n(D)$. It is easy to see that N_{j_0} is a minimal left ideal (namely, an irreducible representation). Then we see that for any other choice of j_0 , say j_1 , we get an isomorphic irreducible representation N_{j_1} . Then this gives the following classical result:

Theorem 1.3.7: Representation of Simple Algebra

Let A be a simple k -algebra. Then all representation of A is isomorphic to the direct sum of irreducible representation

Proof:

use corollary 1.2.20.

Corollary 1.3.8: Representation of Simple Algebra

Let A be a simple k -algebra. Then:

1. All representations of A are semi-simple (completely reducible in Serre's words)
2. Two representation of A which have the same dimension are isomorphic

Proof:

immediate from the above

Let us now turn towards the tensor product of simple algebras.

Lemma 1.3.9: Centralizer of Tensor Product of Algebras

Let A, A' be two k -algebras, B, B' be sub k -algebras of A, A' (respectively) with C, C' their centralizers on A, A' (respectively). Then the centralizer of $B \otimes B'$ in $A \otimes A'$ is $C \otimes C'$.

Proof:

Let us first find the centralizer of $B \otimes 1$. For this, let a'_i be a basis of A' . Any $x \in A \otimes A'$ can be written uniquely in the form:

$$x = \sum_i a_i \otimes a'_i \quad \text{for appropriate } a_i \in A$$

If x commutes with $b \otimes 1$, then $a_i b = b a_i$ that is $a_i \in C$ for all i . We thus see that the centralizer of $B \otimes 1$ is $C \otimes A'$. Similarly, that of $1 \otimes B'$ is $A \otimes C'$. It follows that the centralizer of $B \otimes B'$ is equal to the intersection of the centralizers of $B \otimes 1$ and $1 \otimes B'$, namely

$$(C \otimes A') \cap (A \otimes C') = C \otimes C'$$

as we sought to show.

Thus, the centre of the tensor product of two algebras is the tensor product of the centre of the two algebras. Applying this to the special case of $A = M_n(k) \otimes D$ for some division ring D , then:

$$Z(A) = k \otimes Z(D)$$

Corollary 1.3.10: Centre of simple algebra is a Field

Let A be a simple k -algebra, finitely generated over k . Then $Z(A)$ is a field.

Proof:

Consider $Z(A)$. As A is a finitely generated k -algebra, $Z(A)$ is a finitely generated k -algebra. For any $0 \neq z \in Z(A)$, consider zA . As z is central, we have that zA is two-sided ideal. As A is simple, it must be that $zA = A$, that is z is there exists a $w \in A$ such that $za = 1$. Therefore, $Z(A)$ is a division ring. It is by definition commutative, and hence it is a field. Since z was an arbitrary nonzero element, $Z(A)$ is a division ring. It is by definition commutative, and hence it is a field.

We shall next require to see the interaction between the tensor product of central simple algebras. We require the following lemma:

Theorem 1.3.11: Ideal Generation Tensor Product

Let A be a k -algebra and K' a left division ring with center k . Then every ideal $N \subseteq A \otimes K'$ is generated (as a left module over K') by its intersection with $A \otimes 1$.

Note that neither A nor K' are assumed finite over k

Proof:

We will first note that we can make K' operate on the left on $A \otimes K'$ by:

$$k'(a \otimes k'') = a \otimes k'k''$$

If N is an ideal of the algebra $A \otimes K'$, it is in particular a K' -module subspace of $A \otimes K'$.

Let a_i be a basis of A over k ; the elements $a_i \otimes 1$ equally form a basis of the K' -module $A \otimes K'$. Let x be a primitive element of N with respect to this basis (Bourbaki, Alg.II, paragraph 5):

$$x = \sum_i a_i \otimes k'_i$$

Consider the element $x \cdot m$ ($m \in K'$); we have $x \cdot m \in N$ (N is an ideal), and on the other hand:

$$x \cdot m = \sum_i a_i \otimes k'_i m$$

It then follows from the properties of primitive elements that we have:

$$k'_i m = m k'_i \text{ for all } i \quad (m \in K')$$

Since one of the k'_i is equal to 1, we have $m = n$, and the preceding equality simply means that k'_i commutes with all $m \in K'$, i.e., that $k'_i \in k$. We can then write: $x = a \otimes 1$, by setting: $a = \sum_i k'_i a_i$. We have thus shown that every primitive element x is in $A \otimes 1$; the theorem follows immediately since every vector space is generated by its primitive elements.

Theorem 1.3.12: Tensor Product CSA

Let A, A' be simple k -algebras finitely generated over k , where one of them is a CSA. Then $A \otimes A'$ is a simple k -algebra

Proof:

Suppose that $A' = M_n(k) \otimes K'$ is central, in other words that the center of the left field K' is reduced to k . It will suffice to show that $A \otimes K'$ is a simple algebra. Indeed, if this is demonstrated, $A \otimes K' = M_m(k) \otimes K''$, and $A \otimes A' = M_m(k) \otimes M_n(k) \otimes K'' = M_{mn}(k) \otimes K'' = M_{nm}(K'')$. The algebra $A \otimes A'$ will thus be isomorphic to a matrix algebra for which simplicity is immediately verified. Then the general case of $A \otimes K'$ being simple follow from theorem 1.3.11, completing the proof.

Corollary 1.3.13: Tensor Product of Two CSA's

The tensor product of two CSA is a CSA

Proof:

In the previous proof, it was enough for one of them to be a CSA.

Corollary 1.3.14: Opposite Algebra Tensoring

Let A be a CSA over k and A^{op} the opposite algebra. Then:

$$A \otimes A^{\text{op}} \cong M_{n^2}(k)$$

Proof:

Let us denote by E the algebra A considered as a vector space over k . We can make A naturally on E on the left or right by multiplication to create a left or right A -module. This amounts to identifying A and A^{op} as sub-algebras of the algebra $\text{End}_k(E)$ of k -endomorphisms of E . Since the algebras A and A^{op} commute in $\text{End}_k(E)$, we can define a canonical homomorphism of $A \otimes A^{\text{op}}$ into $\text{End}_k(E)$ which extends the isomorphisms: $A \rightarrow \text{End}_k(E)$ and $A^{\text{op}} \rightarrow \text{End}_k(E)$. The algebra $A \otimes A^{\text{op}}$ being simple, this homomorphism is an isomorphism; moreover, as;

$$[A \otimes A^{\text{op}} : k] = [A : k]^2 = n^2 = [\text{End}_k(E) : k]$$

this isomorphism maps $A \otimes A^{\text{op}}$ onto $\text{End}_k(E) = M_{n^2}(k)$, as we sought to show.

1.3.1 Brauer Group

Let A be a CSA over k . As A is simple, we get by Artin-Wedderburn that it is isomorphic to $M_n(D) \cong M_n(k) \otimes D$. For example, if $n = 1$ and $k = \mathbb{R}$ we get (by a classical theorem from algebraic topology) that $A \cong \mathbb{R}$ or $A \cong \mathbb{H}$ (the quaternions). If $n > 1$, we get $M_n(\mathbb{R})$ and $M_n(\mathbb{H})$ (which are no longer division algebras). We see that the division ring is one of the “characterizing” feature of

the CSA (the other being its matrix ring), especially that up to Morita-equivalence we get the same category. We thus have the following natural equivalence we may define:

Definition 1.3.15: Brauer Equivalence

Let $k\text{-CSA}$ be a collection of central simple algebras over k . Then define the equivalence relation $A \sim A'$ where the two are equivalent if and only if their associated division rings are isomorphic.

Proposition 1.3.16: Brauer Group

The Brauer equivalence on a collection of CSA's forms a commutative group under the tensor product. This group will be denoted $\text{Br}(k)$.

In Serre, this group is denoted G_k .

Proof:

By corollary 1.3.13, it is closed under the tensor product, the equivalence class containing k is certainly the identity, and corollary 1.3.14 shows that every equivalence class has an inverse. Associativity and commutativity is a classical result from tensor products.

Example 1.3.17: Brauer Groups

1. Let $k = \bar{k}$. Then $\text{Br}(\bar{k}) = 0$ by corollary 1.2.20.
2. If $k = \mathbb{R}$, then we shall show in the next section that $\text{Br}(\mathbb{R}) = \mathbb{Z}/2\mathbb{Z}$, namely we shall have $\mathbb{R}$ and $\mathbb{H}$, the quaternions.
3. Let k be a finite field. Then $\text{Br}(k) = 0$, as all finite division rings are fields.
4. If $k = \mathbb{Q}_p$, then $\text{Br}(k) \cong \mathbb{Q}/\mathbb{Z}$ (Serre references Algebren - Chap. VII - 2nd paragraph).
5. If $k = \mathbb{Q}$, $\text{Br}(\mathbb{Q})$ will be a subgroup of

$$\mathbb{Z}/2\mathbb{Z} \oplus (\mathbb{Q}/\mathbb{Z})^{\oplus \mathbb{N}}$$

which can be thought of as a result similar to that in number theory where we study the “global field” $\mathbb{Q}$ by looking at all its localizations (note that $\mathbb{Z}/2\mathbb{Z}$ comes from $\text{Br}(\mathbb{R})$!) Serre references Deuring for the proof (see Sere p.8)

Theorem 1.3.18: Field Extensions and CSA

Let A be a CSA over k , L/k a field extension (either finite or infinite). Then $A \otimes L$ is simple

Proof:

If A is a division ring, then all ideals of $A \otimes L$ will be given with in part with the intersection of the ideals of L , which is simply (0) , making $A \otimes L$ simple.

If A is any CSA, write $A \cong M_n(k) \otimes D$. Then:

$$A \otimes L \cong M_n(k) \otimes (D \otimes L) \stackrel{!}{=} M_n(k) \otimes M_m(D')$$

where for $\stackrel{!}{=}$, D' is the division ring naturally given by $D \otimes_k L$! But this gives the result.

Corollary 1.3.19: Index of CSA

Let A be a simple, central and finite algebra over k . Then $[A : k]$ is a perfect square.

Proof:

Let L be an algebraically closed extension of k . The algebra $A \otimes L$ can be endowed with an algebra structure over L , and we have:

$$[A \otimes L : L] = [A : k]$$

But $A \otimes L$ is a simple algebra, central over L , and finite over L . It then follows from example 1.3.17[1] that it is a matrix algebra over L , and $[A \otimes L : L]$ is indeed a perfect square.

Looking at the field extension of a k -algebra, it is natural to consider that $- \otimes_k L$ gives a homomorphism of Brauer groups $\varphi_{k,L} : \text{Br}(k) \rightarrow \text{Br}(L)$, namely due to the following result from linear algebra:

$$(A \otimes_k L) \otimes_L (A' \otimes_k L) \cong (A \otimes_k A') \otimes_k L$$

The kernel $\ker \varphi_{k,L}$ consists of all k -algebra's that become trivial (matrix algebras) when extending to L . Serre calls these algebras the algebras that *admit L as a field of decomposition*. (or admit a decomposition by L). More recently, the term *splitting field* is adopted due to the parallels to number theory. Naturally, if $L = \bar{k}$, then every algebra admits a decomposition as $\text{Br}(\bar{k}) = 0$. We only require a finite extension:

Theorem 1.3.20: Splitting Field is Finite

The Brauer group $\text{Br}(k)$ is the union of $\ker \varphi_{k,L}$ going over each L/k where L is a finite field extension. In particular, there exists a finite extension L/k such that for a CSA A over k , $[A \otimes_k L] = [1]$ in $\text{Br}(L)$.

Proof:

By example 1.3.17[1] we have that for a CSA A over $\bar{k}$, $A \otimes_k \bar{k} \cong M_n(D)$. Then the right hand side is generated by finitely many elements of the left hand side. We need only take the elements that map to the generators of the right hand side and to choose the appropriate field elements to get find L/k , as we sought to show.

2

Representation of Finite Groups

All modules will be left R -modules for a not necessarily commutative ring R . Furthermore, all actions will be non-trivial actions unless specified otherwise.

2.1 Representations of Finite Groups

Similarly to how we studied groups and by how they acted on themselves through group-actions (ex. how a group acted on itself by conjugation gave us Sylow's Theorem), we will now try to study groups using *linear algebra*. Many really difficult results in classical group theory will become much easier when studied through this new “representation” of groups (and hence why it's called *representation theory*). Furthermore, vector-spaces afford many more structures than a module structures (ex. a complex vector-space is the starting point for the Nullstellensatz, Complex Geometry, Analysis, and much more!) Hence, one large advantage of Representation theory will be the intersection of many mathematical disciplines which contribute towards our study (an they all feedback on each other)

In this part of the group, G will always be a finite group¹, k will always be a field (usually $\mathbb{C}$, but can be any field of any character), $\text{Aut}_{k\text{-Vect}}(V) = \text{GL}(V)$ represents all automorphisms from the k -vector space V to itself, which when $\dim(V) < \infty$ is equivalent to the group of all nonsingular linear transformations from V to itself $\text{GL}_n(k)$ where $n = \dim(V)$.

Definition 2.1.1: Representation

Let G be a finite group and V a vector space over k . Then a *linear representation* is any homomorphism $\varphi : G \rightarrow \text{GL}(V)$. The *degree* of the representation is the dimension of V . A *matrix representation* of G is any homomorphism $\varphi : G \rightarrow \text{GL}_n(k)$

¹Not sure about where to address infinite groups

Usually, when we say representation we will mean *linear representation*. If a vector-space V has a representation of G , we will often call it a G -representation. To be as rigorous as possible we would say we have a representation of G on V over k , if the context requires it. Furthermore, most of the time we will have map from G to $\text{GL}(V)$ to be *injective*; non-injective representations will be more common when representing infinite groups that are “trivial” on some subgroup (see [NatePadicRep]). Just like for group actions, A representation is *faithful* if it is injective.

As usual, we can curry or uncurry an action²: we can take $\varphi : G \rightarrow \text{GL}(V)$ and make it $G \times V \rightarrow V$ where $g : V \rightarrow V$ with $v \mapsto g \cdot v$ is linear over k for all g . It is left for the curious to see that this is indeed the proper uncurrying of this action.

Using the definition of $\varphi : G \rightarrow \text{GL}(V)$, it is easy to define a representation homomorphism mirroring the definition of group-actions and modules (which are ring actions on abelian groups):

Definition 2.1.2: Representation Homomorphism

Let V, W be two vectors spaces over k with a linear representation by G . Then $\varphi : V \rightarrow W$ a *representation homomorphism* if φ k -linear and for any $g \in G$:

$$\varphi(g \cdot v) = g \cdot \varphi(v)$$

such a homomorphism will be called G -linear.

From this perspective, it might be hard to see how our previous discussion on semi-simple rings might give us any insight. We thus introduce a perspective better suited to the results of chapter 1: every representation of G on V over k is in bijective correspondence with all $k[G]$ -modules over V , where $k[G]$ represents the *group ring* as seen in section ???. This second perspective will also make it easier to define a couple of examples, and so we will delay giving examples till the correspondence has been established. As a reminder, a group ring $k[G]$ over a ring k (which in our case will always be a field) is the set of all formal sums:

$$\sum_{g_i \in G} a_i g_i$$

where addition and multiplication are defined in the natural way³:

$$\begin{aligned} \sum_i^n a_i g_i + \sum_i^n b_i g_i &= \sum_i^n (a_i + b_i) g_i \\ \left(\sum_i^n a_i g_i \right) \left(\sum_i^n b_i g_i \right) &= \sum_k^n \left(\sum_{\substack{i,j \\ g_i g_j = g_k}} a_i b_j \right) g_k \end{aligned}$$

Another common notation for groups rings is kG , where the brackets are dropped. A couple of important facts about group rings should be recalled (or the excises ref:HERE can be completed)

1. The group G naturally injects into $k[G]$ by identifying g_i with $1g_i$ which is the formal sum $\sum_j \delta_{ij} g_j$ (where δ_{ij} is the Kronecker delta).

²See remark ??

³A word on this really being a natural way?

2. The field k naturally injects in $k[G]$ by corresponding an element $a \in k$ with $ag_1 = ae$ where $g_1 = e$ is the identity. If it is clear from context, we will sometimes simply write $a \in k[G]$ if it is understood $a = ae = ag_1$.
3. There is a natural k action on $k[G]$ defined by

$$b \cdot \left(\sum_i a_i g_i \right) = \sum_i (ba_i) g_i$$

due to this, $k[G]$ is a vector space over k with the elements of G as a basis, and hence is finite dimensional if $|G| < \infty$.

4. By definition, The elements of k identified as kg_1 commute with all the elements of $k[G]$, hence it is in the center of $k[G]$: $k \subseteq Z(k[G])$. Thus, $k[G]$ is in fact a k -algebra.
5. A group ring the adjoint given by the forgetful functor from **Ring** to **Grp**, hence is free with respect to groups, justifying the name “formal sum” for it’s elements.

Using these, we show how representations of G on V over k correspond to $k[G]$ -modules: let $\varphi : G \rightarrow \text{GL}(V)$ be a representation, with $G = \{g_1, g_2, \dots, g_n\}$. Thus, each $\varphi(g_i)$ is linear transformation (in particular an automorphism). Define the $k[G]$ action on V by defining:

$$\left(\sum_i^n a_i g_i \right) \cdot v = \left(\sum_i^n a_i \varphi(g_i) \right) (v) = \sum_i^n a_i \varphi(g_i)(v)$$

Then this defines a $k[G]$ -module (as should be verified if it doesn’t seem like it). A particular reminder for those who are checking that $\varphi(g_i g_j) = \varphi(g_i) \circ \varphi(g_j)$, since φ is a group homomorphism onto $\text{GL}(V)$ under composition. Notice how this action in fact extends the action of k on $k[G]$ since $a \cdot v = av$ (similarly to how the $F[x]$ action extended the action of F on V in section ??).

Conversely, let’s say V is an $k[G]$ -module. Seeking to make V a vector space over k and finding a group homomorphism $\varphi : G \rightarrow \text{GL}(V)$, the first thing is that V is already a K vector-space by limiting to the action of $k \subseteq k[G]$. Furthermore, for each $g \in G$, we can obtain a map $\varphi : V \rightarrow V$ by defining:

$$\varphi(g)(v) = g \cdot v$$

Then it is easy to check that $\varphi(g)$ is a linear transformation: for $\alpha, \beta \in k$, $v, w \in V$

$$\begin{aligned} \varphi(g)(\alpha v + \beta w) &= g \cdot (\alpha v + \beta w) \\ &= \alpha(g \cdot v) + \beta(g \cdot w) \\ &= \alpha\varphi(g)(v) + \beta\varphi(g)(w) \end{aligned}$$

and by the associativity axiom of modules, we get

$$\varphi(g_i g_j)(v) = (\varphi(g_i) \circ \varphi(g_j))(v)$$

Thus, $\varphi : G \rightarrow \text{GL}(V)$ is indeed a group homomorphism, and we have:

$$\left\{ V \text{ an } k[G]\text{-module} \right\} \xleftrightarrow{\quad} \left\{ \begin{array}{c} V \text{ a vector space over } k \\ \text{and} \\ \varphi : G \rightarrow \text{GL}(V) \text{ a representation} \end{array} \right\}$$

Note that this also means that this also shows that linear representations also have kernel objects, since modules have kernels. Hence, the kernel of a G -representation homomorphism between V and W is G -invariant.

Definition 2.1.3: Affording Representation

Let V be a $k[G]$ -module. Then given the previous discussion, we say that V *affords* the representation φ of G .

Given this new perspective, we can also characterize sub-actions. If we have a $k[G]$ -module V , then a $k[G]$ -submodule $W \subseteq V$ would have to be a k -submodule, and hence a subspace, but it also has to be closed under the action of g on W , i.e. $g \cdot W \subseteq W$. Hence, any $k[G]$ -submodule is a G -stable subspace of V . Conversely, if $W \subseteq V$ is G -stable, then it is closed under the action of G and is a subspace, and hence a $k[G]$ -submodule. Hence

$$\left\{ W \subseteq V \text{ a } k[G]\text{-submodule} \right\} \longleftrightarrow \{ W \subseteq V \text{ a } G\text{-stable subspace} \}$$

Next, we should know when two representations are equivalent. Recall that when studying a linear transformation T on V to itself, a basis for V can be found that gives a “canonical form” to the matrix representation of T . Changing basis for V kept V the same, but changed the matrix representation of T up to similarity (i.e. changed the isomorphism from $\text{GL}(V)$ to $\text{GL}_n(k)$). The following is the analogous term:

Definition 2.1.4: Equivalent Representation

Let φ and ψ be two representations of G over k . Then we say these two representations are *equivalent* (or *similar*) if the $k[G]$ -module affording them are isomorphic as modules. If two representations are not equivalent, they are called *inequivalent* (or *not similar*).

When the field is known^a, if V and W are equivalent (i.e. isomorphic as $k[G]$ -modules), then we write $V \cong_G W$.

^aIn most cases, the field will be $\mathbb{C}$

Note that two different representations can mean two different vector-spaces over k . If $\varphi : G \rightarrow \text{GL}(V)$ and $\psi : G \rightarrow \text{GL}(W)$ are equivalent representations, it would make sense if there is a relation between V and W . If $T : V \rightarrow W$ is the $k[G]$ -module isomorphism between them, It is a k -module isomorphism, and hence $V \cong W$ as vector spaces, implying $\dim(V) = \dim(W)$. Furthermore, by definition of T , $T(g \cdot v) = g \cdot T(v)$. By definition of the action, we have that $T(\varphi(g)(v)) = \psi(g)(T(v))$, i.e.

$$T \circ \varphi(g) = \psi(g) \circ T \quad \Leftrightarrow \quad T \circ \varphi(g) \circ T^{-1} = \psi(g) \quad \forall g \in G$$

That is, T is the *simultaneous change of basis* for all $\varphi(g)$, $g \in G$. In terms of matrices, there must exist a fixed invertible matrix P such that for all $g \in G$,

$$P\varphi(g)P^{-1} = \psi(g)$$

The function T (or matrix P) are said to *intertwine* φ and ψ (it gives the rules on how to relate φ and ψ).

Example 2.1.5: Representations

1. Let G be any finite group, V any 1 dimensional vector-space over k . Define the $k[G]$ -module structure on V by

$$g \cdot v = v$$

for all $g \in G, v \in V$. This called the *trivial representation* of G . This module affords the representation $\varphi : G \rightarrow \text{GL}(V)$ by $\varphi(g) = I$ (the corresponding matrix representation from G to $\text{GL}_1(k)$ sends $g \mapsto I$).

2. Let G be any finite group and k any field. Let $V = k[G]$ and consider it as a $k[G]$ -module over itself. Then $\dim_k(V) = |G| = n$, with the basis being the elements of G . Then we see that each $g \in G$ permute the basis elements:

$$g \cdot g_i = gg_i = g_j \in G$$

Thus, the matrix representation for the action by the element g is an $n \times n$ matrix with 1 in the ij component if $g \cdot g_j = g_i$ (hence n 1's) and zero everywhere else. This representation of g is called the *regular representation* of G . Note how the underlying field didn't matter: the coefficients in the matrices can only ever be 1 or 0, and we would simply say "Let φ be the regular representation of G " without specifying which field the group-ring is.

3. Let $G = S_n$ ($n \in \mathbb{N}_{>0}$) and k be any field. Then let V be a vector-space over k of dimension n , $(e_1, e_2, \dots, e_n)$ a basis for V , and define the action:

$$\sigma \cdot e_i = e_{\sigma(i)}$$

that is, σ permute the basis of V . Through this, we see that we afford a faithful representation $\varphi : S_n \hookrightarrow \text{GL}(V)$. Furthermore, since any finite group is a subgroup of S_n for an appropriate n , we see that a group of order n we can define $\varphi : G \rightarrow S_n \rightarrow \text{GL}(V)$, giving another perspective on how to embed any finite group into a matrix group. Notice too how we made a $k[G]$ -module given the action of a $k[S_n]$ -module. This is called the *permutation representation* of G .

Note that this means that *any* finite groups acts on a vector space, and by using this construction we see that any finite group G of size n can be embedded into $\text{GL}_n(k)$ showing how much structure the matrix groups embody.

4. Generalizing the last result, if $\psi : H \rightarrow \text{GL}(V)$ is any representation and $\varphi : G \rightarrow H$ is a group homomorphism, then $\psi \circ \varphi$ is a representation of G on V .
5. Recall that in section ?? we defined a *group character* to be a homomorphism from G to $L^\times$. This is in fact a 1-dimensional representation of G , since $\text{GL}_1(L) = L^\times$. For a concrete example, let $G = C_n$ be the cyclic group of order n and $\zeta_n \in \mathbb{C}^\times$ the n th root of unity. Then the map:

$$\chi : G \rightarrow \mathbb{C}^\times \quad a^i \mapsto \zeta_n^i$$

is a faithful representation of G .

6. Let $G = D_8$. We already have the regular representation of dimension 8 and permutation representation of dimension 8, but can we do better? In fact we can. In order to see this, let's first write out D_8 in terms of generators and relations:

$$D_8 = \langle r, s \mid r^4 = s^2 = 1, rs = sr^{-1} \rangle$$

Let $k = \mathbb{R}$ and consider $\mathrm{GL}_2(\mathbb{R})$. If we can find matrices that satisfy these relations, then we would have a faithful representation of degree 2. This can be laborious, but with some work we can find:

$$r \mapsto \begin{pmatrix} \cos(2\pi/n) & -\sin(2\pi/n) \\ \sin(2\pi/n) & \cos(2\pi/n) \end{pmatrix} \quad s \mapsto \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Let $G = Q_8$. We will show how different fields can produce different representations. First, recall the following two generator and relation set for Q_8 :

$$Q_8 = \langle i, j \mid i^4 = j^4 = 1, i^2 = j^2, i^{-1}ji = j^{-1} \rangle$$

$$Q_8 = \langle i, j, k \mid i^2 = j^2 = k^2 = ijk = 1 \rangle$$

If $k = \mathbb{C}$, and we work over $\mathrm{GL}_2(\mathbb{C})$ we can use the first representation and find:

$$i \mapsto \begin{pmatrix} \sqrt{-1} & 0 \\ 0 & -\sqrt{-1} \end{pmatrix} \quad j \mapsto \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

can you find one if $k = \mathbb{R}$ and $\mathrm{GL}_4(\mathbb{R})$? (Hint: think of the Hamiltonian's $\mathbb{H}$)

7. Let $H \trianglelefteq G$ which is further an abelian p -group for some prime p . Then $V = H$ is a vector-space over $\mathbb{F}_p$ with the action of $a \cdot v = v^a$. Then define the action of G on H by conjugation: $g \cdot h = ghg^{-1}$. This action is well-defined since $gh^a g^{-1} = (ghg^{-1})^a$. The kernel of this action is $C_G(H)$. Such an action is called a *module representation*.
8. Let V be the $k[S_n]$ -module which affords the permutation representation. We will find 2 S_n -invariant subspaces: Let $N \subseteq V$ be the subspace defined by:

$$N = \{ \alpha_1 e_1 + \cdots + \alpha_n e_n \mid \alpha_1 = \cdots = \alpha_n \}$$

Then this is a subspace that is also S_n -invariant. As practice, show that if $n \geq 3$, then N is the unique 1-dimension subspace which is S_n -stable. This submodule is called the *trace* submodule of V .

Another natural S_n -stable subspace is the following: take $I \subseteq V$ such that

$$I = \{ \alpha_1 e_1 + \cdots + \alpha_n e_n \mid \alpha_1 + \cdots + \alpha_n = 0 \}$$

notice that $I = \ker(\varphi)$ where $\varphi : V \rightarrow \mathbb{C}$, $\varphi(\alpha_1 e_1 + \alpha_2 e_2 + \cdots + \alpha_n e_n) = \sum_i \alpha_i$, and so I is $n - 1$ dimensional. Since φ is called the *augmentation map* (was in section 7.3 of D&F), I is called the *augmentation module*.

9. If V is a regular representation of G so that $V = k[G]$, then V has the same two natural G -invariant subspaces:

$$N = \{ \alpha_1 e_1 + \cdots + \alpha_n e_n \mid \alpha_1 = \cdots = \alpha_n \}$$

$$I = \{ \alpha_1 e_1 + \cdots + \alpha_n e_n \mid \alpha_1 + \cdots + \alpha_n = 0 \}$$

these are both in fact *two-sided ideals*, even if G is not abelian. Since they appear naturally, they are usually given a name: N is called the *trace ideal* and I is called the *augment ideal*.

Corollary 2.1.6: Schur's Lemma For Representations

Let V, W be irreducible G -representations (over $\mathbb{C}$) with representations ρ_V and ρ_W . Then:

1. If $V \not\cong_G W$, then there does not exist a non-trivial G -linear map between V and W
2. Let's say $V = W$, are finite dimensional vector-spaces with $\rho_V = \rho_W$, then the only non-trivial G -linear maps from V to W are scalars: $\lambda I : V \rightarrow W$, $\lambda I(v) = \lambda v$.

Proof:

1. Since V and W are irreducible G -representations, they are simple $k[G]$ -modules, and hence we apply the original Schur's lemma.
2. Let $V = W$ of the same dimension with matching actions $\rho_V = \rho_W$, and let $f : V \rightarrow W$. Since f is a linear transformation over $\mathbb{C}$, there always exists an eigenvalue, say λ so that $f(x) = \lambda x$. Consider:

$$\varphi = f - \lambda I$$

so that $\varphi(x) = 0$. Since the difference of two G -linear maps is G -linear, then $\ker(\varphi)$ is a G -invariant subspace. But V is irreducible, and $\ker(\varphi)$ is non-empty, and hence must be all of V , and so $\varphi(V) = 0$, telling us that $0 = f(x) - \lambda I(x)$, i.e.

$$f(x) = \lambda x$$

completing the proof.

Recall that if V is a simple R -module over a semi-simple ring R , then $\text{End}_R(V) \cong_{\text{Ring}} D^{\text{op}}$ is a division ring, and that any finite-dimensional division $\mathbb{C}$ -algebra is in fact equal to $\mathbb{C}$. This lemma now gives an explicit isomorphism between these two!

Since we are working over a vector-space, it is often convenient when possible to bring in an inner product that give further relations between our objects. The key the following inner product is that it is invariant on G -actions:

Definition 2.1.7: G -Invariant Inner Product

Let V be $k[G]$ -module. Then a G -invariant inner product is an inner product $\langle \cdot, \cdot \rangle$ along with the additional property of:

$$\langle g \cdot v, g \cdot w \rangle = \langle v, w \rangle$$

Instead of giving examples of this inner product, we will soon show if V is a semi-simple $k[G]$ -module, we can transform any inner product into a G -invariant inner product (hint: can you see how to use the “averaging” idea in Machke's Theorem to do so? The answer is given in corollary 2.1.12, page 49).

Now, the study of linear representations in themselves can be difficult; we are essentially studying all possible $\mathbb{C}[G]$ -modules. We would rather be able to decompose them into smaller pieces, and show that any representation of G is somehow built from these simpler pieces (similarly to our exploration of semi-simple modules). The rest of this chapter is dedicated to seeing if we can bring

Artin-Wedderburn's theorem and all of its consequences to understanding linear representations of finite groups:

Definition 2.1.8: Decomposing Representations

A representation is called *simple*, *reducible*, *indecomposable*, *decomposable*, or *semisimple* if the $k[G]$ -module affording it has the corresponding property

Recall that indecomposable in general does not imply irreducible. The example under definition 1.1.1 could be in fact a representation of $\mathbb{Z}$ defined by:

$$n \mapsto \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$$

which is reducible (since it is in block upper-triangular form) but not indecomposable (since it's not in block-diagonal form). Notice that any 1-dimensional representation is automatically an irreducible, since any $k[G]$ -submodule of k would be a k -subspace of k , but k is already irreducible as vector-space over k .

To get an idea of how $k[G]$ -submodules interact with the definitions above, let V be an n -dimensional representation of G affording the representation ρ , and $W \subseteq V$ an m -dimensional $k[G]$ -submodule (i.e. m -dimensional G -invariant subspace) with $0 \neq m \neq n$. Then choosing a basis $\{w_1, w_2, \dots, w_m\}$ for W , we can extend it to a basis $\{w_1, w_2, \dots, w_m, v_{m+1}, \dots, v_n\}$. Since W is G -invariant, for any $w \in W$, $\rho(g)(w) \in W$. In matrix representation, $\rho(g)(w^i) = \rho_j^i w^j$, and the coefficients of $v^{m+1}, \dots, v_n$ are zero. Hence we get the matrix representation of $\rho(g)$ as:

$$\begin{pmatrix} \alpha(g) & \beta(g) \\ 0 & \gamma(g) \end{pmatrix}$$

Notice too that:

$$\begin{aligned} \begin{pmatrix} \alpha(gh) & \beta(gh) \\ 0 & \gamma(gh) \end{pmatrix} &= \rho(gh) \\ &= \rho(g)\rho(h) \\ &= \begin{pmatrix} \alpha(g) & \beta(g) \\ 0 & \gamma(g) \end{pmatrix} \begin{pmatrix} \alpha(h) & \beta(h) \\ 0 & \gamma(h) \end{pmatrix} \\ &= \begin{pmatrix} \alpha(g)\alpha(h) & \alpha(g)\beta(h) + \beta(g)\gamma(h) \\ 0 & \gamma(g)\gamma(h) \end{pmatrix} \end{aligned}$$

which shows the the converse is also true: if we find a basis so that the matrices are block upper triangular, then the subspace spanned by the first m are G -invariant. As an example of this, let V be the permutation representation of S_3 . Then the set of all vectors that are invariant to the action of $G = S_3$ is $V^G = \text{span}(e_1 + e_2 + e_3)$. Let $W = \text{span}(e_1 + e_2 + e_3)$. Since $G \curvearrowright W$ is of the form $g \cdot w = w$, we see that W is in fact the *trivial representation*. Since V is semi-simple, we have that $V = V^G \oplus V^\perp$. Making some new basis for V , say $(e_1 + e_2 + e_3, e_2, e_3)$, we see that $S_3 \curvearrowright V$ will have matrix representations that are of the above form:

$$\begin{pmatrix} 1 & * & * \\ 0 & * & * \\ 0 & * & * \end{pmatrix}$$

If V were irreducible, then no such basis would exist. If $V = U \oplus W$, then we would get a block diagonal form. It turns out that all representations of groups over either a character zero ring or fields where $|G| \nmid \text{char}(k)$ are block-diagonal, i.e. are semisimple!

Theorem 2.1.9: Maschke's Theorem

Let G be a finite group and let k be a field whose characteristic does not divide $|G|$ or $\text{char}(k) = 0$. Then if V is a $k[G]$ -module, it is a semi-simple $k[G]$ -module.

The condition that $|G| \nmid \text{ch}(k)$ is equivalent to $|G| \in k^\times$. Note how this theorem is “surprising” as a finite group G certainly cannot be reduced to a direct sum of irreducible (or simple) groups, but as extensions of simple groups, but furthermore (as of writing this book) we do not know how to construct all finite group via extensions of simple groups. Hence, having a very computable decomposition into irreducible representations provides a very potent tool to analyze the structure of finite groups.

Proof:

We will use lemma 1.1.7[3] to show that for any $U \subseteq V$, there exists a complement W such that $V = U \oplus W$

We will find a projection onto U : $\pi : V \rightarrow U$. Thus, we need to show it has the following properties:

1. $\pi(u) = u$ for all $u \in U$
2. $\pi(\pi(v)) = \pi(v)$ for all $v \in V$

As a reminder on why this is sufficient^b, let π be such that the function defined on the $k[G]$ -module that satisfies these properties. Let $W = \ker(\pi)$. Then $W \subseteq V$ is a submodule, and W is the complement of U . First, let $v \in U \cap W$. Then since $v \in U$, $\pi(v) = v$ and since $v \in W$, $\pi(v) = 0$, hence $v = 0$, meaning $U \cap W = 0$. Next, let $v \in V$ be an arbitrary element, and write $v = \pi(v) + (v - \pi(v))$. By definition, $\pi(v) \in U$, and by the 2nd property:

$$\pi(v - \pi(v)) = \pi(v) - \pi(\pi(v)) = \pi(v) - \pi(v) = 0$$

and hence $v - \pi(v) \in W$. Hence $v \in U + W$, and so $V = U + W$, hence $V \cong U \oplus W$. Hence, it suffices to find an $k[G]$ -module projection π .

Let $U \subseteq V$ be a $k[G]$ -submodule. Then it is already a k -subspace, and so it has a vector-space direct sum complement $W_0 \subseteq V$ (take a basis β for U , extend it to a basis $\mathcal{B}$ for V , then take $\mathcal{B} - \beta$). Thus, $V = U \oplus W_0$ as a vector-space. However, W_0 need not be G -stable (i.e. a $k[G]$ -submodule), for example $G = C_2$ on $\mathbb{R}^2$ with the reflection across $y = x$ which has the G -subspace $y = x$ and complement spanned by $(1, 0)$ which is certainly not G -invariant. We will show how to find such the appropriate $k[G]$ -module given W_0 . First, let $\pi_0 : V \rightarrow U$ be the vector-space projection map:

$$\pi_0(u + w_0) = u$$

Here is the key to finding a proper complement: we will “average” π_0 over G as follows:

$$g\pi_0g^{-1} : V \rightarrow U \quad g\pi_0g^{-1}(v) = g \cdot \pi_0(g^{-1} \cdot v)$$

Since π_0 maps V to U , and U is a $k[G]$ -submodule, U is stable under this G -action, hence $g\pi_0g^{-1}$ is well-defined. Note that both g and g^{-1} act as k -linear transformations, so $g\pi_0g^{-1}$

is a k -linear transformation. Furthermore, if $u \in U$, then so must $g^{-1}u$, so by definition of π_0 , $\pi_0(g^{-1}u) = g^{-1}u$. Thus, for all $g \in G$:

$$g\pi_0g^{-1}(u) = u$$

hence, $g\pi_0g^{-1}$ is also a vector-space projection of V onto U !

Let $n = |G|$, and viewing n as an element of k ($n = 1 + \cdots + 1$), since $\text{ch}(k) \nmid |G|$ $n \neq 0$, i.e. $n \in k^\times$. Thus, n has an inverse in k , meaning we can define:

$$\pi = \frac{1}{n} \sum_{g \in G} g\pi_0g^{-1}$$

We can immediately deduce that π has the following properties:

1. By definition, π is still k -linear from V to U .
2. Restricting π to U would restrict each summand in the definition of π , and so as we've pointed out earlier each summand would be the identity map. Hence, we get $1/n$ times n copies of u , meaning $\pi|_U = \text{id}$
3. π is well-defined, hence $\pi(v) \in U$, so $\pi(v)^2 = \pi(v)$

It remains to show that π is in fact a $k[G]$ -module homomorphism (i.e. is $k[G]$ linear). It suffices to show that for any $h \in G$, $\pi(hv) = h\pi(v)$. This is simply a computation:

$$\begin{aligned} \pi(hv) &= \frac{1}{n} \sum_{g \in G} g\pi_0(g^{-1}hv) \\ &= \frac{1}{n} \sum_{g \in G} h(h^{-1}g)\pi_0(g^{-1}hv) \\ &= \frac{1}{n} \sum_{\substack{k=h^{-1}g \\ g \in G}} h(k\pi_0(k^{-1}v)) \\ &\stackrel{!}{=} h \frac{1}{n} \sum_{\substack{k=h^{-1}g \\ g \in G}} (k\pi_0(k^{-1}v)) \\ &= h\pi(v) \end{aligned}$$

where $\stackrel{!}{=}$ comes from the fact that as g goes over all elements of G , then so will $k = h^{-1}g$ (i.e. groups acting on themselves are always transitive actions). Hence, π is indeed a $k[G]$ -homomorphism, and so it is a projection map, completing the proof.

^aIn fact, this is implied by 1

^bsee ref:HERE for the original development of this

Thus, if we take any representation V over k of G , and consider the $k[G]$ -module, then we know through $k[G]$ and G being finite and k a field (so V is a vector-space) that there are only *finitely many* simple $k[G]$ -modules (i.e. finitely many irreducible representations of G), meaning we only

have a finite number of “building blocks” from which we can build) any representation for G ! Even better, the way we put together irreducible representation is through direct sums the simplest of ways of putting together algebraic objects! Then through the Artin-Wedderburn decomposition of V , we can find all *irreducible* representations that make up the representation V of G , giving us a simple finite list of irreducible representations per finite group! Note that if $p = \text{ch}(k) > 0$ and $p \nmid |G|$, then $k[G]$ is *not* semisimple, or if G is infinite, it need not be semi-simple

Example 2.1.10: Non-Simple Group-Ring

For the infinite case, we’ve presented $\mathbb{Z} \curvearrowright \mathbb{Z}^2$ above. For the finite case where $p \nmid |G|$,

1. let $G = C_n$ be the cyclic group of order n . Show that $k[C_p] \cong k[a]/(a^n - 1)$
2. Show that if $\text{char}(k) \nmid |G|$, then $k[C_n]$ is isomorphic product of fields. Since $M_1(k) = k$, by the uniqueness of Artin Wedderburn, this is the Artin-Wedderburn decomposition. Conclude that since $k[C_n]$ is abelian, any Artin-Wedderburn decomposition *has* to be a product of fields.
3. Show that if $\text{char}(k) = p$, then $k[C_p]$ has nilpotent elements, and hence cannot be a product of fields.

If $k = \mathbb{C}$, then Maschke’s Theorem always applies, and we can further use Artin Wedderburn over closed k -algebras (corollary 1.2.20) to deduce a relatively simple structure! Hence, from now on, we will always have $k = \mathbb{C}$, meaning all of our representations of finite groups will be over the complex vector-field of finite dimension.

Corollary 2.1.11: Matrix Representation Consequences

Let $\rho : G \rightarrow \text{GL}(V)$ be a representation over a field k where $\text{char}(k) \nmid |G|$. Then there exists a basis for V such that for all $g \in G$, the matrix $\rho(g)$ is of block-diagonal form:

$$\begin{pmatrix} \rho_1(g) & & & \\ & \rho_2(g) & & \\ & & \ddots & \\ & & & \rho_m(g) \end{pmatrix}$$

where ρ_i is an irreducible representation of G , $1 \leq i \leq m$.

Proof:

Let V be the $k[G]$ -module representing ρ . By Maschke’s Theorem, we may write $V = U_1 \oplus \cdots \oplus U_m$. where U_i is simple $k[G]$ -module. For each U_i , choose a basis β_i . Let $\beta = \bigcup_i \beta_i$. then each matrix representation of $\rho(g)$ is of the form in the corollary, with $\rho(g)|_{U_i} = \rho_i(g)$, completing the proof.

Corollary 2.1.12: Defining G -Invariant Inner Product

Let V be a $k[G]$ -module and $\langle \cdot, \cdot \rangle$ be an inner product. Then this inner product induces a G -invariant inner product

Proof:

If $\langle \cdot, \cdot \rangle$ is an inner product, then:

$$\langle v, w \rangle_G = \frac{1}{|G|} \sum_{g \in G} \langle gv, gw \rangle$$

is a well-defined inner product (as can be checked), and for any $h \in G$

$$\begin{aligned} \langle hv, hw \rangle_G &= \frac{1}{|G|} \sum_{g \in G} \langle ghv, ghw \rangle \\ &= \frac{1}{|G|} \sum_{k=gh \in G} \langle kv, kw \rangle \\ &= \langle v, w \rangle_G \end{aligned}$$

In other words, there is always an inner product on our space in which the group actions keeps all elements of the same length and the same angle between them, showing how we can interpret a group as somehow rotating and reflecting an object in a vector-space. Now that we know any representation of G is composed in some way by finitely many irreducible representations, it would be useful to be able to quickly deduce the Artin-Wedderburn decomposition of $\mathbb{C}[G]$:

Proposition 2.1.13: Properties Of Representations

Let $k = \mathbb{C}$ and G be finite. Let $S_1, S_2, \dots, S_r$ be the irreducible G -representations (up to isomorphism). Then:

1. the number of irreducible G -representations is equal to the number of conjugacy classes in G
2. For every representation V of G ,

$$V \cong S_1^{\oplus \ell_1} \oplus \dots \oplus S_r^{\oplus \ell_r}$$

where $\ell_i > 0$ and is unique

3. $\mathbb{C}[G] \cong \bigoplus_{i=1}^n \left(S_i^{\oplus \dim_k S_i} \right)$ as regular representation
4. $|G| = \sum (\dim_k S_i)^2$

Proof:

part (3) and (4) come from corollary 1.2.20.

For part (1), we first use Artin-Wedderburn to conclude that $Z(\mathbb{C}[G]) = \prod_i^r \mathbb{C}$, so $\dim_{\mathbb{C}}(Z(\mathbb{C}[G])) = r$, so the dimension of the center is equal to the number of irreducible G -representations.

Now, let $\mathcal{K}_1, \dots, \mathcal{K}_s$ represent the conjugacy classes (recall they partition G). Define:

$$\sigma_i = \sum_{g \in \mathcal{K}_i} g$$

Note that any pair of elements σ_i and σ_j have no common terms since the conjugacy classes partition G . Furthermore, by definition, for any $g \in G$, $g\sigma_i g^{-1} = \sigma_i$, and so $\sigma_i \in Z(\mathbb{C}[G])$. If the σ_i 's form a basis for $Z(\mathbb{C}[G])$, then we would have $s = t$. As a $\mathbb{C}$ -vector-space, they are already linearly independent since the group elements form a basis, so it suffices to show they span. If $\sum_{g \in G} \lambda_g a_g \in Z(\mathbb{C}[G])$. Then by definition for all $h \in G$,

$$\begin{aligned} \Leftrightarrow h \left(\sum_{g \in G} \lambda_g g \right) &= \left(\sum_{g \in G} \lambda_g g \right) h \\ \Leftrightarrow \left(\sum_{g \in G} \lambda_g g \right) &= h^{-1} \left(\sum_{g \in G} \lambda_g g \right) h \\ \Leftrightarrow \sum_{g \in G} \lambda_g g &= \left(\sum_{g \in G} \lambda_g (h^{-1} g h) \right) \\ \stackrel{!}{\Leftrightarrow} \sum_{g \in G} \lambda_g g &= \sum_{g \in G} \lambda_{hgh^{-1}} g \\ \Leftrightarrow \lambda_{hgh^{-1}} &= \lambda_g \quad \forall g \in G \end{aligned}$$

since h arbitrary, we see that the coefficients of elements in the same conjugacy class remain with an element that's in that conjugacy class. Equivalently, the function $G \rightarrow \mathbb{C}$, $g \mapsto \lambda_g$ is constant on conjugacy classes. Thus:

$$Z(\mathbb{C}[G]) = \text{span} \left\{ \sum_{g \in G} f([g])g \right\}$$

where f is a function from the conjugacy classes of G to $\mathbb{C}$. Hence $s = t$.

for (2), Since $\mathbb{C}[G]$ is a semi-simple ring, V is a semi-simple $\mathbb{C}[G]$ -module, giving us existence of the decomposition by AW. For uniqueness, first note that:

$$\text{End}_{\mathbb{C}[G]}(S_i) = \mathbb{C}$$

since $\mathbb{C}$ is the only [finite-dimensional?] division $\mathbb{C}$ -algebra. Then, we get by Schur's lemma:

$$\text{Hom}_{\mathbb{C}[G]}(S_i, V) \stackrel{\text{Schur}}{\cong} \text{Hom}_{\mathbb{C}[G]}(S_i, S_i)^{\oplus l_i} \cong \mathbb{C}^{l_i}$$

Corollary 2.1.14: Abelian Groups And Degree 1 Decomposition

1. If A is an abelian group, then every irreducible complex representation of A is 1-dimensional (i.e. a homomorphism from A to $\mathbb{C}^\times$), and A has $|A|$ inequivalent irreducible complex representations.
2. The number of inequivalent (irreducible) 1-dimensional complex representations of any finite group G is equal to $|G/G'|$

Proof:

1. Since A is abelian, so is $\mathbb{C}[A]$. By the Artin-Wedderburn decomposition, it must be a product of matrix rings, which since $\mathbb{C}[A]$ abelian is so must each matrix ring, which is only true if and only if $n_i = 1$. Then, by corollary 1.2.20, $\mathbb{C}[A]$ is a product of fields. It follows that any irreducible representation (i.e. simple $\mathbb{C}[A]$ -module) is 1-dimensional. Conversely, if A were not abelian, $\mathbb{C}[A]$ cannot be a product of fields, hence one of the dimensions of the matrix rings *must* be greater than 1
2. Let G be an arbitrary group. We want to show that there is a bijective correspondence between the 1-dimension representations of G and the 1-dimensional representations of G/G' . $\rho : G/G' \rightarrow \mathbb{C}^\times$ is a 1-dimensional representations of G/G' , then we have the natural map $G \rightarrow G/G'$, which shows that each 1-dimensional representations of G/G' is also a 1-dimensional representations of G . Conversely, if $\rho : G \rightarrow \mathbb{C}^\times$ is a 1-dimensional representations of G , then by the universal property, this factors through:

$$\begin{array}{ccc} G/G' & \longrightarrow & \mathbb{C}^\times \\ \uparrow & \nearrow & \\ G & & \end{array}$$

showing that any 1-dimensional representation of G is associated to a 1-dimensional representations of G/G' , showing there can't be more 1-dimensional representations of G than there are of G/G' , completing the proof.

A more general technique can be adopted from this proof: If $H \trianglelefteq G$ and V is an irreducible representation of G/H , then V is an irreducible representation of G .

Example 2.1.15: Decomposition Of Modules

1. DnF p. 847, and a bit about $\dim(V) = 1$ example right under definition

Finally, for those who read chapter ??, the following results are a glimpse at how representation theory is a “generalization” of number theory. First, if $H \leq G$ is a subgroup and (ρ, V) is an irreducible representation, $\rho|_H$ breaks up into the direct sum of representations, not necessarily of the same size or multiplicity. This is similar to how for a prime $\mathfrak{p} \subseteq F$ it may split to multiple primes with varying inertia degree and ramification numbers in a finite separable extension L/F . If L/F is furthermore normal (and hence Galois) then all the inertia degree's and ramification numbers match. In a similar vein:

Theorem 2.1.16: Clifford's Theorem

G be a group, $N \trianglelefteq G$ a normal subgroup, K a field, $\pi : G \rightarrow \text{GL}_n(K)$ be an irreducible representation. Then the restriction $\pi|_N$ breaks up into a direct sum of irreducible representations of N of equal dimensions (i.e. it is semisimple and each representation is of the same dimension). These irreducible representations of N lie in one orbit for the action of G by conjugation on the equivalence classes of irreducible representations of N (i.e. they are isotypic components with the same multiplicity).

Proof:

Let W be an irreducible subrepresentation of $V|_H$. Since H is normal in G , for $g \in G$, H acts irreducibly on gW by $(h, gw) \mapsto hgh^{-1}hw$. The subspace $\sum_{g \in G} gW$ is a nonzero subrepresentation of V . Since V is irreducible, it is equal to V . Since a representation which is a sum of irreducible sub-representations is semisimple, $V|_H$ is semisimple. A similar argument follows for the 2nd assertion, completing the proof.

Exercise 2.1.1

1. If χ_1 and χ_2 are both 1 dimensional representations, then $\chi_1 \cong \chi_2 \Leftrightarrow \chi_1 = \chi_2$. Use the property of equivalent representations to see this.
2. Let V be a non-trivial irreducible G -representation (i.e. not $\chi(g) = 1$ for all $g \in G$). Show that there are $\dim_{\mathbb{C}}(V^G) = 0$
3. By Cayley's Theorem, all finite groups are linear. Do there exist infinite non-linear groups? (Hard: Do there exist finitely generated non-linear groups?)
4. Show that if ρ is irreducible and $N \trianglelefteq G$ is normal, if N fixes a point v , N acts trivially on all of V .

2.2 Character Theory

We know now we can decompose a G -representation V into the direct sum of r different types of irreducible G -modules. How do we know proceed to find these? These can be quite unwieldy: if we have a 10 or 100-dimensional representation of a group G , then we have a matrix with 100 and 10'000 entries! We'd then have to multiply these and figure out what relations they have between each other, quite a hassle! Furthermore, how do we do this in such way that is independent of equivalent representations? A convenient basis might be nice, but to find a nice basis for every $\rho(g) \in \rho(G)$, we'd have to find a basis that can *simultaneously* diagonalize every element in $\rho(G)$! If there was some invariant we can find on a representation that we can use to keep track of the decomposition information between representations that is invariant of representation, that would be useful to this endeavour. The following does exactly this (Move the discussion of trace here to build-up to it? Give some of D&F examples to demonstrate it does get quite ugly sometimes?)

Definition 2.2.1: Class Function and Character

Let V be a representation of G over an arbitrary field F . Then

1. A function $f : G \rightarrow F$ that is constant on conjugacy classes is called a *class function* of G
2. If φ is a representation of G afforded by the $F[G]$ -module V , then a function:

$$\chi_V : G \rightarrow \mathbb{C} \quad g \mapsto \text{tr}(g : V \rightarrow V) = \text{tr}(\varphi(g))$$

is called the *character* of φ , where $\text{tr}(\varphi(g))$ is the trace of φ with respect to some basis for V . A character is called *irreducible* if φ is irreducible (and reducible otherwise). The *degree* of a character is the degree of any representation affording it.

For ease of reference, recall these properties of the trace:

1. $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$ and $\text{tr}(cA) = c\text{tr}(A)$
2. $\text{tr}(A) = \text{tr}(A^t)$
3. $\text{tr}(AB) = \text{tr}(BA)$

Note that in general, a character is *not* a homomorphism from G into the additive or multiplicative structure of k . Since $\text{tr}(AB) = \text{tr}(BA)$, $\text{tr}(ABA^{-1}) = \text{tr}(B)$, we see that the trace (and hence the character) is independent of choice of basis, and so equivalent representations have the same character, meaning the character is indeed an invariant. Furthermore, for any representation φ of G ,

$$\varphi(g^{-1}xg) = \varphi(g)^{-1}\varphi(x)\varphi(g)$$

hence, we can take the trace and use the 3rd property above to show that characters are an example of *class functions*. Furthermore, since the trace is linear, we can extend the domain of a character to be the trace over any element in $\mathbb{C}[G]$. If $F = \mathbb{C}$, then the collection of class functions form a complex vector-space, since the sum of two class functions is a class function, and multiplying by a scalar $z \in \mathbb{C}$ does not change that a class function is invariant.

Example 2.2.2: Character

1. Given the trivial representation of G (i.e. any 1-dim vector space where $g \mapsto I$), then if φ is the representation affording the trivial representation, it has $\chi_\varphi(g) = 1$ for all g . This is called the *trivial character*.
2. If we have a degree 1-representation, then the character and the representation are the same. Thus for abelian group, complex representations and their characters are the same. For example, take the representation $\varphi : C_p \rightarrow \mathbb{C}^\times$, $\varphi(a^i) = \zeta_p^i$. Then:

$$\chi(a^i) = \zeta_p^i$$

3. Let $\Pi : G \rightarrow S_n$ be a permutation representation of G and φ the linear of G of degree n on V where:

$$\varphi(g)(e_i) = e_{\Pi(g)(i)}$$

with respect to the given basis, $\varphi(g)$ has 1 in the ii -entry if $\Phi(g)$ fixes i , and zero otherwise. Thus, we can say that the character χ is:

$$\chi(g) = \#\{\text{number of fixed points of } g \text{ on } \{1, 2, \dots, n\}\}$$

If in particular the permutation representation of G into S_n is by left multiplication, then the character is called the *permutation character*.

4. Consider any regular representation of G , φ . Then the character is:

$$\chi(g) = \begin{cases} 0 & g \neq 1 \\ |G| & g = 1 \end{cases}$$

This character is called the *regular character* of G (note how this character is not a group homomorphism into k or $k^\times$)

5. Take the representation $\varphi : D_n \rightarrow \text{GL}_2(\mathbb{R})$ from example 2.1.5. Then:

$$\chi(r) = 2 \cos(2\pi/n) \quad \chi(s) = 0 \quad \chi(1) = 2$$

where the last one is since $1 \in G$ must map to the identity matrix $I \in \text{GL}_2(\mathbb{R})$.

From now on, all our representation will be complex.

Lemma 2.2.3: Properties Of Character

Let $\rho_V : G \rightarrow \text{GL}(V)$ be a complex representation. Then:

1. $\chi_V(1) = \dim_{\mathbb{C}}(V)$
2. $\rho_V(g)$ is diagonalizable for all $g \in G$ and $\chi(g)$ is integral over the integers: $\chi(g) \in \overline{\mathbb{Z}}^{\mathbb{C}}$
3. If $g \in G$ has order n , then $\chi_V(g)$ is a sum of $\dim(V)$ number of n th roots of 1
4. $|\chi_V(g)| \leq \chi_V(1)$ for all $g \in G$
5. $\chi_V(g^{-1}) = \overline{\chi_V(g)}$

Proof:

1. We showed this in the above examples.
2. The criterion we need from linear algebra is that $A \in M_n(\mathbb{C})$ is diagonalizable if and only if the minimal polynomial has no repeated roots, which can be implied if $M^k = I$ for some k , which it certainly is in our case since every element is finite.
3. If $g^n = 1$, then $\rho_V(g)^n = 1$, so by (2)

$$\rho_V(g) \sim \begin{pmatrix} z_1 & & \\ & \ddots & \\ & & z_k \end{pmatrix}$$

where $z_i^n = 1$ for all $1 \leq i \leq k$. Furthermore, this tells us that these trace values in the matrix representation are always integral over the integers, being the sum of integral elements.

4.

$$|\chi_V(g)| = \left| \sum z_i \right| \leq \sum |z_i| = \dim(\chi_V(1))$$

5. Since $\rho_V(g^{-1}) = \rho_V(g)^{-1}$ and

$$\rho_V(g)^{-1} \sim \begin{pmatrix} z_1^{-1} & & \\ & \ddots & \\ & & z_k^{-1} \end{pmatrix}$$

$\chi_V(g^{-1}) = \sum_i z_i^{-1}$, but $\|z_i\| = 1$, so $z_i^{-1} = \overline{z_i}$, so

$$\chi_V(g^{-1}) = \sum_i z_i^{-1} = \sum_i \overline{z_i} = \overline{\chi_V(g)}$$

Now, if we have some G -representation V , it has some character χ . In the following, we will show how we can always break-down a character to be the sums of characterse on the irreducible G -representations:

Proposition 2.2.4: Linearly Independent Characters

Let G be a finite group and V a semi-simple $k[G]$ -module, and φ the representation that V affords. Then:

1. The character of V is the sum of the characters of the componens appearing in the direct sum decomposition.
2. The irreducible characters of G are linearly independent set in the set of class functions. Furthermore, the irreducible characters of G form a basis of the vector-space of class functions.

Proof:

For the 1st point, By the Artin-Wedderburn decoposition, we have:

$$V \cong (S_1)^{n_1} \oplus \dots \oplus (S_k)^{n_k}$$

where S_i are simple $k[G]$ -modules (irreducible G -representations). If we focus on the case where $M = S_1 \oplus S_2$, then we can choose a basis for S_1 and S_2 which will give us:

$$\varphi(g) = \begin{pmatrix} \varphi_1(g) & 0 \\ 0 & \varphi_2(g) \end{pmatrix}$$

where φ_1, φ_2 are the representations afforded by M_1 and M_2 respectively. Then if ψ_1 and ψ_2 are the characters of φ_1 and φ_2 an χ is the character of φ then we get:

$$\chi = \psi_1 + \psi_2$$

using induction get's us the result.

For the 2nd point, first notice that the class functions defined by being 1 on a conjugacy class and 0 otherwise forms a basis for class functions of dimension r . It thus suffices to show that the r irreducible characters are independent. To that end, define:

$$\widetilde{\chi}_V : \mathbb{C}[G] \rightarrow \mathbb{C} \quad x \mapsto \text{tr}(x : V \rightarrow V)$$

Note that this map is $\mathbb{C}$ -linear since $\mathbb{C}$ commutes with $\mathbb{C}[G]$, and $\widetilde{\chi}_V|_G = \chi_V$. Now, say $S_1, \dots, S_r$ are simple $\mathbb{C}[G]$ -modules and:

$$\sum_i^r \lambda_i \chi_{S_i} = 0 \quad \lambda_i \in \mathbb{C}$$

Then this is equivalent to:

$$\sum_i^r \lambda_i \widetilde{\chi}_{S_i}(g) = 0 \quad \forall g \in G$$

and since the maps are $\mathbb{C}$ -linear, by linearity:

$$\sum_i^r \lambda_i \widetilde{\chi}_{S_i}(x) = 0 \quad \forall x \in \mathbb{C}[G]$$

which by plugging into the definition we get:

$$\sum_i^r \lambda_i \text{tr}(x : S_i \rightarrow S_i) = 0 \quad \forall x \in \mathbb{C}[G]$$

Next, by Artin Wedderburn, we have the decomposition:

$$\mathbb{C}[G] \cong M_{n_1}(\mathbb{C}) \times \dots \times M_{n_r}(\mathbb{C})$$

where each $M_{n_i}(\mathbb{C}) \cong S_i^{\oplus n_i}$, with $S_i = \mathbb{C}^{\oplus n_i}$. Using this, take the element:

$$x_i = (0, \dots, I, \dots, 0)$$

where I is the identity matrix^a. Then $x_i : S_j \rightarrow S_j$ is 0 if $i \neq j$, and the identity matrix if $i = j$. Thus, the trace is:

$$\text{tr}(x_j : S_j \rightarrow S_j) = \begin{cases} 0 & i \neq j \\ n_i & i = j \end{cases}$$

and since n_j is at least 1, we must have that

$$\lambda_1 = \dots = \lambda_r = 0$$

where we took advantage of the fact we are in characteristic 0

^aIf the field had finite characteristic, it is better to take the elementary matrix E_{11}

2.2.5 Remark We can prove this directly without using Artin-Wedderburn, which we will

Corollary 2.2.6: Detecting Isomorphic Representations Using Characters

Let V_1, V_2 be representations that are semi-simple modules.

1. Every character of G is uniquely of the form

$$\sum_i^r \lambda_i \chi_i \quad \lambda_i \geq 0$$

where χ_i are irreducible characters.

2. the representations V_1 and V_2 are isomorphic if and only if $\chi_{V_1} = \chi_{V_2}$

Proof:

1. This comes straight from the fact that irreducible characters form a basis, and that any G -representation V will be some direct sum of a finite list of known simple G -modules
2. If $V_1 \cong_G V_2$, there is a matrix P that simultaneously diagonalizes so that if A and B are basis matrices, $A = PBP^{-1}$. But the trace is independent of conjugation, and so the characters are the same. For the other way around, treating a representation as a $k[G]$ -module, then by Maschke's theorem, we know the list simple $k[G]$ -modules and the actions of $k[G]$ on them. Since any $k[G]$ -module can be broken down into a direct summand of simple $k[G]$ -modules, and the two characters are the same, the list of $k[G]$ -modules and their multiplicity are the same.

Thus, we have translated the problem of finding the irreducible representations of a group G to finding the irreducible characters of G ! Since a character of a representation is determined by the irreducible characters, we give a way to systematize the tracking of irreducible characters:

Definition 2.2.7: Character Table

Let $\varphi : G \rightarrow \text{GL}(V)$ be a representation of G and χ_V its character. Then we call the *character table* a table with the rows being irreducible characters and columns being an element in the conjugacy class. Common practice is to make the first column the identity conjugacy class and the first row the trivial representation

Example 2.2.8: Character Table

Consider $G = S_3$. Then we can start creating its character table:

	e	$(1\ 2)$	$(1\ 2\ 3)$
1	1	1	1
χ_{sgn}	1	-1	1
$\vdots$	$\vdots$	$\vdots$	$\vdots$

Since G has 3 conjugacy classes, exactly 3 irreducible characters of G . How would we go about finding it? The coming propositions work towards this.

To be able to find the missing character, and more generally find new characters for a character table, we may look to construct new character given old one, and hope that we break down those characters into a linear combination of irreducible characters. In a sense, we are finding a way to create representations without doing it “by hand”. We will do this in two ways: the first is by taking advantage of group-actions, since they are relatively easy to define and their representations are matrices with 1’s and 0’s. Since all groups embed into a symmetric group, this can be done systematically. We then see how to work directly with known representations to create new representations.

Definition 2.2.9: Permutation Representation

Let G act on a finite set X . Define $\mathbb{C}X$ to be the set of formal sums:

$$\mathbb{C}X = \left\{ \sum_{x \in X} \lambda_x x \mid \lambda_x \in \mathbb{C} \right\}$$

So that we get a $\mathbb{C}$ -algebra with $\dim_{\mathbb{C}}(\mathbb{C}X) = |X|$. Then we can make this a G -representation by defining:

$$g \cdot \left(\sum \lambda_x x \right) = \sum \lambda_x (g \cdot x)$$

The 3 usual permutation representations are $G \curvearrowright G$ by left multiplication (which gives the regular representation) or conjugation, and $G \curvearrowright \{*\}$ trivially (which gives the trivial representation).

Lemma 2.2.10: Character Of Permutation Representation

Let G act on X , $G \curvearrowright X$. Then

$$\chi_{\mathbb{C}X}(g) = \# \{x \in X \mid gx = x\}$$

Proof:

Fix $g \in G$. Then the set of x ’s is a basis for the representation. To make the computation simpler, we will find a convenient order for them. In particular, by choosing any $x_1 \in X$, we can find the orbit of the element:

$$x_1 \xrightarrow{g} x_2 \longrightarrow \cdots \longrightarrow x_k \quad x_{k+1} \xrightarrow{g} \cdots$$

so the basis representation will look like:

$$\begin{pmatrix} 0 & & & \\ 1 & 0 & & \\ 0 & 1 & \ddots & \\ & & 1 & 0 \end{pmatrix}$$

so we get that the trace is 0 if the order of the orbit is greater than 1, hence:

$$\chi_{\mathbb{C}X}(g) = \#\{g\text{-orbits of size 1}\}$$

An immediate consequence of this is that we have another way of computing the character of the regular representation

$$\chi_{\mathbb{C}G}(g) = \begin{cases} |G| & g = e \\ 0 & g \neq e \end{cases}$$

which makes sense, since the $\dim_{\mathbb{C}}(\mathbb{C}G) = |G|$.

Next, we turn towards a more systematic way of constructing characters by creating them out of previous ones, that is, given some representations V and W of a group G , we can make new representations by creating new representations from V and G :

Definition 2.2.11: Constructing Representations from smaller ones

Let V, W be G -representations. Then

1. we can make a $V \oplus W$ representation by:

$$g \cdot (v, w) = (g \cdot v, g \cdot w) \quad [(\rho_V(g)v, \rho_W(g)w)]$$

2. we can make $V \otimes_{\mathbb{C}} W$ a representations by:

$$g \cdot (v \otimes w) \mapsto (gv) \otimes (gw)$$

3. we can make $\text{Hom}_{\mathbb{C}}(V, W)$ a representations by:

$$g \cdot f : V \rightarrow W \quad v \mapsto g \cdot f(g^{-1} \cdot v)$$

4. we can make $V^* = \text{Hom}_{\mathbb{C}}(V, \mathbb{C})$ a representaiton by:

$$g \cdot f : V \rightarrow \mathbb{C}, \quad v \mapsto f(g^{-1} \cdot v)$$

Note that the last construction is the same as the 3rd if we take $\mathbb{C}$ as the trivial representation. Before we show what the character's of these new representations are, we first find a relatively simpler way of representing the hom representation of V and W :

Lemma 2.2.12: Hom-Tensor Representation Dual

Let V, W be G representations. Then:

$$\text{Hom}_{\mathbb{C}}(V, W) \cong_G V^* \otimes_{\mathbb{C}} W$$

as G -representations.

Note that this is not true in general, even for vector-spaces. It is important that the spaces are finite-dimensional. In this way, this is a stronger result then the hom-tensor adjoint presented in ?? for the case of finite-dimensional vector-spaces.

Proof:

By the universal property:

$$\theta : V^* \otimes_{\mathbb{C}} W \rightarrow \text{Hom}_{\mathbb{C}}(V, W) \quad \lambda \otimes w \mapsto (v \mapsto \lambda(v)w)$$

you should check that this is well-defined (note that it is bilinear).

Now, pick basis $e_1, e_2, \dots, e_n$ for V , $f_1, f_2, \dots, f_m$ for W , and the natural dual $e_1^*, \dots, e_n^*$ for V^* . Then we get a basis for $V^* \otimes W$:

$$(e_i^* \otimes f_j) \quad 1 \leq i \leq n, 1 \leq j \leq m$$

Now, notice that:

$$\theta(e_i^* \otimes f_j) \quad v \mapsto e_i^*(v)f_j \sim \begin{pmatrix} 0 & \cdots & 0 & \cdots & 0 \\ \vdots & & \vdots & & 0 \\ 0 & & 1 & & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \cdots & 0 & \cdots & 0 \end{pmatrix} = E_{ij}$$

which is a matrix with 1 in the ij position and zero everywhere else. This is clearly a basis, hence θ sends a basis to a basis, and hence θ is a vector-space isomorphism. We next check that θ is G -linear:

$$\begin{aligned} \theta(g \cdot (\lambda \otimes w)) &= \theta(g\lambda \otimes gw) \\ &\equiv v \mapsto (g\lambda)(v)gw \\ &= v \mapsto (g\lambda)(g^{-1}v)w \\ &= g(\lambda(g^{-1}v)w) \\ &\equiv g(\theta(\lambda \otimes w)g^{-1}v) \\ &= g \cdot \theta(\lambda \otimes w) \end{aligned}$$

Lemma 2.2.13: Creating New Characters

1. $\chi_{V \oplus W} = \chi_V + \chi_W$
2. $\chi_{V \otimes W} = \chi_V \cdot \chi_W$
3. $\chi_{\text{Hom}_{\mathbb{C}}(V, W)} = \overline{\chi_V} \chi_W$
4. $\chi_{V^*} = \overline{\chi_V}$

Proof:

1. obvious

2. It is enough to show that if $f : V \rightarrow V$ and $g : W \rightarrow W$ are linear functions, then:

$$\text{tr}(f \otimes h) = \text{tr}(f)\text{tr}(h)$$

Now if we pick basis (e_i) for V and (f_j) for W , then $f(e_i) = \sum a_{ij}e_j$ and $h(f_k) = \sum b_{lk}f_l$. Then:

$$(f \otimes h)(e_i \otimes f_j) = \left(\sum_j a_{ij}e_j \right) \otimes \left(\sum_l b_{kl}f_l \right) = \sum_{j,l} a_{ij}b_{lk}(e_j \otimes f_l)$$

Notice the coefficient of $e_i \otimes f_j$ is $\sum_{i,k} a_{ii}b_{kk} = (\sum_i a_{ii})(\sum_k b_{kk}) = \text{tr}(f)\text{tr}(h)$

3. Recall that for $f \in V^*$,

$$(gf)(v) = f(g^{-1} \cdot v)$$

that is

$$\rho_{V^*}(g)(f) = f \circ \rho_V(g^{-1})$$

that is, it is the transpose:

$$\rho_{V^*}(g) = (\rho_V(g^{-1}))^t$$

So, since the transpose matrix has the same trace, we have

$$\chi_{V^*}(g) = \chi_V(g^{-1}) = \overline{\chi_V(g)}$$

we then apply lemma 2.2.12 and part (3) to get:

$$\chi_{\text{Hom}_{\mathbb{C}}(V,W)} = \overline{\chi_V} \cdot \chi_W$$

4. $\chi_{V^*} = \chi_{\text{Hom}(V,\mathbb{C})} = \overline{\chi_V} \cdot 1 = \overline{\chi_V}$.

With this, we can create new characters! However, we have yet to establish how we may find the irreducible components of these characters. For example, if we take $V \otimes V$, we may find that $\chi_{V \otimes V}$ is a linear combination of some characters already present in our table, but then we have a left-over part. How would we determine if the left-over is irreducible? We will show this by defining an *inner product* on the vector-space of class equations, and show that the irreducible form an *orthogonal basis*!

Definition 2.2.14: Inner Product On Class functions

Let $\chi_1, \chi_2 : G \rightarrow \mathbb{C}$ be class functions. Then define:

$$\langle \chi_1, \chi_2 \rangle = \frac{1}{|G|} \sum_{g \in G} \chi_1(g) \overline{\chi_2(g)} \in \mathbb{C}$$

Then this is a [Hermitian] *inner product*.

It should be verified that this is indeed an inner product. Since class functions are by definition constant on conjugacy classes, we only need to run through representatives of conjugacy classes

the order of each conjugacy class:

$$\langle \chi_1, \chi_2 \rangle = \frac{1}{|G|} \sum_{\substack{\mathcal{K} \in \text{conj}(G) \\ g \in \mathcal{K}}} |\mathcal{K}| \chi_1(g) \overline{\chi_2(g)} \in \mathbb{C}$$

where we pick a single representative $g \in \mathcal{K}$ and sum over the number of conjugacy classes, i.e. $|\text{conj}(G)|$. Looking at the inner product like this is desirable to make computation easier (ex. if taking the inner product of a character with itself, we would simply take the square of each element in a row multiplied by the size of its conjugacy class). Notice how this inner product mirrors the G -invariant inner product we induced in corollary 2.1.12 if we let $\langle \cdot, \cdot \rangle$ be the usual conjugate dot product on $\mathbb{C}^n$. (TBD)

Lemma 2.2.15: Inner Product With Trivial Character

Let V be a G -representation. Then if $\mathbf{1}$ is the trivial representation,

$$\langle \chi_V, \chi_{\mathbf{1}} \rangle = \dim_{\mathbb{C}}(V^G)$$

where $V^G = \{v \in V \mid g \cdot v = v \ \forall g \in G\}$ is the subspace fixed by G .

Proof:

Define the linear map $\pi : V \rightarrow V$ where

$$v \mapsto \frac{1}{|G|} \sum_{g \in G} g \cdot v$$

then:

$$\pi|_{V^G} = \text{id}$$

and

$$\text{im}(\pi) \subseteq V^G$$

which implies $\pi^2 = \pi$, i.e. a projection. By linear algebra:

$$V = \ker(\pi) \oplus \text{im}(\pi)$$

where on the kernel, $\pi = 0$, and on the image $\pi = \text{id}$. Thus,

$$\frac{1}{|G|} \sum_g \chi_V(g) = \frac{1}{|G|} \sum_g \text{tr}(g : V \rightarrow V) = \text{tr}(\pi) = \dim_{\mathbb{C}}(V^G)$$

but the left hand side is $\langle \chi_V, \chi_{\mathbf{1}} \rangle$, completing the proof.

Corollary 2.2.16: Row Orthogonality

If V and W are any representations of G , then

$$\langle \chi_V, \chi_W \rangle = \dim_{\mathbb{C}}(\text{hom}_{\mathbb{C}[G]}(W, V))$$

In particular, if V and W are irreducible, then

$$\langle \chi_V, \chi_W \rangle = \begin{cases} 1 & V \cong W \\ 0 & \text{otherwise} \end{cases}$$

that is, irreducible characters are orthogonal to each other.

Proof:

We reduce to the previous lemma:

$$\begin{aligned} \langle \chi_V, \chi_W \rangle &= \frac{1}{|G|} \sum_{g \in G} \chi_V(g) \overline{\chi_W(g)} \\ &= \frac{1}{|G|} \sum_{g \in G} \chi_{\text{Hom}_{\mathbb{C}}(W, V)}(g) \\ &= \langle \chi_{\text{Hom}_{\mathbb{C}}(W, V)}, 1 \rangle \\ &= \dim_{\mathbb{C}}(\text{Hom}_{\mathbb{C}}(W, V)^G) \end{aligned}$$

Note that $f \in \text{Hom}_{\mathbb{C}}(W, V)$ is G -invariant, so $g \cdot f = gf g^{-1} = f$ for all $g \in G$ (recall the definition of $g \cdot f$ on $\text{Hom}_{\mathbb{C}}(W, V)$). Thus this is true iff

$$gf = fg$$

Thus

$$gf(w) = f(gw) \quad \forall g \forall w \in W$$

so f is $\mathbb{C}[G]$ -linear so f is in the $\text{Hom}_{\mathbb{C}[G]}(W, V)$.

There is also a more concrete proof worth remarking: since characters are linear combinations of irreducible characters, the direct product of irreducible modules translates to the sum of an irreducible character with the multiplicity of the irreducible module affording the representation being the multiplicity of the irreducible characters. Hence:

$$\begin{aligned} \langle \chi_V, \chi_W \rangle &= \left\langle \sum_i m_i \chi^{(i)}, \sum_j m_j \chi^{(j)} \right\rangle \\ &= \sum_{i,j} m_i m_j \langle \chi^{(i)}, \chi^{(j)} \rangle \end{aligned}$$

where we have now reduced it to $\langle \chi^{(i)}, \chi^{(j)} \rangle$. Expanding the inner product and using the hom-set definition, we can apply Schur's Lemma and get:

$$\langle \chi^{(i)}, \chi^{(j)} \rangle = \delta_{ij}$$

Thus, our formula becomes:

$$\begin{aligned}\langle \chi_V, \chi_W \rangle &= \left\langle \sum_i m_i \chi^{(i)}, \sum_j m_j \chi^{(j)} \right\rangle \\ &= \sum_{i,j} m_i m_j \langle \chi^{(i)}, \chi^{(j)} \rangle \\ &= \sum_{i,j} m_i m_j \delta_{ij}\end{aligned}$$

To find $\langle \chi^{(i)}, \chi^{(i)} \rangle$ for an irreducible character, seeing it through multiplicities, then we know that an irreducible character corresponds to an irreducible representation, which necessarily has to then have multiplicity one, so if in particular:

$$\langle \chi^{(i)}, \chi^{(i)} \rangle = m_i^2 = 1 \cdot 1 = 1$$

Thus, the irreducible characters produce an orthogonal basis of class functions!

Corollary 2.2.17: Consequence of Orthogonality

Let $V = \bigoplus_i^r S_i^{l_i}$ where each S_i irreducible and $S_i \not\cong S_j$. Then

1. the multiplicity of l_i of S_i is: $l_i = \langle \chi_V, \chi_{S_i} \rangle$
2. $\langle \chi_V, \chi_V \rangle = \sum_i l_i^2$
3. $\text{hom}_G(W, V) \cong \text{hom}_G(V, W)^*$

Proof:

1. Use the Artin-Wedderburn decomposition and the fact that direct sums of modules become sums of characters
2. this is just taking advantage of the summing formula
3. This is taking advantage of the fact that $\langle \chi, \psi \rangle = \overline{\langle \psi, \chi \rangle}$.

The second line is useful enough to be given its own name:

Definition 2.2.18: Character Norm

Let V be a G -representation and let $\langle \cdot, \cdot \rangle$ be the inner product on the class functions of G . Then for any class function $\psi = \sum_i \lambda_i \chi_i$,

$$\|\psi\| = \left(\sum_i \lambda_i^2 \right)^{1/2}$$

We will use all this information to deduce new characters:

Example 2.2.19: Orthogonality Relation

Let's go back to our table for S_3 :

	e	$(1\ 2)$	$(1\ 2\ 3)$
1	1	1	1
χ_{sgn}	1	-1	1
$\vdots$	$\vdots$	$\vdots$	$\vdots$

Since S_3 has 3 conjugacy classes, it can at most have one more irreducible character. What can this character be? We may start by looking at the regular representation of S_3 and take its character χ_V . Then:

$$\begin{aligned}\langle \chi_V, 1 \rangle &= 1 \\ \langle \chi_V, \chi_{\text{sgn}} \rangle &= 0\end{aligned}$$

showing the regular character χ_V has one copy of the trivial representation (which makes sense, since $\chi_V(e) = 1$). Since

$$\chi_V(e) = 3 \quad \chi_V((1\ 2)) = 1 \quad \chi_V((1\ 2\ 3)) = 0$$

then we can take a “compliment” character by subtracting the trivial character, which defines a character which gives:

$$\chi^\perp(e) = 2 \quad \chi^\perp(1\ 2) = 0 \quad \chi^\perp(1\ 2\ 3) = -1$$

So, if there were to be another irreducible character, it would be of this form. But does there exist a degree 2 irreducible representation with this character? In fact, there has to: we saw that the number of irreducible representations is equal to the number of conjugacy classes, meaning the number of characters must be equal to the number of irreducible representations.

More generally, there can be a few approaches to finding it. One is to tensor two characters we know and see if it is irreducible, or one of its linear components is irreducible. Another is to find the compliment of a representation of an existing character (since $k[G]$ -modules are semi-simple). This compliment method is in general less desirable, and usually harder, but is manageable if working with permutation representations. To demonstrate, let V be the degree three permutation representation of S_3 with basis $\{e_1, e_2, e_3\}$. Then if $\mathbb{1}$ is the trivial action, $V^{\mathbb{1}} = \text{span}(e_1 + e_2 + e_3)$. Thus, by Maschke's theorem, $V^{\mathbb{1}}$ has a compliment. To see it, we simply have to extend $e_1 + e_2 + e_3$ to an orthogonal basis for V , for example, choose $(e_1 + e_2 + e_3, e_2 - e_1, e_3 - e_1)$. Then we can calculate what the action ρ does on this basis. For example:

$$\begin{aligned}\rho(1\ 2)(e_1 + e_2 + e_3) &= e_1 + e_2 + e_3 \\ \rho(1\ 2)(e_2 - e_1) &= e_1 - e_2 = -(e_2 - e_1) \\ \rho(1\ 2)(e_3 - e_1) &= e_3 - e_2 = -(e_2 - e_1) + (e_3 - e_1)\end{aligned}$$

If we write out ρ for an element in each conjugacy class, we will find that its trace will be

$$\chi^\perp(e) = 2 \quad \chi^\perp((1\ 2)) = 0 \quad \chi^\perp((1\ 2\ 3)) = -1$$

showing we have found the desired irreducible representation! Finally, by Artin-Wedderburn, we have that $\mathbb{C}S_3 \cong \mathbb{1} \oplus \text{sgn} \oplus V^\perp$

Now that we know the character table for S_3 , we can construct any other character of representations of S_3 , and hence any other representation of S_3 . For example, we know that $V^\perp \otimes V^\perp$ has as character $\chi_{V^\perp} \cdot \chi_{V^\perp}$. Using this, we can compute that:

$$\begin{aligned}\langle V^\perp \otimes V^\perp, \mathbf{1} \rangle &= 1 \\ \langle V^\perp \otimes V^\perp, \text{sgn} \rangle &= 1 \\ \langle V^\perp \otimes V^\perp, V^\perp \rangle &= 1\end{aligned}$$

showing that

$$\chi_{V^\perp \otimes V^\perp} = \chi_{\mathbf{1}} + \chi_{\text{sgn}} + \chi_{V^\perp}$$

^aWhich in this case, is just the regular representation

Recall from linear algebra that if the rows are orthogonal in a square matrix, then so would be the column (and vice-versa). We bring to light this relation for the character table. As in linear algebra, it comes down to proper normalizing of a different basis: recall that the class equations of the form $\delta_K : G \rightarrow \mathbb{C}^\times$ where:

$$\delta_K(g) = \begin{cases} 1 & g \in K \\ 0 & \text{otherwise} \end{cases}$$

this naturally forms a basis for the class equations too. The following theorem will also show how to go between this basis and the irreducible character basis:

Theorem 2.2.20: Column Orthogonality

If $\chi_1, \chi_2, \dots, \chi_r$ are the irreducible characters of G and $g_1, g_2, \dots, g_r$ represent the conjugacy classes of G , then

$$\sum_i \chi_i(g_s) \overline{\chi_i(g_t)} = \delta_{st} \frac{|G|}{|\text{conj}(g_s)|}$$

which by Orbt-stabalizer,

$$\sum_i \chi_i(g_s) \overline{\chi_i(g_t)} = \delta_{st} |C_G(g_s)|$$

If $g_s, g_t \in \text{conj}(g)$, then we get that

$$\sum_i \chi_i(g_s) \overline{\chi_i(g_t)} = \sum_i |\chi_i(g_s)|^2 = |C_G(g_s)|$$

in particular, we must have:

$$\sum_i |\chi_i(e)|^2 = |G|$$

Proof:

Define $f_s : G \rightarrow \mathbb{C}$,

$$g \mapsto \begin{cases} 1 & g \in \text{conj}(g_s) \\ 0 & \text{otherwise} \end{cases}$$

which is natural basis for the class functions. since it's a class function, we know we can write it

as the sum of orthogonal irreducible characters:

$$f_s = \sum_i^r \lambda_{is} \chi_i \quad \lambda_{is} \in \mathbb{C}$$

since these are orthonormal, we know that:

$$\lambda_{is} = \langle f_s, \chi_i \rangle$$

so computing:

$$\begin{aligned} \lambda_{is} &= \langle f_s, \chi_i \rangle \\ &= \frac{1}{|G|} \sum_{g \in G} f_s(g) \overline{\chi_i(g)} \\ &= \frac{1}{|G|} \sum_{g \in \text{conj}(g_s)} \overline{\chi_i(g)} \\ &= \frac{|\text{conj}(g_s)|}{|G|} \overline{\chi_i(g_s)} \end{aligned}$$

so

$$f_s = \frac{|\text{conj}(g_s)|}{|G|} \sum_i^r \overline{\chi_i(g_s)} \chi_i$$

we know evaluate at g_t .

Since both the rows and columns of a character table are all orthogonal to each other, the character table is in fact invertible!

Example 2.2.21: Using Column Orthogonality

Let $G = S_4$. We want to find its character table. We first pick what conjugacy classes to index the columns by, say:

conj. class	1	(1 2)	(1 2 3)	(1 2)(3 4)	(1 2 3 4)
size	1	6	8	3	6

Next, we can immediately deduce that χ_1 and χ_{sgn} are irreducible characters, giving us the current beginning of a character table:

	1	6	8	3	6
	1	(1 2)	(1 2 3)	(1 2)(3 4)	(1 2 3 4)
χ_1	1	1	1	1	1
χ_{sgn}	1	-1	1	1	-1

To find more characters, we can first take the permutation character of $\mathbb{C}X$. The character of each conjugacy class is the number of fixed points, giving us:

	1	6	8	3	6
	1	(1 2)	(1 2 3)	(1 2)(3 4)	(1 2 3 4)
$\chi_{\mathbb{C}X}$	4	2	1	0	0

We can see that the trivial character is part of linear decomposition of $\chi_{\mathbb{C}X}$, since:

$$\langle \chi_{\mathbb{C}X}, \chi_1 \rangle = (1(4 \cdot 1) + 6(2 \cdot 1) + 8(1 \cdot 1))/24 = 24/2 = 1$$

and so $\chi_{\mathbb{C}X} - \chi_1$ is a 3 dimensional character, let's label this χ_3 . Then $\|\chi_3\|^2 = 1$, showing that it's irreducible, hence we expand our table:

	1	6	8	3	6
	1	(1 2)	(1 2 3)	(1 2)(3 4)	(1 2 3 4)
χ_1	1	1	1	1	1
χ_{sgn}	1	-1	1	1	-1
$\chi_{\mathbb{C}X} - \chi_1 = \chi_3$	3	2	0	-1	-1

To find the next character, we can multiply χ_{sgn} and χ_3 (note we can do χ_3 with itself, but it will be 9 dimensional, which we'll show in the next proposition cannot be an irreducible representation). Let $\chi_4 = \chi_{\text{sgn}} \cdot \chi_3$. Then $\|\chi_4\|^2 = 1$, meaning it's irreducible, and so we add another row to our table:

	1	6	8	3	6
	1	(1 2)	(1 2 3)	(1 2)(3 4)	(1 2 3 4)
χ_1	1	1	1	1	1
χ_{sgn}	1	-1	1	1	-1
χ_3	3	2	0	-1	-1
$\chi_{\text{sgn}} \cdot \chi_3 = \chi_4$	3	-1	0	-1	1

Finally, since $|G| = \sum_i \chi_i(e)^2$, we can deduce that $\chi_5(e) = 2$. We can't do this same trick exactly for the other columns since we will get the value up to a complex number z with $\|z\| = 1$. To find the rest of the character values, we can either use column orthogonality. For example, to find $\chi_5(1\ 2)$, notice that the inner product of the first and second column is:

$$1 \cdot 1 + (-1) \cdot 1 + 3 \cdot 1 + 3 \cdot (-1) = 0$$

and so $\chi_5(2)$ must be 0. For the next column, we'd get :

$$1 \cdot 1 + 1 \cdot 1 + 3 \cdot 0 + 3 \cdot 0 = 2$$

and so, since $2 \cdot -1 = -2$, we get that $\chi_5(1\ 2) = -1$. Continuing this way, we get that the rest of the table will be:

	1	6	8	3	6
	1	(1 2)	(1 2 3)	(1 2)(3 4)	(1 2 3 4)
χ_1	1	1	1	1	1
χ_{sgn}	1	-1	1	1	-1
χ_3	3	2	0	-1	-1
χ_4	3	-1	0	-1	1
χ_5	2	0	-1	2	0

We can verify this is indeed irreducible by taking $\|\chi_5\|^2 = 1$.

Another method would be to use the regular representation, where we get:

$$\chi_{\mathbb{C}[G]} = \sum_i \chi_i(1) \chi_i$$

These can be verified by taking advantage of the fact that the “column norm” should give you $C_G(g)$ (i.e. the number on top of the table).

As a fun aside, it can be shown that the represented by χ_4 is the action of the rotation of a cube that is, if $C \subseteq \mathbb{R}^3$ are the vertex points of the cube, then χ_3 is the character of $S_4 \curvearrowright \text{GL}_3(\mathbb{R})$ of all rotations C .

Notice how the dimension of each irreducible representation divided the order of the group. This is in fact true in general. It is not a trivial fact, the only “order” information we’ve shown so far is that the sums of the squares of the dimensions of the irreducible representation sum to $|G|$. It is also relatively easy to show that if $W \subseteq V$ is an irreducible $k[G]$ -module then the dimension of W is $< |G|$. The following is thus a much stronger result:

Proposition 2.2.22: Dimension Theorem of Irreducible Representations

Let V be an irreducible G -representation. Then:

$$\dim_{\mathbb{C}}(V) \mid |G|$$

that is, the dimension of V divides the order of G .

Proof:

Let $g_1, g_2, \dots, g_r$ be representatives of the conjugacy class. Recall the following trick from proposition 2.1.13:

$$e_i = \left(\sum_{g \in \text{conj}(g)} g \right) \in Z(\mathbb{C}[G])$$

Note that e_i in fact lies in $e_i \in Z(\mathbb{Z}[G])$, since $ge_i g^{-1} = e_i$ (by definition), and so extending linearly all $xe_i x^{-1} = e_i$. Now we have a commutative ring that is a finitely generated $\mathbb{Z}$ -module (one way to see this is as we’ve done in proposition 2.1.13, another way is to notice that $\mathbb{Z}[G]$ is a finitely generated $\mathbb{Z}$ -module and $\mathbb{Z}$ is Noetherian, so the center is finitely generated)

Thus, by ??, $\mathbb{Z} \subseteq Z(\mathbb{Z}[G])$ is an *integral extension*. Next, we have a ring homomorphism:

$$\varphi : Z(\mathbb{C}[G]) \rightarrow \text{End}_{\mathbb{C}[G]}(V) \stackrel{!}{\cong} \mathbb{C} \quad z \mapsto (z : V \rightarrow V)$$

where $\stackrel{!}{=}$ comes from corollary 1.2.19 and Schur’s lemma, and $z(v) = zv$. Hence, each element of $\varphi(Z(\mathbb{C}[G]))$ is $\mathbb{C}$ -linear, and must be invertible, and so φ is in fact a map from $Z(\mathbb{C}[G]) \rightarrow \text{Aut}_{\mathbb{C}}(V) = \text{GL}(V)$, which is simply a restriction of the original G -action ρ to $Z(\mathbb{C}[G])$. Hence $\rho(e_i) = \lambda_i I$ for appropriate $\lambda_i = \varphi(e_i)$, meaning $\chi(e_i) = \text{tr}(e_i : V \rightarrow V) = n\varphi(e_i)$ where $n = \dim(V)$ since by what we’ve just shown, $\varphi(e_i)$ acts as scalar multiplication on the representation V .

Now, since $e_i \in Z(\mathbb{C}[G])$ is integral over $\mathbb{Z}$, $\varphi(e_i) \in \mathbb{C}$ is integral over $\mathbb{Z}$, that is $\varphi(e_i) \in \overline{\mathbb{Z}}^{\mathbb{C}} \subseteq \mathbb{C}$ (i.e. $\varphi(e_i)$ is algebraic). Next, we can express $\chi(e_i)$ in a different way:

$$\sum_{g \in \text{conj}(g_i)} \chi_V(g) = |\text{conj}(g_i)| \cdot \chi_V(g_i) = \chi(e_i)$$

so:

$$\frac{|\text{conj}(g_i)|}{n} \chi_V(g_i) = \varphi(e_i) \in \overline{\mathbb{Z}}^{\mathbb{C}}$$

which is telling us that if we divide it by n , we still get an algebraic integer! Taking a nice linear combination and multiplying by $\overline{\chi_V(g_i)}$ we get:

$$\sum_i^r \frac{\text{conj}(g_i)}{n} \chi_V(g_i) \overline{\chi_V(g_i)}$$

Then we've just shown that $\frac{\text{conj}(g_i)}{n} \chi_V(g_i)$ is an algebraic integer, and $\overline{\chi_V(g_i)}$ is an algebraic integer (being the sum of roots of unity). Now, notice this is the form of the inner product, giving us:

$$\frac{|G|}{n} \langle \chi_V, \chi_V \rangle = \frac{|G|}{n} \in \overline{\mathbb{Z}}$$

as a V irreducible. Thus, we have that $\frac{|G|}{n}$ is some algebraic integer. We further have that it is certainly a rational number ($|G|, n \in \mathbb{N}$), but then we have:

$$\frac{|G|}{n} \in \overline{\mathbb{Z}} \cap \mathbb{Q} \stackrel{!}{=} \mathbb{Z}$$

where $\stackrel{!}{=}$ comes from the fact that if R is a Unique Factorization Domain, $\alpha \in \text{Frac}(R)$, α is integral over R if and only if $\alpha \in R$, completing the proof.

This will make the computation of characters of abelian groups particularly simple! (see rep theory of finite groups: An introductory Approach p. 47!)

Exercise 2.2.1

1. Let G act on the n -dimensional vector-space V by basis permutation. Show that $\dim(V^G)$ equals the number of orbits of G .

2.2.1 More Character Tables

In this section, we go over a few more examples of character tables in order to see some techniques used to find them. For any abelian group, the character table is straight forward. As a really simple example, consider $\mathbb{Z}/3\mathbb{Z}$:

	1	1	1
	1	2	3
χ_1	1	1	1
χ_2	1	ζ_3	ζ_3^2
χ_3	1	ζ_3^2	ζ_3

If we take $\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$, we will get a 9 by 9 table. The table below fills in a few of the characters (note that these were found by taking $\chi_i(g_i)\chi_j(g_j)$)

$\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$	1 (1, 1)	1 (2, 1)	1 (3, 1)	1 (1, 2)	1 (2, 2)	1 (3, 2)	1 (1, 3)	1 (2, 3)	1 (3, 3)
$\chi_{(1,1)}$	1	1	1	1	1	1	1	1	1
$\chi_{(2,1)}$	1	ζ_3	ζ_3^2	1	ζ_3	ζ_3^2	1	ζ_3	ζ_3^2
$\chi_{(2,2)}$	1	ζ_3	ζ_3^2	ζ_3	ζ_3^2	1	ζ_3^2	1	ζ_3
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Note that the root ζ_3 might change if we take the direct product of cyclic groups of different orders, for example the character table of $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ has roots of unity of order 3 and 6 (as you can verify).

More generally, if G is abelian, we break G down into a product of cyclic groups and construct the table like above. Thus, we focus on non-abelian groups. We have already found the character table for S_3 :

S_3	e	(1 2)	(1 2 3)
χ_1	1	1	1
χ_{sgn}	1	-1	1
χ_3	2	0	-1

and the character table for S_4 :

S_4	1 1	6 (1 2)	8 (1 2 3)	3 (1 2)(3 4)	6 (1 2 3 4)
χ_1	1	1	1	1	1
χ_{sgn}	1	-1	1	1	-1
χ_3	3	2	0	-1	-1
χ_4	3	-1	0	-1	1
χ_5	2	0	-1	2	0

Next, let $G = D_4$. Then: $D' = \{e, r^2\}$, so G has 4 degree 1-irreducible representations. We also found in example 2.1.5 that D_4 has a degree 2 irreducible representation. These are all of the representations, since $1 + 1 + 1 + 1 + 2^2 = 8$. To find the 1-dimensional irreducible representations, notice that $D_4/D'_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, meaning we can find the irreducible representations of $\mathbb{Z}/2\mathbb{Z}$, which are:

$\mathbb{Z}/2\mathbb{Z}$	0	1
χ_0	1	1
χ_1	1	-1

and take the tensor product to create:

$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$	(0, 0)	(1, 0)	(0, 1)	(1, 1)
$\chi_{(0,0)}$	1	1	1	1
$\chi_{(1,0)}$	1	-1	1	-1
$\chi_{(0,1)}$	1	-1	-1	1
$\chi_{(1,1)}$	1	1	-1	-1

and using this, we can complete the table for D_4 :

D_4	e	r	r^2	σ	σr
χ_1	1	1	1	1	1
χ_2	1	1	1	-1	-1
χ_3	1	-1	1	1	-1
χ_4	1	1	1	-1	1
χ_5	2	0	-2	0	0

More generally, for D_n , the character table will consist of characters of degree 1 and 2.

Let's now let $G = Q_8$. Taking the presentation:

$$Q_8 = \langle i, j | i^4 = 1, i^2 = j^2, i^{-1}ji = j^{-1} \rangle$$

with $k = ij$ and $i^2 = -1$, we get the conjugacy classes can be represented by $1, -1, i, j, k$, with sizes 1, 1, 2, 2, 2. Since $Q_8/Q_8' \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, there are 4 degree 1 characters. The final irreducible character thus *must* have degree 2; let's label it χ_5 . We have in fact explicitly found this character before in example 2.1.5, but we can find it without actually manually finding a representation. Let φ be an irreducible representation of degree 2. Since $-1 \in Z(Q_8)$, by Schur's lemma $\varphi(-1)$ is a 2×2 scalar matrix of order 2, hence it is $\pm I$, and so $\chi_5(-1) = \pm 2$. Next, if $\chi_5(i) = a$, $\chi_5(j) = b$, and $\chi_5(k) = c$, then since χ_5 is orthogonal, we have that:

$$1 = \langle \chi_5, \chi_5 \rangle = \frac{1}{8}(2^2 + (\pm 2)^2 + 2a\bar{a} + 2b\bar{b} + 2c\bar{c})$$

and since $a\bar{a}, b\bar{b}, c\bar{c} \in \mathbb{R}^+$, they must all be zero. Furthermore, we must have that:

$$0 = \langle \chi_1, \chi_5 \rangle = \frac{1}{8}(2(\pm 2) + 0 + 0 + 0 + 0)$$

implying that $\chi_5(-1) = -2$. Thus, we have that the character table is:

Q_8	1	i	-1	j	k
χ_1	1	1	1	1	1
χ_2	1	1	1	-1	-1
χ_3	1	-1	1	1	-1
χ_4	1	1	1	-1	1
χ_5	2	0	-2	0	0

Notice that this character table is identical to the character table of D_8 . Hence, identical character tables does not imply isomorphic groups! Finally, we could have found the last row of the above table by applying column orthogonality to find the missing values, once we knew $\chi_5(1) = 2$.

Next, let $G = A_4$. Since we found the character table for S_4 , we can use it to find the character table of A_4 . We first note that A_4 has 4 conjugacy classes (represented by 1, (1 2)(3 4), (1 2 3) and (1 3 2)), and $[A_4 : A_4'] = 3$, meaning A_4 has 3 characters of degree 1, and the only other character must be of degree 3. Next, using the orthogonality relation, notice that χ_3 in the table for S_4 when restricted to A_4 is still irreducible (this is certainly not generally the case, for ex. the degree 2 representation of S_3 will become reducible since any proper subgroup of S_3 is abelian). Overall, we get:

A_4	1	(1 2)(3 4)	(1 2 3)	(1 3 2)
χ_1	1	1	1	1
χ_2	1	1	ζ_3	ζ_3^3
χ_3	1	1	ζ_3^2	ζ_3
χ_5	3	-1	0	0

Finally, we will do the character table for S_5 :

S_5	1	(1 2)	(1 2 3)	(1 2 3 4)	(1 2 3 4 5)	(1 2)(3 4)	(1 2)(3 4 5)
χ_1	1	1	1	1	1	1	1
χ_2	1	-1	1	-1	1	1	-1
χ_3	4	2	1	0	-1	0	-1
χ_4	4	-2	1	0	-1	0	1
χ_5	5	-1	-1	1	0	1	-1
χ_6	5	1	-1	-1	0	1	1
χ_7	6	0	0	0	1	-2	0

TBD

2.2.2 Burnside's Theorem

We now move towards Burnside's Theorem, which tells us that any group of order $p^a q^b$ for primes p, q and numbers $a, b \in \mathbb{N}$ is a solvable group! This proof is relatively simple in the context of representation theory, but it took over 70 years before a proof was found using just group theory! This is a good example of the power perspective: simply switching to the context of representation theory and seeing groups as embedded into $GL(V)$ let us prove a claim that from another perspective is much much harder!

Lemma 2.2.23: Burnside build-up Lemma

Suppose that V is an irreducible representation of dimension n (i.e. $\chi_1(V) = n$). If $\gcd(n, |\text{conj}(g)|) = 1$, then:

$$|\chi_V(g)| = 0 \text{ or } n$$

Proof:

By the above proof, we have:

$$\frac{\text{conj}(g)}{n} \chi_V(g) \in \overline{\mathbb{Z}}^{\mathbb{C}}$$

where $\chi_V(g) \in \overline{\mathbb{Z}}^{\mathbb{C}}$ as well, so by Bézout ($1 = am + b|\text{conj}(g)|$)

$$\frac{1}{n} \chi_V(g) \in \overline{\mathbb{Z}}^{\mathbb{C}}$$

Let $\alpha = \frac{1}{n} \chi_V(g)$. Furthermore, notice that:

$$|\alpha| = \left| \frac{\chi_V(g)}{\chi_V(1)} \right| \leq 1$$

Now, let N be the Galois closure of $\mathbb{Q}(\alpha)/\mathbb{Q}$ so that $G = \text{Gal}(N/\mathbb{Q})$ is the Galois group of N . Let

$$\beta = \prod_{\sigma \in G} \sigma(\alpha)$$

Then certainly β is in the fixed field of the Galois group, $\beta \in N^G = \mathbb{Q}$. Notice that each $\sigma(\alpha)$ satisfies the same monic polynomials as that of α , and so $\sigma(\alpha) \in \overline{\mathbb{Z}}^{\mathbb{C}}$, and so $\beta \in \overline{\mathbb{Z}}^{\mathbb{C}}$, but then $\beta \in \overline{\mathbb{Z}}^{\mathbb{C}} \cap \mathbb{Q} = \mathbb{Z}$.

Now, recall that:

$$\alpha = \frac{\chi_V(g)}{n} = \frac{\zeta_1 + \zeta_2 + \dots + \zeta_n}{n}$$

which are roots of unity, which means that:

$$\sigma(\alpha) = \frac{\sigma(\zeta_1) + \dots + \sigma(\zeta_n)}{n}$$

so $|\sigma(\zeta_1)| \leq 1$, so $|\beta| \leq 1$. Since β is an integer, we either get that $|\beta| \in \{0, 1\}$. If $|\beta| = 0$, then $\alpha = 0$, and if $|\beta| = 1$, then $|\alpha| = 1$, and so plugging back into our original equation we get that:

$$\chi_V \in \{0, n\}$$

as we sought to show.

We can also treat this lemma as another check that we have given the correct character table (verify it for the character table of S_4 we found in example 2.2.21).

Theorem 2.2.24: Burnside's Theorem

Let $p \neq q$ be prime numbers, $a, b \in \mathbb{N}$. If $|G| = p^a q^b$, then G is solvable.

Notice how this immediately implies that the smallest possible order of a non-abelian simple group is 30, since $30 = 2 \cdot 3 \cdot 5$, which matches with the fact that the smallest non-abelian group is A_5 which has order 60! Recall also that if $|G| = p^a$, then we would need that $a = 1$ (namely, any p -group has a non-trivial center)

Proof:

Take the composition series:

$$1 = G_0 \trianglelefteq G_1 \trianglelefteq \dots \trianglelefteq G_k = G$$

where G_i/G_{i-1} . Then we'll show that if G is simple, then G is *abelian*. For the sake of contradiction, suppose that G is non-abelian and simple with order $p^a q^b$ (where we may as well assume $a, b \in \mathbb{N}_{>0}$). Let $\chi_1, \chi_2, \dots, \chi_r$ be the irreducible representation for the group G , and $\chi_1 = \chi_1$. Let $Z_i = \{g \in G \mid |\chi_i(g)| = \chi_i(1), \forall i\}$. The condition in side doesn't happen so often, we can equivalently say that $\rho_i(g)$ is a scalar by using the triangle inequality. Thus,

$$Z_i \trianglelefteq G$$

If $Z_i = G$, then G acts by scalars on V_i . But then every subspace of V_i is a subrepresentation, so the dimension of all the irreducible is 1. And so each character is of the form:

$$\chi_i : G \rightarrow \mathbb{C}^\times$$

Since G is simple, each i is injective or trivial. If each i is injective, then G is abelian, a contradiction, and if it is trivial we get $i = 1$ (the trivial representation). Thus, it must be that:

$$Z_i = 1 \quad \forall i > 1$$

Now, take the sylow q -subgroup Q : (notice we are using that $a, b > 0$)

$$1 \subsetneq Q \subsetneq G$$

So Q is a p -group, and so has a non-trivial center: $Z(Q) \neq 1$. Take $h \in Z(Q) \setminus \{e\}$. Hence $Q \leq C_G(h)$, so $[G : C_G(h)][G : Q] = p^a$, where $[G : C_G(h)] = |\text{conj}(h)|$. Now recall we've shown that:

$$\chi_1 ||G| = p^a q^b$$

Now, if $p \nmid \chi_i(1)$, then by the previous line we have that $\chi_1(1) | q^b$. So, $\gcd(\chi_i(1), |\text{conj}(h)|) = 1$, and so by lemma 2.2.23

$$\chi_i(h) = 0 \quad \text{or} \quad \chi_i(1)$$

But if $|\chi_i(h)| = \chi_i(1)$, $h \in Z_i$, and so $i = 1$

So, if $p \nmid \chi_i(1)$ and $i > 0$, then $\chi_i(h) = 0$. Using this, we can get our contradiction: as h is not the trivial element, we get:

$$\begin{aligned} 0 &= \sum_i \overline{\chi_i(h)} \chi_i(1) \\ &= \sum_i \chi_i(h) \chi_i(1) \\ &\stackrel{!}{=} 1 + \sum_{\substack{i>1 \\ p \mid \chi_i(1)}} \underbrace{\chi_i(1)}_{\in p\mathbb{Z}} \underbrace{\chi_i(h)}_{\in \overline{\mathbb{Z}}^c} \\ &= 1 + p\alpha \quad \text{some } \alpha \in \overline{\mathbb{Z}}^c \end{aligned}$$

where the one in the $\stackrel{!}{=}$ equality came from when $i = 1$. Thus,

$$a = -1/p \in \overline{\mathbb{Z}}^c \cap \mathbb{Q} = \mathbb{Z}$$

a contradiction, completing the proof.

(huge list of character tables: commented)

2.3 Induced Representation

This is another way of constructing representations. We saw that if $H \leq G$ and V is a G representation, then we can make V an H -representation through the inclusion map $H \hookrightarrow G \rightarrow \text{GL}(V)$. We may ask if we may go the other way: given an H representation W , can we find a G -representation? If φ is the representations afforded by H , can we find a Φ such that $\Phi|_H = \varphi$? It turns out we cannot: take $A_3 \subseteq S_3$ and consider a faithful representation of degree 1 (recall it's a cyclic group). Then notice that any 1 dimensional representations of S_3 will have to have A_3 in it's kernel, meaning we can't extend the action of A_3 to an action of S_3 . A little more generally, if we let G be a simple group, $H \leq G$, and φ a representation of H where $\ker(\varphi)$ is a proper nontrivial subgroup of H , then if we somehow extend the representation φ of H to Φ of G , then $\ker(\Phi)$ would have a proper kernel too, contradicting that G is simple. So instead, we will not be extending representation from H to G , but *inducing* a representation in a natural way. This method of inducing a representation will work even in the two above cases, meaning an induced representation will not always be an extension.

Let $H \leq G$, and consider some H -representation V , which can be represented as a $k[H]$ -module V .

Recall that we can “extend scalars” and define a new $k[G]$ -module:

$$k[G] \otimes_{k[H]} V$$

This is (the outer tensor product for representatives, which I haven’t talked about yet, but probably should have it here to motivate the idea)

Definition 2.3.1: Induced Representation

Let $H \leq G$ of a finite group G and let V be a $k[H]$ -module affording the representation φ . Then the $k[G]$ -module $k[G] \otimes_{k[H]} V$ is called the *induced module* of V , the representation is called the *induced representation*, and if χ is the character of φ , then the character of the induced representation is called the *induced character*, and is written as $\text{Ind}_H^G(\psi)$.

Another intuition behind defining the induced module is the following: If W is an H -representation, then let our new module be:

$$\text{Ind}_H^G(W) = \{f \in \text{Hom}_{\text{Set}}(G, W) \mid f(hg) = h \cdot f(g), \forall g \in G, h \in H\}$$

Then $\text{Ind}_H^G(W)$ is clearly a complex vector space under pointwise addition and scalar multiplication, and the G -action on this space would be:

$$\gamma \cdot f : G \rightarrow W \quad g \mapsto f(g\gamma)$$

It should be verified that this is indeed G -linear. In this case, we get that:

$$\text{Ind}_H^G(W) = \text{Hom}_{\mathbb{C}[H]}(\mathbb{C}[G], W)$$

where we may use a variant of lemma 2.2.12 to show that this is the same thing as we’ve just defined. It is also easy to see the dimension of $\text{Ind}_H^G(W)$, namely:

$$\dim_{\mathbb{C}}(\text{Ind}_H^G(W)) = [G : H] \dim_{\mathbb{C}}(W)$$

with the $\mathbb{C}$ -vector space isomorphism being:

$$\text{Ind}_H^G(W) \rightarrow W^{\oplus [G:H]} \quad f \mapsto f(g_1), \dots, f(g_{[G:H]})$$

where we treat $G = \coprod_i Hg_i$ for the cosets Hg_i . By letting $g_1 = e$ being the identity, there is a natural canonical map:

$$\text{Ind}_H^G(W) \rightarrow W \quad f \mapsto f(e)$$

This motivates the following adjoint of the restriction functor:

Theorem 2.3.2: Frobenius Reciprocity

Let $H \leq G$. Then if V is a G -representation and W an H -representation, then:

$$\text{Hom}_G(V, \text{Ind}_H^G W) \cong_{\mathbb{C}} \text{Hom}_H(V|_H, W)$$

where $V|_H$ is the restriction of the action to H

Proof:

We will explicitly construct inverse bijections Φ from left to right and Θ from right to left.

Let $\alpha : \text{Ind}_H^G(W) \rightarrow W$, $f \mapsto f(1)$. Then α is H -linear. Then let

$$\Phi(f) = \alpha \circ f$$

Then $\Phi(f)$ is H -linear (check $\mathbb{C}$ -linearity of Φ). For the other way, let $\psi : V|_H \rightarrow W$ be H -linear,

$$\Theta(\psi) : V \rightarrow \text{Ind}_H^G(W) \quad v \mapsto (g \mapsto \psi(gv))$$

Then this is well-defined. What's left is to check they are inverses of each other

2.3.3 Left Vs Right Adjoint This proof shows that induction is right-adjoint to restriction. It can be shown to also be *left adjoint*. A pair of functors (F, G) where the functor F is both left and right adjoint to a functor G (and hence G is also right/left adjoint to F) is sometimes called a *Frobenius pairing*. The functor Ind is usually only right-adjoint to the restriction restriction functor when G is infinite (The *compact induction* functor is usually left-adjoint to the restriction functor, see [NatePadicRep]).

Corollary 2.3.4: Frobenius Reciprocity On Characters

Let $H \leq G$, χ be the character of a G representation and ψ a character of an H -representation. Then:

$$\langle \chi|_H, \psi \rangle_H = \langle \chi, \text{Ind}_H^G(\psi) \rangle_G$$

Proof:

<https://unapologetic.wordpress.com/2010/12/03/dual-frobenius-reciprocity/>

Example 2.3.5: Induced Representations

1. $\text{Ind}_G^G(W) = W$
2. $\text{Ind}_{\mathbb{1}}^G(\mathbb{C}) = \mathbb{C}[G]$
3. $\text{Ind}_H^G(\mathbb{1}) = \mathbb{C}[G/H]$

Another way of thinking of this result is that if V is an irreducible representation of G and W is an irreducible representation of H , then the number of times W occurs in $\text{Ind}_H^G(W)$ is equal to the number of times W occurs in $V|_H$.

Proposition 2.3.6: Properties Of Induced Representation

Let $K \leq H \leq G$, W be an K -representation, $\text{Ind}_K^H(W)$ the induced H representation, $\text{Ind}_H^G(\text{Ind}_K^H(W))$ the induced G -representation. Then:

1. If χ, χ' are two characters of W ,

$$\text{Ind}_K^H(\chi + \chi') = \text{Ind}_K^H(\chi) + \text{Ind}_K^H(\chi')$$

2. Induction is transitive:

$$\text{Ind}_H^G(\text{Ind}_K^H(W)) = \text{Ind}_K^G(W)$$

Proof:

Theorem 2.3.7: Computing Induced Representation

Let $H \leq G$ of a finite group G , and $g_1, g_2, \dots, g_m$ be representatives for distinct left cosets of H in G . Let V be an H -representation of degree n . Then the induced G -representation $W = k[G] \otimes_{k[H]} V$ is an nm degree representation, and if Φ represented the induced representation, then there exists a basis for W such that W affords the matrix representation defined for each $g \in G$ by:

$$\Phi(g) = \begin{pmatrix} \varphi(g_1^{-1}gg_1) & \cdots & \varphi(g_1^{-1}gg_m) \\ \vdots & \ddots & \vdots \\ \varphi(g_m^{-1}gg_1) & \cdots & \varphi(g_m^{-1}gg_m) \end{pmatrix}$$

where each $\varphi(g_i^{-1}gg_j)$ is an $n \times n$ block, with $\varphi(g_i^{-1}gg_j) = 0$ if $g_i^{-1}gg_j \notin H$.

Proof:

First, $k[G]$ is a free right $k[H]$ -module with decomposition:

$$k[G] = g_1k[H] \oplus \cdots \oplus g_mk[H]$$

Since tensor products commute with direct sums, we get:

$$W = k[G] \otimes_{k[H]} V \cong (g_1 \otimes V) \oplus \cdots \oplus (g_m \otimes V)$$

Since k is in the center of $k[G]$, the above isomorphism is also a k -vector space isomorphism. Thus, if $v_1, v_2, \dots, v_n$ is a basis for V , then $\{g_i \otimes v_j \mid 1 \leq i \leq m, 1 \leq j \leq n\}$ is a basis for W , and hence $\dim_k(W) = nm$.

To show $\Phi(g)$ is of the given form, order the basis into m of size n as:

$$g_1 \otimes v_1, g_1 \otimes v_2, \dots, g_1 \otimes v_n, g_2 \otimes v_1, \dots, g_m \otimes v_n$$

Using this order, we can compute the matrix $\varphi(g)$ for each g acting on W given this basis. Fixing j and g , let $g_j = g_i h$ for some index i and $h \in H$. Then for every k :

$$g(g_j \otimes v_k) = (gg_j \otimes v_k) = g_i \otimes v_k = \sum_{t=1}^n a_{tk}(h)(g_i \otimes v_t)$$

where a_{tk} is the t, k coefficient of the matrix h acting on V with respect to the basis $v_1, v_2, \dots, v_n$. Notice that the action of g on W maps the j th block of n basis vectors of W to the i th block of basis vectors, and then has the matrix $\varphi(h)$ as that block. Since $h = g_i^{-1}gg_j$, this fully describes $\Phi(g)$, as we sought to show.

Corollary 2.3.8: Computing The Induced Character

Let $H \leq G$ where G is a finite group, V be an H -representation with character χ , and $\text{Ind}_H^G(\chi)$ the induced character. Then:

$$\text{Ind}_H^G(\chi) = \sum_{i=1}^m \chi(g_i^{-1}gg_i)$$

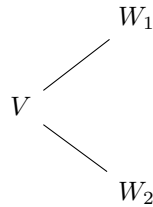
where $\chi(g_i^{-1}gg_i) := 0$ if $g_i^{-1}gg_i \notin H$. Hence, $\text{Ind}_H^G(\chi)(g) = 0$ if g is not conjugate of some element in H in G , in particular if $H \trianglelefteq G$ then $\text{Ind}_H^G(\chi)$ is zero on all elements $G - H$.

Proposition 2.3.9: Induced Index Two Induced Representation

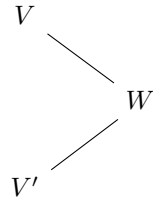
Let $H \leq G$ where $[G : H]$ has index two^a (ex. $A_n \leq S_n$ or $C_n \leq D_{2n}$). Then if V is a finite-dimensional irreducible representation of G , either $V|_H$ is irreducible or $V|_H = W_1 \oplus W_2$ where $W_1 \not\cong W_2$, are both irreducible, and have the same dimension. Moreover, the second option implies that $V \cong V \otimes \chi$ where $\chi : G \rightarrow G/H \cong \{\pm 1\}$ (i.e. a 1-dimensional representation)

^aRecall that this makes H a normal subgroup

In the case where $V|_H$ is not irreducible, we can imagine it splitting into a diagram like so:



while in the case where $V|_H$ is irreducible, what we actually have (and didn't have time to prove), is that we have *two* irreducible representation's of G (V and V') that both get restricted to W :



and then the induced representation of W is the direct sum of V and V' .

Proof:

Let $W \subseteq V|_H$ be an irreducible H -representation. Then Since $H \trianglelefteq G$, gW is an H -stable space: gW is some subspace of V (it satisfies the subspace axioms since the action is linear), and for any $h \in H$, $h \cdot gW = hgW = g'hW$ that is $gW = hg'h^{-1}W$, and since $hg'h^{-1} \in H$, we have that $h \cdot gW = gW$. Now, gW is irreducible: since W is irreducible and $\varphi_g(W) = gW$ is a G -linear isomorphism (with inverse $\varphi_{g^{-1}}(gW) = W$), gW is irreducible. Finally, the structure of gW only depends on that of the coset gH

Thus, fix $\gamma \in G \setminus H$ so that

$$\sum_{g \in G} gW = W + \gamma W \subseteq V$$

where the $\subseteq$ comes from the fact that this is a sum of two subspaces. Now, notice that the left hand side is a G -stable space, but V is irreducible meaning it does not have a G -stable subspace, and therefore, we actually have the equality:

$$W + \gamma W = V$$

meaning the $H + gH = G$ is paralleled in the action! We know have two possibilities: either $W = \gamma W$, and so $V|_H = W$, and we have $\dim V = \dim W$ (which is case (a)).

on the otherhand, if $W \neq \gamma W$, then $W \cap \gamma W = (0)$ (since they are both irreducible), and so $V = W \oplus \gamma W$ (i.e. direct sum as H -representations), and so V is the direct sum of two irreducible H -representations.

Now, by Frobenius reciprocity, we get:

$$\langle W, V|_H \rangle_H = \langle \text{Ind}_H^G(W), V \rangle_G$$

Since W occurs in $V|_H$ by definition, we get that $1 \leq \langle W, V|_H \rangle_H$, while since V is irreducible, meaning by a dimension argument we get $\langle \text{Ind}_H^G(W), V \rangle_G \leq 1$, and hence:

$$1 = \langle W, V|_H \rangle_H = \langle \text{Ind}_H^G(W), V \rangle_G = 1$$

Thus, from the first one, we get that $W \cong \gamma W$ (or else the left hand side would be equal to 2), and we must have $V = \text{Ind}_H^G(W)$ (or else the right hand side would be equal to something greater than 1)

If part (b) holds, then

$$V \oplus \chi|_H \cong V|_H$$

and

$$V \cong \text{Ind}_H^G(W) \cong V \oplus \chi$$

Conversely, if $V \cong V \oplus \chi$, then $\chi_V(g) = \chi_V(g)\chi(g)$ for all $g \in G$. Thus, by simple arithmetics we see that $\chi(g) = 0$ for all $g \notin H$. And so:

$$\|\chi_{V|_H}\|_H^2 = 2\|\chi_V\|_G^2$$

and so it is a direct sum of irreducibles, as we sought to show.

2.4 *Application to Harmonic Analysis

In this section, we take a brief moment to explore the representation of locally compact abelian groups, such as $\mathbb{R}$. This section will assume some understanding of topological groups, and is mainly to motivate for the reader the connection between the representation theory of this chapter with *Fourier analysis* and more broadly *Harmonic analysis*.

Every group G is always a topological group with the discrete topology (see appendix ??). The group G is finite iff it is a compact discrete group. Many groups of interest are either compact (like the torus or compact lie groups), or locally compact (like the real line). The representation theory of these groups rely on the correct framing of the representation in the finite case. For the case of infinite dimension *non-abelian* groups, the build-up is a good deal too complicated for this expository section⁴, however many of the major ideas for the abelian are simple enough to exposit, and even better naturally motivate the Fourier transform and abstract harmonic analysis through the lens of representation theory.

Let G be a finite abelian group so that all complex irreducible representations are 1-dimensional, and hence characters and necessarily a homomorphism. As G is finite, say $n = |G|$, then $\chi(g^n) = \chi(e) = 1$ for all $g \in G$, and so $\chi^n = \text{id}$, showing that $\chi : G \rightarrow \mathbb{C}^*$ restricts to a map $\chi : S^1$. This shows that all characters are *unitary representations*. All characters for a finite group are necessarily unitary representations, while for non-finite groups there can be non-unitary representations⁵. Such groups which have a (general) representation theory that doesn't generalize as well as infinite unitary representations (many representation are indecomposable but not irreducible, for the details see [NateLieGroup]). Thus, for this sections we stick to maps to S^1 with the natural multiplication structure as the subgroup of $\mathbb{C}^*$: label this collection $\hat{G} = \text{Hom}_{\text{Ab}}(G, S^1)$. As the codomain is a group, $\text{hom}_{\text{Ab}}(G, S^1)$ has a group structure given by point-wise multiplication.

Definition 2.4.1: Pontryagin Group

Let G be locally compact abelian group. Then the *Pontryagin dual* of G is the group:

$$\text{Hom}_{\text{Top}}(G, S^1)$$

with the operation being point-wise multiplication.

This is just another way of representing the characters of the group. Let us stick to the special case where G is a finite abelian group and re-frame the previous theory using the Pontryagin group:

⁴see [NateGeoRepTheory] for the EYNTKA series on geometric representation theory

⁵ $\text{SL}_2(\mathbb{R})$ acting on $\mathbb{R}^2$ need not be unitary

Proposition 2.4.2: Pontryagin Dual

Let G be a finite abelian group, $\hat{G} = \text{Hom}_{\text{Ab}}(G, S^1)$ the group of characters under point-wise multiplication. Then there is an isomorphism:

$$G \cong \hat{\hat{G}}$$

Furthermore, there is a natural isomorphism:

$$G \cong \hat{\hat{G}}$$

If G is infinite, it is rarely the case that $G \cong \hat{G}$ (think $\widehat{L^p} \cong L^q$). Even when G is compact this G need not be isomorphic to $\hat{G}$. What is most important is that $\hat{\hat{G}}$ contains the necessary information to recover G , as the double-dual recovers G . We can think of $\widehat{(-)}$ as an operation that goes back and forth from a group to a unitary representation, and as we are working with unitary representations this collection also forms a group⁶. We can thus imagine going back and forth between these two:

$$\text{Group} \xrightleftharpoons{\widehat{(-)}} \text{Unitary Rep.}$$

Proof:

As G is finite abelian,

$$G \cong \mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z} \times \cdots \times \mathbb{Z}/n_k\mathbb{Z}$$

where $n_1 | n_2 | \cdots | n_k$. Choose generators, $a_1, \dots, a_k$. Define the indicator characters on the generators:

$$\chi_i(a_j) = \begin{cases} e^{\frac{2\pi i}{n_i}} & i = j \\ 1 & i \neq j \end{cases}$$

These certainly generate $\hat{G}$. Define the map on the generators:

$$\Phi(a_i) = \chi_i$$

Clearly this is an isomorphism.

For the double-dual, take the natural map:

$$\Phi(g_i) = \text{ev}_{g_i}$$

where as usual $\text{ev}_{g_i}(\chi) = \chi(g_i)$. This is easy to check that is an isomorphism, completing the proof.

Now, when studying spaces we study all (continuous) functions on a space. In our case, our space is finite abelian groups. Let G have the discrete topology so that the set of continuous functions from G to $\mathbb{C}$ is all set functions. We want a $\mathbb{C}$ -algebra isomorphism between $\mathbb{C}[G]$ and $\text{Hom}_{\text{Top}}(G, \mathbb{C})$.

⁶the collection of representations also form a ring called the *representation ring* and has fascinating ties to K-theory and intersection theory, but it is again too much for this section and the connections are discussed in [NateAlgGeo] of the EYTNA series

These two sets are clearly bijective via the natural mapping:

$$\sum_i a_i g_i \mapsto (g_i \mapsto a_i)$$

And this map preserves addition if addition is defined point-wise in the hom-set:

$$\begin{aligned} & \Phi \left(\sum_i a_i g_i + \sum_i b_i g_i \right) \\ &= \Phi \left(\sum_i (a_i + b_i) g_i \right) \\ &\mapsto (g_i \mapsto a_i + b_i) \\ &= (g_i \mapsto a_i) + (g_i \mapsto b_i) \\ &= \Phi \left(\sum_i a_i g_i \right) + \Phi \left(\sum_i b_i g_i \right) \end{aligned}$$

But it does *not* preserve multiplication if the operation is pointwise multiplication:

$$\begin{aligned} & \Phi \left(\sum_{g \in G} a_i g \sum_{g \in G} b_i g \right) \\ &= \Phi \left(\sum_{g \in G} \sum_{g_k g_\ell = g} (a_k b_\ell) g \right) \\ &\mapsto (g_i \mapsto \sum_{g_k g_\ell = g_i} (a_k b_\ell)) \\ &\neq (g_i \mapsto a_i b_i) \\ &\neq (g_i \mapsto a_i) \cdot (g_i \mapsto b_i) \\ &= \Phi \left(\sum_i a_i g_i \right) \cdot \Phi \left(\sum_i b_i g_i \right) \end{aligned}$$

This is because the points g_i act on each other. There is a natural multiplication function defined on $\text{Hom}_{\text{Top}}(G, \mathbb{C})$ called *convolution*:

Definition 2.4.3: Convolution

Let $p, q \in \text{Hom}_{\text{Top}}(G, \mathbb{C})$. Then their *convolution* is define pointwise as:

$$(p * q)(g) = \sum_{g_i g_j = g} p(g_i) q(g_j) = \sum_{h \in G} p(h) q(h^{-1} g)$$

Lemma 2.4.4: Isomorphism With Convolution

Let G be a finite abelian group. Then:

$$(\mathbb{C}[G], +, \cdot) \cong_{k\text{-}\mathbf{Alg}} (\mathrm{Hom}_{\mathrm{Top}}(G, \mathbb{C}), +, *)$$

where the left hand side is the usual group ring and the right hand side has convolution as multiplication

Proof:

refer to the above calculations.

Now, $(\mathrm{Hom}_{\mathrm{Top}}(G, \mathbb{C}), +, *)$ is a $|G|$ -dimensional $\mathbb{C}$ vector-space with the natural basis given by the indicator function:

$$\delta_h(g) = \begin{cases} 1 & g = h \\ 0 & g \neq h \end{cases}$$

Namely for any $f : G \rightarrow \mathbb{C}$ there is a unique decomposition given the functionals defined by each $g \in G$:

$$f = \sum_{h \in G} f(h) \delta_h \quad \text{so that} \quad f(g) = \sum_{h \in G} f(h) \delta_h(g) = f(g) \cdot 1 = f(g)$$

The key feature of this basis is that it acts like evaluation when taking the inner product. Define:

$$\langle f, g \rangle = \sum_{h \in G} f(h) \overline{g(h)}$$

Then $\{\delta_g\}_{g \in G}$ and:

$$\langle f, \delta_{g_i} \rangle = f(g_i) \cdot \overline{\delta_{g_i}(g_i)} = f(g_i) \cdot 1 = f(g_i) = a_i$$

This inner product can be thought as expressly chosen so that the collection δ_{g_i} forms an orthonormal basis, or conversely given this standard inner product, the indicator function will always be an orthonormal basis. By Pontryagin duality, $G \cong \hat{G}$, but this group isomorphism does not *a-priori* preserve the orthogonality of the basis given by G . The key insight of Fourier analysis is that it does!

Proposition 2.4.5: Character Form Orthogonal Basis: Fourier Basis

The characters $\hat{G}$ form an orthogonal basis for $\mathrm{Hom}_{\mathrm{Top}}(G, \mathbb{C})$. Normalizing by $1/\sqrt{|G|}$ forms an orthonormal basis.

Proof:

If $i = j$:

$$\begin{aligned}\langle \chi_i, \chi_i \rangle &= \sum_{g \in G} \chi_i(g) \overline{\chi_i(g)} \\ &= \sum_{g \in G} |\chi_i(g)|^2 \\ &= \sum_{g \in G} 1^2 \\ &= |G|\end{aligned}$$

if $i \neq j$, Then as $\overline{\chi_j} = \chi_j^{-1}$ (as we are mapping to the unit circle), we get that $\chi' = \chi_i \chi_j^{-1}$ is a non-trivial character, so there is some h such that $\chi'(h) \neq 1$. Let

$$\langle \chi_i, \chi_j \rangle = S = \sum_{g \in G} \chi'(g)$$

Then:

$$\chi'(h)S = \chi'(h) \sum_{g \in G} \chi'(g) = \sum_{g \in G} \chi'(h)\chi'(g) = \sum_{g \in G} \chi'(h+g)$$

As $h + G = G$, this is just a re-arranging of terms, hence:

$$\chi'(h)S = \sum_{g \in G} \chi'(g) = S \quad \text{so} \quad S(\chi'(h) - 1) = 0$$

Now, by assumption $\chi'(h) \neq 1$, so $\chi'(h) - 1 \neq 0$, so as $\mathbb{C}$ is an integral domain it must be that $S = 0$, implying:

$$\langle \chi_i, \chi_j \rangle = 0$$

as we sought to show.

This inner product can be thought of as extracting the “phase” information.

Example 2.4.6: Phase-Space Basis

Let $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} = \{(0,0), (1,0), (0,1), (1,1)\}$. A character $\chi : G \rightarrow \mathbb{C}^*$ is determined by its values on the generators $g_1 = (1,0)$ and $g_2 = (0,1)$. Since both generators have order 2, we must have $\chi(g_1)^2 = \chi(g_1^2) = \chi(0,0) = 1$ and likewise for g_2 . Thus, the character’s values on the generators must be either 1 or -1 . This gives $2 \times 2 = 4$ possible characters, which we can index by pairs $(k_1, k_2) \in \{0, 1\}^2$.

The four characters χ_{k_1, k_2} are defined by their values on the generators:

$$\chi_{k_1, k_2}(1, 0) = (-1)^{k_1} \quad \text{and} \quad \chi_{k_1, k_2}(0, 1) = (-1)^{k_2}$$

From this, the value for any element $(a, b) \in G$ is $\chi_{k_1, k_2}(a, b) = (-1)^{k_1 a + k_2 b}$.

This gives the following character table, which explicitly lists the functions that form our new “basis”:

$\hat{G}$	(0,0)	(1,0)	(0,1)	(1,1)
χ_{00}	1	1	1	1
χ_{10}	1	-1	1	-1
χ_{01}	1	1	-1	-1
χ_{11}	1	-1	-1	1

We can verify the orthogonality relation from ?? directly. Let us compute $\langle \chi_{10}, \chi_{11} \rangle$:

$$\begin{aligned}
 \langle \chi_{10}, \chi_{11} \rangle &= \sum_{g \in G} \chi_{10}(g) \overline{\chi_{11}(g)} \\
 &= \chi_{10}(0,0)\chi_{11}(0,0) + \chi_{10}(1,0)\chi_{11}(1,0) + \chi_{10}(0,1)\chi_{11}(0,1) + \chi_{10}(1,1)\chi_{11}(1,1) \\
 &= (1)(1) + (-1)(-1) + (1)(-1) + (-1)(1) \\
 &= 1 + 1 - 1 - 1 = 0
 \end{aligned}$$

Definition 2.4.7: Fourier Transform (Gelfand Transform)

Let G be a finite abelian group with the discrete topology. Then the *Fourier transform* is:

$$(\hat{}) : \text{Hom}_{\text{Top}}(G, \mathbb{C}) \mapsto \text{Hom}_{\text{Top}}(\hat{G}, \mathbb{C}) \quad f \mapsto (\chi \mapsto \langle f, \chi \rangle)$$

where the image of f is often written as:

$$\hat{f}(\chi) = \langle f, \chi \rangle = \sum_{g \in G} f(g) \overline{\chi(g)}$$

The name *Gelfand transform* comes from the fact that the Fourier transform is traditionally about real periodic functions, or Lebesgue measurable functions. It wasn't until later where mathematicians like Folland and Gelfand generalized it to the context of locally compact abelian groups, where the above is the special case where G is finite.

Theorem 2.4.8: Fourier Reciprocity

The map given in definition 2.4.7 is a unitary $\mathbb{C}$ -algebra isomorphism, with point-wise multiplication on $(\text{Hom}_{\text{Top}}(\hat{G}, \mathbb{C}), +, \cdot)$.

Recall that unitary means $\langle \hat{f}, \hat{g} \rangle = \langle f, g \rangle$. Often, the $\mathbb{C}$ -algebra homomorphism is called the *Convolution Theorem*, the unitary property is called the *Plancherel's Theorem*, and invertibility (i.e. bijectivity) is called the *Fourier Inversion Theorem*.

Proof:

1. 1. *Linearity:* Let $f, h \in \text{Hom}_{\text{Top}}(G, \mathbb{C})$ and $a, b \in \mathbb{C}$.

$$\begin{aligned}
 \widehat{af + bh}(\chi) &= \langle af + bh, \chi \rangle && \text{by definition} \\
 &= \sum_{g \in G} (af + bh)(g) \overline{\chi(g)} \\
 &= a \sum_{g \in G} f(g) \overline{\chi(g)} + b \sum_{g \in G} h(g) \overline{\chi(g)} && \text{by linearity of sums} \\
 &= a \langle f, \chi \rangle + b \langle h, \chi \rangle \\
 &= a \hat{f}(\chi) + b \hat{h}(\chi)
 \end{aligned}$$

Since this holds for all $\chi \in \hat{G}$, the transform is a linear map.

2. *$\mathbb{C}$ -algebra Homomorphism (The Convolution Theorem):* We must show that the Fourier transform converts convolution into pointwise multiplication.

$$\begin{aligned}
 \widehat{f * h}(\chi) &= \langle f * h, \chi \rangle \\
 &= \sum_{k \in G} (f * h)(k) \overline{\chi(k)} && \text{by definition of inner product} \\
 &= \sum_{k \in G} \left(\sum_{g \in G} f(g) h(g^{-1}k) \right) \overline{\chi(k)} && \text{by definition of convolution} \\
 &= \sum_{g \in G} f(g) \sum_{k \in G} h(g^{-1}k) \overline{\chi(k)} && \text{swapping order of summation}
 \end{aligned}$$

Let $\ell = g^{-1}k$, so $k = g\ell$. As k runs over G , so does ℓ . The inner sum becomes:

$$\sum_{\ell \in G} h(\ell) \overline{\chi(g\ell)} = \sum_{\ell \in G} h(\ell) \overline{\chi(g)} \overline{\chi(\ell)} = \overline{\chi(g)} \sum_{\ell \in G} h(\ell) \overline{\chi(\ell)} = \overline{\chi(g)} \hat{h}(\chi)$$

Substituting this back into the main expression:

$$\begin{aligned}
 \widehat{f * h}(\chi) &= \sum_{g \in G} f(g) \left(\overline{\chi(g)} \hat{h}(\chi) \right) \\
 &= \hat{h}(\chi) \sum_{g \in G} f(g) \overline{\chi(g)} \\
 &= \hat{h}(\chi) \hat{f}(\chi)
 \end{aligned}$$

Thus, $\widehat{f * h} = \hat{f} \cdot \hat{h}$, proving the map is an algebra homomorphism.

3. *Isomorphism (The Fourier Inversion Theorem):* We show the map is invertible by constructing its inverse. Let $\mathcal{F}(f) = \hat{f}$. We claim the inverse is given by $\mathcal{F}^{-1}(\hat{f})(g) =$

$$\frac{1}{|G|} \sum_{\chi \in \hat{G}} \hat{f}(\chi) \chi(g).$$

$$\begin{aligned} \frac{1}{|G|} \sum_{\chi \in \hat{G}} \hat{f}(\chi) \chi(g) &= \frac{1}{|G|} \sum_{\chi \in \hat{G}} \left(\sum_{h \in G} f(h) \overline{\chi(h)} \right) \chi(g) && \text{by definition of } \hat{f} \\ &= \frac{1}{|G|} \sum_{h \in G} f(h) \sum_{\chi \in \hat{G}} \chi(g) \overline{\chi(h)} && \text{swapping order of summation} \\ &= \frac{1}{|G|} \sum_{h \in G} f(h) \sum_{\chi \in \hat{G}} \chi(g-h) \end{aligned}$$

The inner sum is an example of the *Second Orthogonality Relation* and is a standard exercise the reader should do. In particular, the sum of all characters evaluated at a group element (giving all powers of a certain root of unity) is $|G|$ if $x = e$ and 0 otherwise. Thus $\sum_{\chi \in \hat{G}} \chi(g-h) = |G| \delta_{g,h}$.

$$\frac{1}{|G|} \sum_{h \in G} f(h) (|G| \delta_{g,h}) = \sum_{h \in G} f(h) \delta_{g,h} = f(g)$$

Since we can recover f uniquely from $\hat{f}$, the map is an isomorphism.

4. *Unitary Property (Plancherel's Theorem)*: Note that here is where it is important for the inner product to be normalized. Let us calculate the inner product for the given definition.

$$\begin{aligned} \langle \hat{f}, \hat{h} \rangle_{\hat{G}} &= \sum_{\chi \in \hat{G}} \hat{f}(\chi) \overline{\hat{h}(\chi)} \\ &= \sum_{\chi \in \hat{G}} \left(\sum_{g \in G} f(g) \overline{\chi(g)} \right) \overline{\left(\sum_{k \in G} h(k) \overline{\chi(k)} \right)} \\ &= \sum_{\chi \in \hat{G}} \left(\sum_{g \in G} f(g) \overline{\chi(g)} \right) \left(\sum_{k \in G} \overline{h(k)} \chi(k) \right) \\ &= \sum_{g \in G} \sum_{k \in G} f(g) \overline{h(k)} \sum_{\chi \in \hat{G}} \chi(k) \overline{\chi(g)} && \text{re-arranging terms} \\ &= \sum_{g \in G} \sum_{k \in G} f(g) \overline{h(k)} (|G| \delta_{g,k}) && \text{by Second Orthogonality} \\ &= |G| \sum_{g \in G} f(g) \overline{h(g)} \\ &= |G| \langle f, h \rangle_G \end{aligned}$$

This proves *Plancherel's Theorem*. To make the map strictly unitary, one must normalize the transform. If we define the unitary transform $\mathcal{U}(f) = \frac{1}{\sqrt{|G|}} \hat{f}$, then it follows immediately that $\langle \mathcal{U}(f), \mathcal{U}(h) \rangle = \langle f, h \rangle$, completing the proof.

Example 2.4.9: Fourier Transform

Take $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ from example 2.4.6. Let us now compute the Fourier transform of a concrete function. Let $f = \delta_{(1,0)}$, so that $f(1,0) = 1$ and is zero elsewhere. This is a basis vector in the standard “position” basis. We find its coordinates in the character basis by computing $\hat{f}(\chi) = \langle f, \chi \rangle$ for each character.

$$\begin{aligned}\hat{f}(\chi) &= \langle \delta_{(1,0)}, \chi \rangle = \sum_{g \in G} \delta_{(1,0)}(g) \overline{\chi(g)} \\ &= 1 \cdot \overline{\chi(1,0)} + 0 + 0 + 0 = \chi(1,0)\end{aligned}$$

Using the character table, we can compute this value for each χ :

- $\hat{f}(\chi_{00}) = \chi_{00}(1,0) = 1$
- $\hat{f}(\chi_{10}) = \chi_{10}(1,0) = -1$
- $\hat{f}(\chi_{01}) = \chi_{01}(1,0) = 1$
- $\hat{f}(\chi_{11}) = \chi_{11}(1,0) = -1$

Thus, the original function $f = \delta_{(1,0)}$, which is a spike at one point in the “position” basis, has been transformed into a new function $\hat{f}$ on $\hat{G}$ with values $(1, -1, 1, -1)$.

Recall the inversion formula from the proof of theorem 2.4.8: $f(g) = \frac{1}{|G|} \sum_{\chi \in \hat{G}} \hat{f}(\chi) \chi(g)$ (here $|G| = 4$). Let’s check the value at $g = (1,0)$:

$$\begin{aligned}f(1,0) &= \frac{1}{4} \left(\hat{f}(\chi_{00}) \chi_{00}(1,0) + \hat{f}(\chi_{10}) \chi_{10}(1,0) + \hat{f}(\chi_{01}) \chi_{01}(1,0) + \hat{f}(\chi_{11}) \chi_{11}(1,0) \right) \\ &= \frac{1}{4} ((1)(1) + (-1)(-1) + (1)(1) + (-1)(-1)) \\ &= \frac{1}{4} (1 + 1 + 1 + 1) = 1\end{aligned}$$

Now let’s check the value at $g = (0,1)$, where we expect 0:

$$\begin{aligned}f(0,1) &= \frac{1}{4} \left(\hat{f}(\chi_{00}) \chi_{00}(0,1) + \hat{f}(\chi_{10}) \chi_{10}(0,1) + \hat{f}(\chi_{01}) \chi_{01}(0,1) + \hat{f}(\chi_{11}) \chi_{11}(0,1) \right) \\ &= \frac{1}{4} ((1)(1) + (-1)(1) + (1)(-1) + (-1)(-1)) \\ &= \frac{1}{4} (1 - 1 - 1 + 1) = 0\end{aligned}$$

Let us now connect this to Representation Theory. So far, we have established that for a finite abelian group G , the Fourier transform is a unitary $\mathbb{C}$ -algebra isomorphism between the algebra of functions on G with convolution, $(\text{Hom}_{\text{Top}}(G, \mathbb{C}), *)$, and the algebra of functions on the dual group $\hat{G}$ with pointwise multiplication. The representation theory perspective shows that Fourier analysis is the simplest case of a Artin-Wedderburn decomposition. The key is to view the $\mathbb{C}$ -group algebra $\mathbb{C}[G]$ with the natural G -action given by left multiplication (convolution), namely $\rho : G \rightarrow \text{GL}(\mathbb{C}[G])$

given by:

$$\rho(g) : f \mapsto g * f$$

From this viewpoint, the study of the $\mathbb{C}$ -algebra $(\mathbb{C}[G], +, *)$ and the $\mathbb{C}[G]$ -module $\mathbb{C}[G]$ is the same thing that is the study of the regular representation of G . Now, this representation can be broken down into the direct sum of 1-dimensional irreducible representations. Namely, for a finite abelian group, the set of irreducible representations is exactly the dual group: $\text{Irr}(G) = \hat{G}$ and so we get:

$$\mathbb{C}[G] \cong \bigoplus_{\chi \in \hat{G}} V_{\chi}$$

where each V_{χ} is a 1-dimensional subspace corresponding to the character χ .

This decomposition is made concrete by the Fourier transform. The Fourier Inversion Formula, $f = \frac{1}{|G|} \sum_{\chi \in \hat{G}} \hat{f}(\chi) \chi$, is the literal expression of this decomposition. It writes the function f as a sum of components, where each term $\hat{f}(\chi) \chi$ is the projection of f onto the 1-dimensional irreducible subspace V_{χ} . Then the fourier coefficient $\hat{f}(\chi)$ is the coordinate of f in the irreducible component corresponding to χ .

Note that we can define projection functions $p_{\chi} = \frac{1}{|G|} \sum_{g \in G} \overline{\chi(g)} g$ that we can construct in $\mathbb{C}[G]$ that project onto the irreducible subspace V_{χ} . Convolving any function f with p_{χ} annihilates all other “frequency components”, leaving the part of f that lies in V_{χ} .

Hence, the Fourier basis *the* unique basis that diagonalizes the natural action of the group on its own algebra of functions.

If G is a *non-abelian* finite group, the concept of a dual group $\hat{G}$ and a simple transform onto it ceases to be useful (it is now a set with no natural group structure). However, the representation theory perspective remains: the generalized Fourier transform (given by the *Peter-Weyl Theorem*) decomposes a function f on the group into a sum of *matrix coefficients* corresponding to these higher-dimensional matrix representations.

2.4.1 Generalization to Locally Compact Abelian Groups

What happens when we wish to analyze functions on abelian groups like $(\mathbb{R}, +)$, $(\mathbb{Z}, +)$, or the circle group $(S^1, \cdot)$? These are examples of *Locally Compact Abelian (LCA)* groups. Moving from the finite to the infinite setting requires us to replace our finite sums with integrals, but to integrate on an arbitrary group, we first need a notion of “volume” or “measure”. The essential new ingredient is the *Haar measure*. For any LCA group G , there exists a measure μ that gives a consistent way to integrate functions on the group.

Definition 2.4.10: Haar Measure

A *Haar measure* on an LCA group G is a measure μ on the Borel σ -algebra of G that is left-translation-invariant. That is, for any measurable set $E \subseteq G$ and any element $g \in G$:

$$\mu(gE) = \mu(E)$$

This measure is unique up to a positive scaling constant. For abelian groups, it is also right-translation-invariant.

See [NateRealAnalysis] for details on how to construct such a measure on any LCA that is unique up to scaling. This measure is the direct generalization of taking the order of the group.

Example 2.4.11: Haar Measure

1. For a finite group with the discrete topology, the Haar measure is the counting measure, and $\mu(G) = |G|$.
2. For $(\mathbb{R}, +)$, the Haar measure is the standard Lebesgue measure dx .
3. For $(S^1, \cdot)$, the Haar measure can be taken as $d\theta$.
4. For $(\mathbb{Z}, +)$, the Haar measure is again the counting measure.

With the Haar measure, we can redefine our main objects of study by replacing every sum over G with an integral with respect to $d\mu$.

1. *The Group Algebra $L^1(G)$* : The space of all functions becomes too large. The natural replacement for the convolution algebra $\mathbb{C}[G]$ is the space of absolutely integrable functions, $L^1(G, d\mu)$. For two functions $f, h \in L^1(G)$, their convolution is defined by the integral:

$$(f * h)(g) = \int_G f(x)h(x^{-1}g)d\mu(x)$$

This space $(L^1(G), +, *)$ forms a Banach algebra.

2. *The Dual Group $\widehat{G}$* : It was proven that the correct codomain should be S^1 (see [ref:HERE](#) for the details, Poincare duality paper). The dual group is the set of all *continuous* group homomorphisms into S^1 .

$$\widehat{G} = \text{Hom}_{\text{Top}}(G, S^1)$$

The dual group $\widehat{G}$ is itself an LCA group. It is crucial to note that, unlike the finite case, it is rarely true that $G \cong \widehat{\widehat{G}}$. For example:

$$\widehat{\mathbb{R}} \cong \mathbb{R}, \quad \widehat{\mathbb{Z}} \cong S^1, \quad \widehat{S^1} \cong \mathbb{Z}$$

This highlights why it is essential to treat $\widehat{G}$ as a distinct space. The theory that governs this relationship is *Pontryagin Duality*, which states that the double-dual is canonically isomorphic to the original group: $\widehat{\widehat{G}} \cong G$.

3. *The Fourier Transform*: The definition of the Fourier transform is the direct analogue of the finite case, replacing the sum with an integral. For a function $f \in L^1(G) \cap L^2(G)$, its Fourier transform is a function $\hat{f}$ on the dual group $\widehat{G}$:

$$\hat{f}(\chi) = \int_G f(g)\overline{\chi(g)}d\mu(g)$$

Similarly, the inner product that defines the Hilbert space structure on $L^2(G)$ is:

$$\langle f, h \rangle = \int_G f(g)\overline{h(g)}d\mu(g)$$

With these definitions, the entire suite of theorems proven in the finite case carries over:

Theorem 2.4.12: Fourier Reciprocity on LCA Groups

Let G be an LCA group with Haar measure μ . Then there exists a unique Haar measure ν on the dual group $\widehat{G}$ such that the Fourier transform, appropriately normalized, is a unitary $\mathbb{C}$ -algebra isomorphism. The main results are:

1. *The Fourier Inversion Theorem:* For a sufficiently well-behaved function f (e.g., $f \in L^1 \cap L^2$ such that $\hat{f} \in L^1$), the original function can be recovered by an integral over the dual group:

$$f(g) = \int_{\widehat{G}} \hat{f}(\chi) \chi(g) d\nu(\chi)$$

2. *Plancherel's Theorem:* The Fourier transform extends uniquely from $L^1 \cap L^2$ to a unitary isomorphism of Hilbert spaces, $\mathcal{F} : L^2(G, \mu) \rightarrow L^2(\widehat{G}, \nu)$. This means it preserves the inner product:

$$\langle \hat{f}, \hat{h} \rangle_{L^2(\widehat{G})} = \langle f, h \rangle_{L^2(G)}$$

3. *The Convolution Theorem:* For $f, h \in L^1(G)$, the Fourier transform maps convolution to pointwise multiplication:

$$\widehat{f * h}(\chi) = \hat{f}(\chi) \cdot \hat{h}(\chi)$$

Proof:

see [cite:HERE](#) (it's not in my analysis book, and I didn't type it up in Pontryagin notes)

The correspondence between the finite and general LCA settings can be summarized as follows:

Finite Abelian Case	LCA Case
Group G	LCA Group G
Order $ G $	Haar Measure μ
Group Algebra $\mathbb{C}[G]$	Banach Algebra $L^1(G, \mu)$
Hilbert Space $\text{Hom}(G, \mathbb{C})$	Hilbert Space $L^2(G, \mu)$
Summation $\sum_{g \in G}$	Integration $\int_G d\mu(g)$
Dual Group $\widehat{G} = \text{Hom}(G, \mathbb{C}^\times)$	Dual Group $\widehat{G} = \text{Hom}_{\text{Top}}(G, S^1)$
Inner Product $\sum f(g) \overline{h(g)}$	Inner Product $\int f(g) \overline{h(g)} d\mu(g)$
Fourier Transform $\sum f(g) \chi(g)$	Fourier Transform $\int f(g) \chi(g) d\mu(g)$

Further interesting facts can be found here: <https://mathoverflow.net/questions/37021/is-fourier-analysis-a-spe>

2.4.2 The Algebraic-Geometric Meaning of the Dual Group

In algebraic geometry, a fundamental principle is that the geometry of a space can be recovered from its algebra of functions. For an affine space k^n , the corresponding algebra is the polynomial ring $k[x_1, \dots, x_n]$. The points of the space correspond to the maximal ideals of this algebra. Specifically, for any point $p = (a_1, \dots, a_n) \in k^n$, the evaluation map:

$$\text{ev}_p : k[x_1, \dots, x_n] \rightarrow k \quad f \mapsto f(p)$$

is an algebra homomorphism whose kernel, $m_p = (x_1 - a_1, \dots, x_n - a_n)$, is a maximal ideal. The quotient $k[x_1, \dots, x_n]/m_p \cong k$ recovers the value of the function at that point.

We have established a similar setup for a finite abelian group G , with the convolution algebra $(\text{Hom}_{\text{Top}}(G, \mathbb{C}), *, +)$ taking the role of the function algebra. A natural question arises: can we recover the “points” of our space from the maximal ideals of this algebra? Let’s first attempt a direct analogy. We might assume the points are the elements of the group G itself. For a point $g_i \in G$, we can define the point-evaluation map, which is a linear map:

$$\text{ev}_{g_i} : \text{Hom}_{\text{Top}}(G, \mathbb{C}) \rightarrow \mathbb{C} \quad f \mapsto f(g_i)$$

The kernel of this map is the vector subspace of functions that are zero at g_i . However, for this subspace to be an ideal, it must be closed under convolution. Let $f \in \ker(\text{ev}_{g_i})$ (so $f(g_i) = 0$) and let h be any function in our algebra. Consider their convolution:

$$(h * f)(g_i) = \sum_{k \in G} h(k) f(k^{-1}g_i)$$

Just because $f(g_i) = 0$ does not imply that the sum on the right-hand side is zero. The value of f at other points, $f(k^{-1}g_i)$, can be non-zero. Therefore, the kernel of the point-evaluation map is *not* an ideal with respect to convolution. This implies the points of G do not correspond to the maximal ideals of its convolution algebra.

The key insight is that the correct “geometric points” for the convolution algebra are not the elements of G , but the characters of the dual group $\widehat{G}$.

Proposition 2.4.13: Characters and Maximal Ideals

Let G be a finite abelian group. The maximal ideals of the convolution algebra $(\text{Hom}_{\text{Top}}(G, \mathbb{C}), +, *)$ are in bijective correspondence with the characters $\chi \in \widehat{G}$.

Proof:

For each character $\chi \in \widehat{G}$, define a Fourier evaluation map:

$$\widehat{\text{ev}}_\chi : \text{Hom}_{\text{Top}}(G, \mathbb{C}) \rightarrow \mathbb{C} \quad f \mapsto \widehat{f}(\chi)$$

This map takes a function and returns its Fourier coefficient at the frequency χ . By 2.4.8 $\widehat{\text{ev}}_\chi$ is a $\mathbb{C}$ -algebra homomorphism. Since $\widehat{\text{ev}}_\chi$ is a surjective algebra homomorphism onto a field ($\mathbb{C}$), its kernel, which we denote by I_χ , is a maximal ideal:

$$I_\chi = \ker(\widehat{\text{ev}}_\chi) = \{f \in \text{Hom}_{\text{Top}}(G, \mathbb{C}) \mid \widehat{f}(\chi) = 0\}$$

By the First Isomorphism Theorem for rings, we have:

$$\text{Hom}_{\text{Top}}(G, \mathbb{C})/I_\chi \cong \mathbb{C}$$

This shows that every character $\chi \in \widehat{G}$ defines a maximal ideal. Conversely, After applying the Fourier transform we get the usual multiplication and ring structure, and then an approach similar to the Gelfand-Kolmogorov theorem gives the desired result.

For the infinite case, the maximal ideals of the Banach algebra $L^1(G, \mu)$ arise in this same way from homomorphisms into $\mathbb{C}$ (see the *Gelfand-Mazur Theorem*).

Thus, with the convolution algebra $\mathbb{C}[G]$ as our primary object, the natural geometric space that emerges is not the group G itself, but its Pontryagin dual, $\widehat{G}$. The Fourier transform is the mechanism that allows us to evaluate the functions of our algebra at the points of this “true” underlying space. This perspective, generalized by Gelfand, is the foundation of non-commutative geometry.

Concepts	Classical Algebraic Geometry
Algebra	Polynomial Ring $(k[x_1, \dots, x_n], \cdot)$
Product	Polynomial Multiplication
Geometric Space	Affine Space k^n
A "Point" p	A tuple $(a_1, \dots, a_n) \in k^n$
Evaluation Map	$\text{ev}_p : f \mapsto f(p)$
Maximal Ideal for p	$\ker(\text{ev}_p) = (x_i - a_i)$

Concepts	Harmonic Analysis on G
Algebra	Group Algebra $(\mathbb{C}[G], *)$
Product	Convolution
Geometric Space	The Dual Group $\widehat{G}$
A "Point" p	A character $\chi \in \widehat{G}$
Evaluation Map	Fourier “Evaluation” $\widehat{\text{ev}}_\chi : f \mapsto \widehat{f}(\chi)$
Maximal Ideal for p	$\ker(\widehat{\text{ev}}_\chi) = I_\chi$

Representation of Lie Algebra's and Quiver Theory

(Currently, this chapter really focuses on Quiver theory. I will put my exploration of Lie Algebra Representations here)

(This motivation is if you know some differential geometry) Quiver's have their origin in the classification of Lie groups. A Lie group is a group $(G, \cdot)$ G is a *smooth manifold* and $\cdot : G \times G \rightarrow G$ is a *smooth map*. The classification of these types of groups (in particular, the continuous ones, since discrete lie groups is just the classification of all groups) is a problem completed by Killing. Out of his classification, there came out 5 exotic Lie groups lying outside of the other classifications. Why did these Lie groups come up? The answer of this question might come from a change in perspective: Any Lie group has a Lie Algebra – the tangent space at the identity of the group which also has an extra operation called the *lie braked* which allows to capture in some of the smooth properties of G into $T_e G$. These Lie brackets allow us to define a linear map $\text{ad}_x : T_e G \rightarrow \mathbb{C}$, $\text{ad}_x(y) = [x, y]$. The image of these operations are called the *ideals* of a lie algebra (not unlike rings, since an ideal $I \subseteq L$ is the set of all elements where for all $x \in L$ and $a \in I$, $[x, a] \in I$). Using this fact, we can ask for the classification of *simple lie algebras*. As an example of one:

$$\mathfrak{sl}(3, \mathbb{C}) = \left\{ \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \mid a + e + i = 0 \right\}$$

which is closed under Lie braked $[A, B] = AB - BA$ (but not closed under multiplication). Now, since these are linear maps, they can be diagonalized. If we take $h_1, \dots, h_k \in \mathfrak{sl}(3, \mathbb{C})$ to be diagonal elements (elements that are zero everywhere except possibly the diagonal), then $\text{ad}_{h_1}, \dots, \text{ad}_{h_k}$ are all simultaneously diagonalizable. This leads us to be able to define *root systems*, which allows for *root decomposition* and *root systems*. This leads to two important theorems for classification of *complex lie algebras*:

(Cartan's Theorems)

1. Up to isomorphism, there is just one complex semisimple Lie algebra for each root system
2. With 5 exceptions, every finite dimensional semisimple Lie algebra over $\mathbb{C}$ is isomorphic to one of the classical Lie Algebras:

$$\mathfrak{sl}(n, \mathbb{C}) \quad \mathfrak{so}(n, \mathbb{C}) \quad \mathfrak{sp}(2n, \mathbb{C})$$

and 5 exceptional Lie algebras, labeled: e_6, e_7, e_8, f_4, g_2

We will be building up to this result for the first bit of this chapter.

3.1 Lie Algebras

(here)

3.2 Quiver's and their Decomposition

The following is just an accumulation of common terms in graph theory that we will need. The one notable exception is that we will rename oriented graphs to *quivers*

Definition 3.2.1: Quiver

A *quiver* Q is an oriented graph (V, E, s, t) where V represents the vertices (the vertex set), E is the set of edges (the “arrows”) and $s : E \rightarrow V$ with $t : E \rightarrow V$ represents the orientation (s for source, t for target), that is for any $e \in E$, if we have:

$$\bullet \xrightarrow{e} \bullet$$

then $s(e) = i$ and $t(e) = j$.

Usually, we will abuse notation say $Q = (V, E)$ is a quiver and omit s and t . Most of the time, we will draw quivers instead of explicitly define them through a 4-tuple or 2-tuple

Example 3.2.2: Quivers

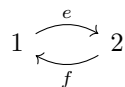
1. If $V = \{1\}$, $E = \{e\}$, and $s(e) = t(e) = 1$, then we can draw this quiver as

$$\begin{array}{c} \curvearrowright \\ 1 \end{array}$$

2. If $V = \{1, 2\}$, $E = \{e\}$, $s(e) = 1$, and $t(e) = 2$, then we can draw this quiver as

$$1 \xrightarrow{e} 2$$

3. If $V = \{1, 2\}$, $E = \{e, f\}$, with $s(e) = 1$, $t(e) = 2$ and $s(f) = 2$, $t(f) = 1$, then we can draw this quiver as



If you are still uncomfortable, with quivers, try to come up with some as a challenge.

Definition 3.2.3: Path and Cycle

Let Q be a quiver.

1. A path γ is a sequence of edges

$$\gamma = e_k e_{k-1} \cdots e_1$$

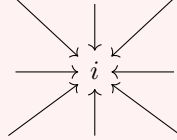
where $t(e_i) = s(e_{i+1})$. for $1 \leq i \leq k-1$.

2. If γ is a path where $s(e_1) = a$ and $t(e_k) = b$, then we say that γ is a path from a to b , and we write $s(\gamma) = a$, $t(\gamma) = b$.
3. The number of number of edges in a path is called the *length* of he path.
4. If γ is a path and $s(\gamma) = t(\gamma)$, we call γ a *cycle*.
5. A cycle of length 1 is called a *loop*

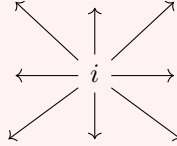
Definition 3.2.4: Sink And Source

Let Q be a quiver

1. Let $e \in E$ and $i \in V$. We say that e contains i if $s(e) = i$ or $t(e) = i$. We say that e is *incoming at i* if $t(e) = i$ and that e is *outgoing at i* if $s(e) = i$
2. We say a vertex $i \in V$ is a *sink* if all edges containing i are incoming. Intuitively, this can be visualized as:



3. We say a vertex $i \in V$ is a *source* if all edges containing i are outgoing. Intuitively, this can be visualized as:



4. Let s be a sink or a source. Then define sQ to be a new quiver where all the arrows at s are reversed.

3.2.1 Quiver Representation and Varieties**Definition 3.2.5: Quiver Representation**

Let Q be a quiver. Then a *quiver representation V of Q* consists of

1. a vector space $V(i)$ over a fixed field $\mathbb{F}$ for each vertex of Q
2. a linear transformation $T_e : V(i) \rightarrow V(j)$ for each $e \in E$

Let $\text{Rep}_{\mathbb{F}}(Q)$, or $\text{Rep}(Q)$ be the collection of all quiver representations of Q .

Example 3.2.6: Quiver Representations

1. Let

$$1 \xrightarrow{e} 2$$

be a quiver. Then any linear transformation $T : V \rightarrow W$ is a representation of this quiver

2. Let

$$1 \xrightarrow{e} 2 \quad \begin{array}{c} \curvearrowright \\ f \end{array}$$

be a quiver. Then we can take any linear transformation $T : V \rightarrow W$ and an identity map $\text{id} : W \rightarrow W$. Note that f need not be represented by the identity: any linear transformation from W to itself is valid.

3. Take

$$1 \xrightarrow{e} 2 \xrightarrow{f} 3$$

Then one class of possible representations of this quiver consists of vector-spaces $V(1)$, $V(2)$, $V(3)$ where the sequence is exact at $V(2)$. Another representation can have T_e and T_f be scalar multiplications. Yet another can be the set of vector spaces and linear transformations T_e and T_f where $T_e, T_f \neq \text{id}$ but $T_f \circ T_e = \text{id}$. There are many more possibilities

Since each vertex has a representation, we can capture a notion of dimension of a quiver representation

Definition 3.2.7: Quiver Dimension

Let Q be a quiver and V a representation of Q . Then the *dimension* or *quiver dimension* of V is the tuple that is denoted:

$$\dim_Q V = (\dim V(i))_{i \in I}$$

where each i is a vertex. If there are finitely many vertices, then we can biject them to natural numbers and have:

$$\dim_Q V = (\dim V(1), \dim V(2), \dots, \dim V(n))$$

Definition 3.2.8: Quiver Subrepresentation

Let Q be a quiver and V be a quiver representation. Then a *subrepresentation* W of V consists of a subspace $W(i) \subseteq V(i)$ for each $i \in V$ and a linear transformations restricted to the new domain, $T_e|_{W(i)} = S_e$ for each $e \in E$, such that each $S_e, S_e(W_i) \subseteq W_j$

the last condition is so that each T_e is still well-defined. As an exercise, see if you can come up with quivers and subrepresentations (you can keep it simple and consider a quiver $\bullet \rightarrow \bullet$).

Definition 3.2.9: Quiver Morphisms

Let Q be a quiver and let V and W be representations of Q . Then we say that $\varphi : V \rightarrow W$ is a *quiver morphism* if for every edge e with representation $T_e : V(i) \rightarrow V(j)$ in V and representation $S_e : W(i) \rightarrow W(j)$ in W , there exists linear transformations $\varphi(i)$ and $\varphi(j)$ such that the following diagram commutes:

$$\begin{array}{ccc} V(i) & \xrightarrow{T_e} & V(j) \\ \downarrow \varphi(i) & & \downarrow \varphi(j) \\ W(i) & \xrightarrow{S_e} & W(j) \end{array}$$

collection of all quiver morphisms between V and W is denoted $\text{Hom}_Q(V, W)$ or $\text{Hom}(V, W)$ if it is unambiguous in context.

Proposition 3.2.10: Structure Of $\text{Hom}(V, W)$

Let $V, W \in \text{Rep}(Q)$. Then $\text{Hom}(V, W)$ forms a vector-space structure over the base field $\mathbb{F}$.

Proof:

left as an exercise

Example 3.2.11: Quiver Morphism

1. Let $V = \mathbb{F}^1 \xrightarrow{1} \mathbb{F}^1$ and $W = \mathbb{F}^1 \xrightarrow{0} \{0\}$ be two quiver representations. Then we can define a morphism:

$$\begin{array}{ccc} V & \mathbb{F} & \xrightarrow{1} \mathbb{F} \\ \downarrow \varphi & \downarrow 1 & \downarrow 0 \\ W & \mathbb{F} & \xrightarrow{0} \{0\} \end{array}$$

Does there exist a nonzero morphism $\varphi : W \rightarrow V$?

2. Let $V = \mathbb{R} \xrightarrow{1} \mathbb{R} \xrightarrow{1} \mathbb{R}$ and $W = \mathbb{R}^2 \xrightarrow{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}} \mathbb{R}^2 \xrightarrow{\begin{pmatrix} 1 & 1 \end{pmatrix}} \mathbb{R}$. Then: (commented to compile) is a quiver morphism. Does there exist a nonzero morphism $\varphi : W \rightarrow V$?

What we have technically defined is a *quiver representation morphism*. A quiver morphisms would usually be between two direct graphs where if $Q = (V, E, s, t)$ and $R = (V', E', s', t')$ are two quivers, then a quiver morphism would consist two functions $m_v : V \rightarrow V'$ and $m_e : E \rightarrow E'$ such that $m_v \circ s = s' \circ m_e$ and $m_v \circ t = t' \circ m_e$. This is ostensibly the same concept with different objects. All morphisms between Q and R would be represented as $\text{Hom}_{\mathbf{Quiv}}(Q, R)$. Most of the time, we will care about the representation of quivers, and hence leaving the name of “quiver morphism” to representations. In either case, the collection of all quivers along with the just-defined quiver morphism defines a category $\mathbf{Quiv}$, and the collection of all representations of a quiver $\text{Rep}(Q)$ along with the previously-defined quiver morphism defines a category $\mathbf{Quiv}(Q)$ (you can check these as exercises)

The category $\mathbf{Quiv}(Q)$ has a 0, namely the representation where all vertices are assigned the zero vector-spaces and the linear transformations are the identity. Since 0 exists, we can define kernels. Images also exist in this category:

Definition 3.2.12: Kernels And Images

Let $\varphi : V \rightarrow W$ be a quiver morphism. Then $\ker(\varphi)$ is the subrepresentation of V consisting of $\ker(\varphi(i)) \subseteq V(i)$, and $\text{im}(\varphi)$ is the subrepresentation of W consisting of $\text{im}(\varphi(i)) \subseteq W(i)$

Note that $\ker(\varphi)$ and $\text{im}(\varphi)$ are indeed subrepresentations: the fact that the vertices exist is a direct consequence of linear algebra. It remains to show that the diagram still commutes.

Definition 3.2.13: Quiver Isomorphisms

Let $V, W \in \text{Rep}(Q)$. Then V is isomorphic to W if there exists a quiver morphism $\varphi : V \rightarrow W$ and $\psi : W \rightarrow V$. We denote two isomorphic quivers as $V \cong W$

Example 3.2.14: Quiver Isomorphism

In Jamail's notes. Some pretty cool ones. In particular, the classification of quiver representations of the form $\bullet \xrightarrow{e} \bullet$ with each vector-space being finite dimensional.

Next, we define the dual of sub-representations. Since for every subspace of a vector space $W \subseteq V$ we can define V/W , the notion of a quotient quiver is defined similarly

Definition 3.2.15: Quotient Quiver

Let $V \in \text{Rep}(Q)$ and $W \subseteq V$ a subrepresentation. Then V/W is defined as a new quiver where every vertex of V is replaced by the quotient of $V(i)/W(i)$.

Proposition 3.2.16: 1st Isomorphism for Quivers

Let $V, W \in \text{Rep}(Q)$ and $\varphi : V \rightarrow W$ be a quiver morphism. Then:

$$\text{im}(\varphi) = V / \ker(\varphi)$$

Proof:

left as an exercise

Definition 3.2.17: Product Quiver

Let $V, W \in \text{Rep}(Q)$. Define $V \oplus W \in \text{Rep}(Q)$ as

1. $(V \oplus W)(i) = V(i) \oplus W(i)$
2. $T_e : (V \oplus W)(i) \rightarrow (V \oplus W)(j)$ is given by $T_e = T_{e,v} \oplus T_{e,w}$ (recall that $T \oplus S(a, b) = (T(a), S(b))$)

Definition 3.2.18: Quiver Variety

Let $Q = (I, E, s, t)$ be a finite quiver. and $\underline{d} \in \mathbb{N}^n$. Then a *quiver variety* is:

$$\text{Rep}(\underline{d}, Q) := \prod_{e \in E} M_{d_j, d_i}(\mathbb{F})$$

where $i = s(e)$ and $j = t(e)$. Every point in $\text{Rep}(\underline{d}, Q)$ is a representation of Q with dimension vector $\underline{d}$.

A quiver variety let's us reduce a quiver where each vertex can have a different dimension (i.e. any possible total dimension) to only thinking about quivers with the same total dimension

Proposition 3.2.19: Action On Quiver Variety

Let $Q = (I, E, s, t)$ be a finite quiver with $I = \{1, \dots, n\}$ and $E = \{e_1, \dots, e_n\}$. Let $\underline{d} = (d_1, \dots, d_n) \in \mathbb{N}^n$, and define the group $G(\underline{d}) = \prod_i GL_{d_i}(\mathbb{F})$. Then define the $G(\underline{d})$ action on $\text{Rep}(Q, \underline{d})$ as follows:

$$g = (\varphi_1, \dots, \varphi_n)$$

$$(g \cdot V)_e = (\varphi_j A_e \varphi_i^{-1})_e$$

where $s(e) = i$ and $t(e) = j$.

Proof:

Clearly, if $g = (I)_e$ for each edge, then

$$(g \cdot V)_e = (I_j A_e I_i)_e = (A_e)_e$$

showing that $I \cdot V = V$. Associativity comes for free from associativity of composition of linear maps.

I think the group in the theorem is called a *Weyl Group*.

Theorem 3.2.20: Isomorphism Equivalence

Let $\text{Rep}(Q, \underline{d})$ be a quiver variety and $G(\underline{d})$ be the group as defined in proposition 3.2.19. Then two representations $V, W \in \text{Rep}(Q, \underline{d})$ are isomorphic if and only if there exists a $g \in G(\underline{d})$ such that $g \cdot V = W$. That is, two quivers are isomorphic if they lie in the same orbit:

$$\{G(\underline{d})\text{-orbit of } \text{Rep}(Q, \underline{d})\} \rightleftarrows \{\text{iso classes of rep with dim } \underline{d}\}$$

Proof:

As we've seen in example 3.2.14, two quiver representations (of the same dimension) of $\bullet \xrightarrow{e} \bullet$ are equivalent if T_e and S_e are equivalent, i.e., their range is of the same dimension, i.e. $\varphi_j A_e \varphi_i^{-1} = B_e$ for two invertible matrices φ_i and φ_j . However, this is exactly the condition for the orbits of this group-actions.

3.2.2 Decomposing Quivers

Similarly to all other algebraic structures we've encountered, we have a notion of simple objects:

Definition 3.2.21: Simple Representation

Let Q be a quiver. Then $V \in \text{Rep}(Q)$ is called a *simple* or *irreducible* representation if the only subrepresentations of V are $\{0\}$ and V .

The notion of simple objects is essentially trivial for vectorspaces since $V \cong \mathbb{F}^n$. In other words, every vector-space is decomposable into simple vector-spaces. The analogous notion of decomposability exists for quivers, however, they can be more complicated for quivers

Definition 3.2.22: Indecomposable Quiver

Let Q be a quiver. Then $V \in \text{Rep}(V)$ said to be *decomposable* if there exists non-trivial subrepresentation $V_1, V_2 \subseteq V$ such that:

$$V \cong V_1 \oplus V_2$$

and is said to be *indecomposable* otherwise

Example 3.2.23: Simple Quivers and decomposability

Let $Q = \bullet \xrightarrow{e} \bullet$. Then:

$$0 \xrightarrow{0} \mathbb{F} \quad \mathbb{F} \xrightarrow{0} 0$$

are both simple representations. Note that the left quivers is a subrepresentations of $V = \mathbb{F} \xrightarrow{\text{id}} \mathbb{F}$, and along with $\mathbb{F} \xrightarrow{\text{id}} \mathbb{F}$ and $0 \xrightarrow{0} 0$, $\mathbb{F} \xrightarrow{\text{id}} \mathbb{F}$ has 3 sub-quivers. Thus, V is not a simple quiver.

However, V is *indecomposable*. To see this, notice that if V were decomposable, then we get:

$$\dim_Q(V_1 \oplus V_2) = \dim_Q(V_1) + \dim_Q(V_2)$$

and for the representation to be nontrivial, we would have that one representation would have dimension $(1, 0)$, while the other has $(0, 1)$. However, V has no representation of dimension $(1, 0)$.

This means that the extension of quivers will be more elaborate than split exact sequences.

Theorem 3.2.24: Quiver Representation Decomposition

Let Q be a quiver and V be a representaiton where $\sum_i \dim V(i) < \infty$. Then there exists a finite set of indecomposable subrepresentations $I_1, \dots, I_n$ such that

$$V = I_1 \oplus I_2 \oplus \dots \oplus I_n$$

Proof:

Define the *total dimension* $\text{Tdim}(V) = \sum_i \dim V(i)$. If $\text{Tdim}(V) = 1$, then V must be only contain one vector-space $\mathbb{F}$ with all other vecor-spcae being the trivial vector spaces. Assume for a moment that V has no oriented cycles. Then all maps from edges that are being the trivial map. Clearly, V is indecomposable, and so it is it's own decomposition.

On the other hand, if V has an oriented cycle, then without loss of generality we can consider

$$T = T_n \circ T_{n-1} \circ \dots \circ T_1$$

so that we have $T : \mathbb{F} \rightarrow \mathbb{F}$. Since $\mathbb{F}$ has no non-trivial subspace (being an indecomposable vector-space over $\mathbb{F}$), V is still indecomposable, and so is it's own decomposition.

Using strong induction, assume that all vector-spaces of total dimension $\text{Tdim}(V) = n$ can be decomposed into a direct sum of indecomposable representations, and consider a representation of total dimension $n + 1$. If V is indecomposable, then we're done, so let's assume V is

decomposable so that there exists two non-trivial subrepresentations V_1, V_2 such that

$$V = V_1 \oplus V_2$$

Then since V_1, V_2 are both non-trivial, their total dimension must be strictly greater than and strictly less than that of V and nonzero. But then, by the inductive hypothesis, V_1 and V_2 can be decomposed into a finite direct sum of indecomposable modules. Hence:

$$V = V_1 \oplus V_2 = (I_{1,1} \oplus \cdots \oplus I_{1,n_1}) \oplus (I_{2,1} \oplus \cdots \oplus I_{2,n_2})$$

completing the proof.

Example 3.2.25: Decomposing Representations

1. Let $Q = \bullet \xrightarrow{e} \bullet$ be a quiver. Find all simple and indecomposable representations of Q . Using this knowledge, decompose any representation $V \xrightarrow{T_e} W$.
2. Let $Q = \bullet \xrightarrow{e} \bullet \xrightarrow{f} \bullet$. Find all simple and indecomposable representation of Q . Using this knowledge, decompose any representation $V \xrightarrow{T_e} W \xrightarrow{T_f} L$.
3. Let $Q = \begin{array}{c} \curvearrowright \\ \bullet \end{array}$ be a quiver. Given $\mathbb{F}$ is algebraically closed, find all simple and indecomposable representations of Q (this is heavily reliant on Jordan canonical forms from chapter ??). Using this knowledge, decompose any representation $T : V \rightarrow V$.

3.2.3 Root Systems

Recall that if $(V, \langle \cdot, \cdot \rangle)$ is an inner product space and $v \in V$, then

$$\frac{\langle w, v \rangle}{\langle v, v \rangle} v$$

is the projection of w onto v . Furthermore, define:

$$s_v(w) = w - 2 \frac{\langle w, v \rangle}{\langle v, v \rangle} v$$

which reflects w across the hyperplane orthogonal to v .

Definition 3.2.26: Root System

Let V be an inner product space over $\mathbb{R}$. Then a (*reduced*) *root system* $R \subseteq V$ is a finite collection of non-zero vectors such that

1. $\text{span}(R) = V$
2. For all $\alpha, \beta \in R$, $\eta_{\alpha, \beta} = \frac{2\langle \alpha, \beta \rangle}{\langle \beta, \beta \rangle} \in \mathbb{Z}$. In terms of angles, recall that $\theta = \cos^{-1} \left(\frac{\langle \alpha, \beta \rangle}{\|\alpha\| \|\beta\|} \right)$, so $n_{\alpha\beta} = \|\alpha\| \cos(\theta)$
3. For each $\alpha \in R$, the corresponding reflection $s_\alpha : V \rightarrow V$ has the property that for all $\beta \in R$, $s_\alpha(\beta) \in R$, i.e., R is symmetric under all α reflections.
4. If both $\alpha, c\alpha \in R$ then $c = \pm 1$ (this is true for reduced systems only)

Example 3.2.27: Root Systems

Let $V = \{(x, y, z) \in V \mid x + y + z = 0\}$, where V inherits the usual inner product from $\mathbb{R}^3$. Let

$$R = \{\pm(e_1 - e_2), \pm(e_2 - e_3), \pm(e_1, e_3)\} = \{\alpha, \beta, \gamma\}$$

Then R is a reduced root system. This set spans V because HERE.

Furthermore:

$$\eta_{\pm\alpha, \pm\beta} = \frac{2\langle \pm(e_1 - e_2), \pm(e_2 - e_3) \rangle}{\pm\langle (e_2 - e_3), \pm(e_2 - e_3) \rangle}$$

(finish rest later)

Include these

Notice that any root system can be simplified:

Definition 3.2.28: Polarization And Simple

Let R be a root system. Then if there exists a vector v such that $\langle v, \alpha \rangle \neq 0$ for all $\alpha \in R$ then $R_+ = \{\alpha \in R \mid \langle v, \alpha \rangle > 0\}$ along with $R_- = \{\alpha \in R \mid \langle v, \alpha \rangle < 0\}$ is called the *polarization* of R , and $R = R_+ \sqcup R_-$. The elements of R_+ are called *positive roots*. A positive root is called *simple* if it cannot be expressed as the sum of other positive roots.

Example 3.2.29: Simple Roots

homework, exercise 8 in Jimails notes

Theorem 3.2.30: Simple Root Decomposition

Let R, V be a (reduced) root system with polarization $R = R_+ \sqcup R_-$. Let Π denote the set of simple roots for this polarization. Then:

1. Every $\alpha \in R_+$ is a sum of simple roots
2. If $\alpha, \beta \in R_+$, then $\langle \alpha, \beta \rangle \leq 0$
3. The set Π of simple roots forms a basis for V

Proof:

exercise 9

(words about root systems being isomorphic, which I'm guessing is a map from one vec sp from another s.t. all the properties of root systems are preserved)

Definition 3.2.31: Cartan Matrix

Let R, V be a (reduced) root system with polarization $R = R_+ \sqcup R_-$, let Π denote the set of simple roots for this polarization, and let $r = |\Pi|$. Then define *Cartan matrix* A to be the $r \times r$ matrix defined by:

$$A_{ij} = n_{\alpha_j \alpha_i} = \frac{2\langle \alpha_i, \alpha_j \rangle}{\langle \alpha_i, \alpha_i \rangle}$$

Example 3.2.32: Cartan Matrix

exercise 10

(word to motivate the following visualization)

Definition 3.2.33: Dynkin Diagrams

Let R, V be a (reduced) root system with polarization $R = R_+ \sqcup R_-$, let Π denote the set of simple roots for this polarization, and identify Π with the set $\{1, \dots, k\}$. Then we can construct a graph following these steps:

1. The vertices are given by simple roots Π (or equivalently, by indices $i \in \{1, \dots, k\}$)
2. We join a vertex i and j by n_{ij} edges, which depends on the angle θ where

$$\theta = \frac{\pi}{2} \Rightarrow n_{ij} = 0 \quad \theta = \frac{2\pi}{3} \Rightarrow n_{ij} = 1 \quad \theta = \frac{3\pi}{4} \Rightarrow n_{ij} = 2 \quad \theta = \frac{5\pi}{6} \Rightarrow n_{ij} = 3$$

3. Roots joined by a single edge have the same length. Roots joined by multiple edges have an arrowhead ">" on the edge pointing towards the root with shorter length (imagine this as an inequality between the length of the roots)

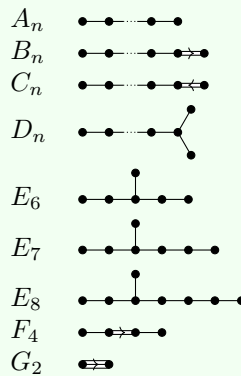
Example 3.2.34: Dynkin Diagrams

1. finding dynkin diagram for many root systems

We know proceed to classify root systems. As with other definition, if R is a root system and $R = R_1 \sqcup R_2$ where R_1, R_2 are proper subsets of R that are root systems, we say that R is reducible, and irreducible otherwise.

Theorem 3.2.35: Classification Of Root Systems

Let R be a (reduced) irreducible root system. Then it's dynkin Diagram is one of the following:

**Proof:**

here

(representing root systems via it's symmetries)

Definition 3.2.36: Weyl Group

Let $R \subseteq V$ be a root system. Let $W \subseteq \text{GL}(V)$ be the group generated by the reflections s_α , $\alpha \in R$. Then W is called the *Weyl group* of R .

Since it is a group generated by reflections, every element acts linearly on V . Furthermore, it perserves lengths and angles, and so it is an element of $O(V)$ (??)

Example 3.2.37: Weyl Group

p. 11 Jimail

Definition 3.2.38: Simple Reflection

Let $R \subseteq V$ be a root system with Weyl group W , and $R = R_+ \sqcup R_-$ is a polarizatoin with simple roots Π .

Then for each $\alpha_i \in \Pi$, the corresponding reflection $s_i = s_{\alpha_i} \in W$ is called a *simple reflection*.

Theorem 3.2.39: Weyl Groups Representing Root Systems

Let $R \subseteq V$ be a root system with Weyl group W , and $R = R_+ \sqcup R_-$ a polarization with simple roots Π . Then:

1. The simple roots s_i (for $i \in \Pi$) generate W
2. $W(\Pi) = R$. Hence, the root system can be recovered from the simple roots; we apply all Weyl group elements to Π to produce the full list of possible roots

Proof:
exercise?

(reduced expressions and longest elements)

Definition 3.2.40: Length Of Weyl Group Elements

Let W be the Weyl group of a root system R , and Π the simple roots. Then the *length* of $w \in W$ is the shortest length of word for w as a product of simple reflections. We denote the length of w by $l(w)$.

In other words, if $w = s_{i_\ell} \cdots s_{i_1}$ and no shorter sequence exists, then $l(w) = \ell$.

Theorem 3.2.41: Generating Roots Using Weyl Group

Let W be the Weyl group of a root system R with polarization $R = R_+ \sqcup R_-$ and Π the corresponding simple roots.

1. There exists a unique element $w_0 \in W$ such that $\ell(w_0) = |R_+|$
2. The element w_0 has the property that $w_0(R_+) = R_-$
3. If $w = s_{i_k} \cdots s_{i_1}$ is an expression for w_0 of length $l(w_0)$, then

$$\begin{aligned}\gamma_1 &= \alpha_{i_1} \\ \gamma_2 &= s_{i_1}(\alpha_{i_2}) \\ &\vdots \\ \gamma_k &= s_{i_k} \cdots s_{i_1}(\alpha_{i_k})\end{aligned}$$

is an enumeration of the positive roots

4. The elements γ_j defined above satisfy the property that the roots $\gamma_j, s_{i_1}(\gamma_j), \dots, s_{i_1}(\gamma_j) \in R_+$ are positive, but the roots $s_{i_j} s_{i_{j-1}} \cdots s_{i_1}(\gamma_j) \in R_-$ are negative.

Proof:

Jimail didn't prove – we don't have enough tools yet

Now that we have a better understanding of Weyl groups, we seek to link some of their structure to structure of simple roots. As we saw, the relations between simple roots can be captured via Dynkin Diagrams. We will seek for elements of the Weyl group which are “compatible” with the Dynkin diagrams. (for some reason), let Γ be a dynkin diagram of type A , D , or E (i.e. a Dynkin diagram with no multiple edges). Let R denote the corresponding root system. For any orientation Ω of Γ , Let (γ, Ω) represent the corresponding quiver. Recall that if i is a sink or source in Ω , we have an operator $s_i^\pm$ that reverses the arrows at i . Denote this operation by

$$(\Gamma, \Omega) \mapsto (\Gamma, s_i^\pm \Omega)$$

where we would put a $+$ if it is a sink-to-source operation, and a $-$ if it is a source-to-sink operation

Definition 3.2.42: Adapted Word

Let Γ be a Dynkin diagram of type A, D , or E (i.e. a diagram with no multiple edges, or “simply laced”), W the corresponding Weyl group, and Ω an orientation of Γ . Let $w \in W$. Then the reduced expression $w = s_{i_k} \cdots s_{i_1}$ for w is *adapted for Ω* if

1. The vertex i_1 is a sink for (Γ, Ω)
2. The vertex i_2 is a sink for $(\Gamma, s_{i_1}^+ \Omega)$
- $\vdots$
3. The vertex i_k is a sink for $(\Gamma, s_{i_{k-1}}^+ \cdots s_{i_1}^+ \Omega)$

i.e. at each step, we can apply the operator $s_{i_j}^+$ to the quiver $(\Gamma, s_{i_{j-1}}^+ \cdots s_{i_1}^+ \Omega)$

Example 3.2.43: Adapted Dynkin Diagrams

p. 15 Jaimal

Theorem 3.2.44: Lusztig's Theorem

Let R be an A, D , or E root system with dynkin diagram Γ and Weyl group W . For any orientation Ω of Γ , there exists an expression for the longest element $w_0 \in W$ which is adapted for Ω

Proof:

exercise 19

3.2.4 Gabriel's Theorem

In this section, we come back to Quiver's. Recall in example 3.2.25 it was asked to show how find the indecomposable quivers with underlying graphs of type A_1 , A_2 , and a cyclic quiver. The result was that A_2 and A_3 (with the given orientation) reduced to finitely many indecomposable representations. However, if there is a cycle, the quiver did not afford such a representation. A quiver which has finite decomposition is naturally much easier to study (TBD). The question then becomes:

Which quivers Q can be decomposed into finitely many indecomposable representations?
Can we classify all such quivers?

The answer of this question is *Gabriel's Theorem*; the goal of this section is to prove this result. Gabriel's Theorem is the key result linking the representation of quivers to Lie algebras, a pivotal tool in studying manifolds with group structures.

(a,b) To make the connection more precise, some terms need to be defined:

Definition 3.2.45: Collection Of Indecomposables

Let Q be a quiver, and let $\text{Rep}(Q)$ be it's representations. Then the set $\text{Ind}(Q) = \{[X] \mid X \in \text{Rep}(Q) \text{ is indecomposable}\}$ is the set of isomorphism classes of indecomposable quivers of $\text{Rep}(Q)$.

We say that a quiver is of *finite type* if $\text{Ind}(Q)$ is finite.

Before continuing with proving the result, it is useful to at least see how such a connection might even be hypothesized:

Example 3.2.46: Motivating Connection

Consider the quiver $1 \longrightarrow 2 \longrightarrow 3$. The underlying graph is A_3 , so let α_1, α_2 , and α_3 denote the simple roots for the root system A_3 . Using these, we can deduce the positive roots are:

$$\{\alpha_1, \alpha_2, \alpha_3, \alpha_1 + \alpha_2, \alpha_2 + \alpha_3, \alpha_1 + \alpha_2 + \alpha_3\}$$

We furthermore saw in example 3.2.25 that the indecomposable representations are:

$$\mathbb{F} \xrightarrow{0} 0 \xrightarrow{0} 0 \quad \mathbb{F} \xrightarrow{1} \mathbb{F} \xrightarrow{0} 0$$

$$0 \xrightarrow{0} \mathbb{F} \xrightarrow{0} 0 \quad 0 \xrightarrow{0} \mathbb{F} \xrightarrow{1} \mathbb{F}$$

$$0 \xrightarrow{0} 0 \xrightarrow{0} \mathbb{F} \quad \mathbb{F} \xrightarrow{1} \mathbb{F} \xrightarrow{1} \mathbb{F}$$

Here is the key insight: notice we can correspond the *dimension vector* of these representations to the positive roots!

Theorem 3.2.47: Gabriel's Theorem

Let Q be a quiver. then Q is of finite type if and only if the underlying graph of Q is a Dynkin Diagram of type A , D , or E . In this case, the dimension vector:

$$\dim_Q : \text{Ind}(Q) \rightarrow R_+ \quad [I] \mapsto \dim_Q(I)$$

bijects the indecomposable representations (up to isomorphism) and positive roots in the corresponding root system.

The proof will be relegated to page pageref:HERE

The theorem hints to us that indecomposable quivers somehow relate to the simple reflections of a Weyl group. However, a Dynkin Diagram is inherently unoriented, while Quivers are by definition an oriented graph. The first thing we need to do to fix is find the equivalent of a “reflection” for quivers.

The idea for the following definition is as follows: Take the quiver and the reflection

$$1 \xrightarrow{e} 2 \quad 1 \xleftarrow{f} 2$$

Now, given a representation assigning $V(1)$, $V(2)$, and T_e , how should we assign values to the reflection? One way of doing it is by taking $T_e : V(1) \rightarrow V(2)$, and mapping it to $S_f : \ker(T_e) \rightarrow V(1)$ where $S_f(x) = 0$. Visually, we see that:

$$\left(\begin{array}{ccc} V(1) & \xrightarrow{T_e} & V(2) \\ \bullet & & \bullet \end{array} \right) \mapsto \left(\begin{array}{ccc} V(1) & \xleftarrow{S_f} & \ker(T_e) \\ \bullet & & \bullet \end{array} \right)$$

this map considers what happens at a sink. Another approach maps by considering the source:

$$\left(\begin{array}{ccc} V(1) & \xrightarrow{T_e} & V(2) \\ \bullet & & \bullet \end{array} \right) \mapsto \left(\begin{array}{ccc} \operatorname{coker}(T_e) & \xleftarrow{S_f} & V(2) \\ \bullet & & \bullet \end{array} \right)$$

where $S_f(x) = \overline{T_e(x)}$. Both of these assignments are in fact functorial, If we have a quiver morphism $\varphi : V \rightarrow W$ represented by:

$$\begin{array}{ccc} V(1) & \xrightarrow{T_e} & V(2) \\ \downarrow \varphi(1) & & \downarrow \varphi(2) \\ W(1) & \xrightarrow{S_e} & W(2) \end{array}$$

Then the appropriate morphism φ maps to is represented in these diagrams:

$$\begin{array}{ccc} V(1) \xleftarrow{T_f|_{\ker(T_e)}} \ker(T_e) & & \operatorname{coker}(T_e) \xleftarrow{\overline{T_f}} V(2) \\ \downarrow \varphi(1) & \downarrow \varphi(1)|_{\ker(T_e)} & \downarrow \overline{\varphi(2)} \\ W(1) \xleftarrow{S_f|_{\ker(S_e)}} \ker(S_e) & & \operatorname{coker}(S_e) \xleftarrow{\overline{S_f}} W(2) \end{array}$$

which both commute (where it was shown that $\varphi|_{\ker(T_e)}(\ker(T_e)) \subseteq \ker(S_e)$ when we showed the category of quivers contains kernels). This will essentially represent how a reflection functor between two quivers will be defined – locally.

The difference between these two functors is where there is a sink or a source. If i is a sink, then for any given quiver representation $V \in \operatorname{Rep}(Q)$, let

$$\Psi_i : \bigoplus_{e:i \rightarrow j} V(j) \rightarrow V(i)$$

represent the collection of all linear transformation, where for each summand we have $T_e : V(i) \rightarrow V(j)$. If i is a source, then let:

$$\Phi_i : V(i) \rightarrow \bigoplus_{e:i \rightarrow j} V(j)$$

where each summand is mapped to T_e .

Definition 3.2.48: Reflection Functor

Let Q be a quiver. Define a functor from $\text{Rep}(Q)$ to $\text{Rep}(s_i^\pm Q)$ as follows:

1. For any $V \in \text{Rep}(Q)$, define $S(V)_i^+ \in \text{Rep}(s_i^+(Q))$ for a sink i , where at each vertex k ,

$$S_i^+(V)(k) = \begin{cases} V(k) & i \neq k \\ \ker(\Psi_i) & i = k \end{cases}$$

and for each linear transformation T_e :

$$S_i^+(T_e) = \begin{cases} T_e & s(k) \neq i \\ \pi_k \circ \iota_i & \text{if } f : i \rightarrow k \text{ is in } s_i^+(Q) \end{cases}$$

where $\iota_i : \ker(\Psi) \rightarrow \bigoplus_{e:j \rightarrow i} V(j)$ is the inclusion map and $\pi_k : \bigoplus_{e:j \rightarrow i} V(j) \rightarrow V(k)$ is the projection map onto the k th component.

2. For any $V \in \text{Rep}(Q)$, define $S(V)_i^- \in \text{Rep}(s_i^-(Q))$ for a source i , where at each vertex k ,

$$S_i^-(V)(k) = \begin{cases} V(k) & i \neq k \\ \text{coker}(\Phi_i) & i = k \end{cases}$$

and for each linear transformation T_e :

$$S_i^-(T_e) = \begin{cases} T_e & s(k) \neq i \\ q_k \circ \iota_i & \text{if } f : i \rightarrow k \text{ is in } s_i^-(Q) \end{cases}$$

where $\iota_i : V(k) \rightarrow \bigoplus_{e:j \rightarrow i} V(j)$ is the inclusion map and $q : \bigoplus_{e:j \rightarrow i} V(j) \rightarrow \text{coker}(\Phi_i)$ is the quotient map

Then these two functors, $S_i^\pm : \text{Rep}(Q) \rightarrow \text{Rep}(s_i^\pm Q)$ are called the *Bernstein-Gelfand-Ponomarev reflection functor*, or *reflection functor* for short.

Example 3.2.49: Reflection Functor

p. 70 Jaimal. The key idea is that these functors give different quiver representations, even if it on A_2 with the only orientation and doing S_1^+ then S_2^- .

A couple of properties of the functors needs to be established. For notational simplicity, let $\text{Rep}^{i,-}(Q)$ denote the full subcategory consisting of representations of Q for which the map

$$\Psi_i : \bigoplus_{e:j \rightarrow i} V(j) \rightarrow V(i)$$

is surjective, while $\text{Rep}^{i,+}(Q)$ denotes the full subcategory consisting of representations of Q for which the map

$$\Phi_i : V(i) \rightarrow \bigoplus_{e:i \rightarrow j} V(j)$$

is injective.

Proposition 3.2.50: Properties Of Reflection Functor

Let Q be a quiver.

1. If i is a sink, then for any $V \in \text{Rep}(Q)$, $S_i^+ \in \text{Rep}^{i,+}(s_i^+(Q))$ (similarly for source)
2. If i a sink, then for any $V \in \text{Rep}(Q)$, $S_j^- S_i^+ \in \text{Rep}^{i,-}(Q)$ (similarly for source)
3. The restriction of the reflection functors are inverse equivalences (adjoints?)

$$S_i^+ : \text{Rep}^{i,-}(Q) \rightarrow \text{Rep}^{i,+}(s_i^+ Q)$$

$$S_i^- : \text{Rep}^{i,+}(Q) \rightarrow \text{Rep}^{i,-}(s_i^- Q)$$

4. If V is simple at vertex i , then $S_i^\pm(V) = 0$.
5. If V is a non-zero indecomposable representation of Q that is not simple at i , then $S_i^\pm(V)$ is a nonzero indecomposable representation of $s_i^\pm(Q)$, and $V \cong S_j^\mp S_i^\pm(V)$.
6. $S_i^\pm(V)$ is a nonzero indecomposable representation of $s_i^\pm(V)$ if and only if $s_i(\dim_Q(V)) \geq 0$
7. If $V \in \text{Rep}^{i,\pm}(Q)$ is a nonzero indecomposable representation, then $\dim_Q(S_i^\pm(V)) = s_i \dim_Q(V)$

Proof:
exercise

Finally, we define a new object with purpose HERE (Apparently, these definitions can be given if we use some homological algebra. Perhaps revamp this after you do the homological algebra section, or keep it at this level with the mention of the version of greater generality for those who want to skip Hom Alg)

Definition 3.2.51: Root Lattice

Let Q be a quiver with vertex set I . Then if the underlying graph of Q is a dynkin diagram, $\mathbb{Z}^I$ is called the *root lattice*. For $(v_i)_{i \in I} \in \mathbb{Z}^I$, we say that $v \geq 0$ if $v_i \geq 0$ for all $i \in I$. Furthermore, we let $\mathbb{Z}_+^I := \{v \in \mathbb{Z}^I \mid v \geq 0\}$

(The following theorem seems to make more sense with some homological algebra, where the domain of the function should actually be the Grothendieck group of a $\text{Rep}(Q)$) (Are objects in the following theorem representations?)

Theorem 3.2.52: Root Lattice Map

here exists a map $\dim_Q : \{\text{iso classes of Rep's}\} \rightarrow \mathbb{Z}_+^I$

Proof:

not proved

Definition 3.2.53: Euler And Tits Form

Let $Q = (I, E)$ be a quiver. Define a bilinear form on $\mathbb{Z}^I$ as follows: for any $v, w \in \mathbb{Z}^I$:

$$\langle v, w \rangle_Q = \sum_i v_i w_i - \sum_e v_{s(e)} w_{t(e)}$$

This is called the *Euler form* of Q . The *symmetrized Euler form* of Q is given by

$$(v, w)_Q = \langle v, w \rangle_Q + \langle w, v \rangle_Q$$

Finally, the *tits form* of Q is the quadratic form $q_Q(v) = \langle v, v \rangle = \frac{1}{2}(v, v)$

Example 3.2.54: Euler And Tits Forms

Let $Q = \begin{array}{ccc} 1 & \xrightarrow{e} & 2 \\ & & \searrow f \\ & & 3 \end{array}$. Then the Euler's form is:

$$\langle v, w \rangle_Q = \sum_{i=1}^3 v_i w_i - v_1 w_2 - v_2 w_3$$

$$q_Q(v) = \sum_{i=1}^3 v_i v_i - v_1 v_2 - v_2 v_3$$

The tits form is also positive definite, since if it weren't, we would be able to pick 3 positive integer's satisfying:

$$v_2(v_1 + v_3) > v_1^2 + v_2^2 + v_3^2$$

However, this is no possible (we can show this with elementary calculus and optimization, or take a couple of space cases and induction).

Now let $Q' = \begin{array}{ccc} 1 & \xleftarrow{e} & 2 \\ & & \searrow f \\ & & 3 \end{array}$. Then:

$$\langle v, w \rangle_{Q'} = \sum_i v_i w_i - v_2 w_1 - v_2 w_3$$

and:

$$q_{Q'}(v) = \sum_i v_i v_i - v_2 v_1 - v_2 v_3$$

then we see that the tits form of Q and Q' agree.

Theorem 3.2.55: Positive Definite Forms And Dynkin Diagrams

Let Γ be a connected graph. The Tits form q_Γ is positive definite if and only if Γ is a Dynkin diagram of type A , D , or E

Proof:

omitted by Jimail

(exercise showing not true for not A, D, E)

Finally, we will state one more theorem we need for the proof

Theorem 3.2.56: Coxeter Element Lemma

Let R be a root system, $\Pi = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ a set of simple roots and W its Weyl group. Let $s_i = s_{\alpha_i}$ denote the simple reflections corresponding to the simple root α_i . let $C = s_{i_n} \cdots s_{i_1}$ where $i_1, i_2, \dots, i_n$ is some ordering of the simple roots (such an element is called a *Coxeter element*).

Then for any $v \in \mathbb{Z}^I$, there exists a $m \geq 0$ such that $C^m(v) \not\geq 0$

Proof:

omitted

Proof:

(of Gabriels theorem) here

3.2.5 Path Algebra of Quivers

(there are some really important insights that feel like they should actually be the priority of this section!! Essentially, we build-up more on how to think about algebras in terms of quotients of free modules, build-up how “path algebras” naturally come into play in representing quivers, and finish off with the Morita equivalence between $R\text{-Mod}$ and $R/I\text{-Mod}$ where $I = \text{Ann}_R(M)$).

Let $Q = (V, E)$ be a quiver. For vertices i, j of Q , Define the set $\text{Path}(i, j) = \text{span}\{\text{paths } \gamma \text{ from } i \text{ to } j\}$ (i.e. consider the path as a basis to generate a vector-space over k , where k is usually understood), which we will call the *path space*. the length of a particular path is the number of edges between each point. The length of the path from i to i is zero. If γ is a path, and if we let i, j represent a path from it to itself, then if γ is a path from i to j , we will say that γi and $j \gamma$ are the same path.

Lemma 3.2.57: Properties Of Path

here

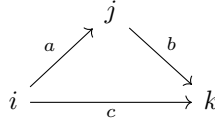
Proof:

1. We want to show that P_i has the structure of a representation $\text{Rep}(Q)$, which consists of a collection of vector-spaces and linear transformation's between them. First, each $P_i(j)$ is a vector-space, as pointed out earlier. Next, we may define a “linear” map $T_e : P_i(j) \rightarrow P_i(k)$ to be $\gamma \mapsto e\gamma$ (i.e. append another edge to the path). Notice that since we are treating paths as basis elements to some freely generated vector-space $P_i(j)$, then we get by construction that $T_e(c \cdot \gamma) = ec \cdot \gamma = ce\gamma = cT_e(\gamma)$. Next, taking $\gamma + \epsilon$ to be a formal sum of two paths, we can naturally extend the action of concatenating to the formal sum:

$$e(\gamma + \epsilon) = e\gamma + e\epsilon$$

which will mirror the linear maps within a representation $\text{Rep}(Q)$. Furthermore, if e, f are neighbouring edges, then $T_{ef} = T_e T_f$. This now mirrors how $\text{Rep}(Q)$ is both a collection of vector-spaces and a collection of maps between them.

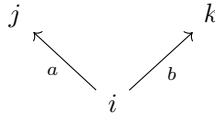
2. We are now going to treat P_i as a representation, since we considered them to be related to them in the above example. We want to prove that the collection of vector-spaces P_i along with the linear transformations T_e are indecomposable, that is we cannot find subrepresentations. This comes from the construction of our map, in particular that we have that for any collection of paths that form a basis for $P_i(j)$, we have that they are each mapped to by some paths through the appropriate T_e for an appropriate edge e . To make it more intuitive to see, consider for example P_i of the below quiver with a “cycle”:



Then $\dim(V(i)) = 1$, $\dim(V(j)) = 1$, and $\dim(V(k)) = 2$, with the path i mapping to ci in $V(k)$ and mapping to bi in $V(j)$, while the path bi of $V(j)$ maps to ibc , in $V(k)$. Then notice that $V(k)$ is “surjected” upon by both $V(i)$ and $V(j)$ combined, and all the maps are injective. If the above quiver were to split, then either the dimension of $V(i)$ or $V(j)$ would have to get smaller (since $V(k)$ would have to diminish a dimension): if $\dim(V(j)) = 0$, then that would break the surjectivity of T_a , while if $\dim(V(i)) = 0$, then the map $T_c \neq T_c|_A \oplus T_c|_B$ for the two subquivers A and B .

If the quiver has a proper cycle (i.e. $P_i(i)$ has more than $\{i\}$ as a basis), then since every map must be injective (mapping onto the appropriate basis) we have that each vector-space is 1-dimensional and an isomorphism, which we’ve shown in chapter 1 with a dimension argument that this representation is indecomposable.

If The quiver does not have a cycle:



Then if for the sake of contradiction P_i were decomposable, then if $\dim(V(i)) = 0$ in one of the decomposition, so would $V(j)$ and $V(k)$, making it a trivial decomposition, while

if $\dim(V(i)) = 1$, both $\dim(V(j))$ and $\dim(V(k))$ would have to be 1 or else the map wouldn't be well-defined, and hence this quiver is indecomposable.

More generally, any quiver Q can be broken down into one of these two cases: If the quiver has a “cycle”, then since every map is injective and must surject onto the vertex where two edges meet, we can repeat the above argument to show that the two scenarios which we may have while trying to decompose a quiver lead to a contradiction, and if the quiver has no cycles, we repeat the argument we've done for the above quiver.

3. If P_i were not a sink, then if e is an edge where that makes i not a sink, say $s(e) = i$ and $t(e) = j$, then we can take a subquiver where the vector-space at $P_i(i) = 0$, and we have a zero-map from $P_i(i)$ to $P_i(j)$, giving us a subquiver.

Conversely, if P_i were a sink, the only path from i would be to i , and so all other vertices would have a 0 dimensional vector-space, while $P_i(i)$ would be 1-dimensional, which makes P_i into a simple representation

4. For notational simplicity, if $\gamma = k_n k_{n-1} \cdots k_1$ is some path, let

$$f_\gamma(v) = f_{k_n}(f_{k_{n-1}}(\cdots f_1(v)))$$

so that $f_{ab}(v) = f_a(f_b(v))$.

For the proof, pick any $v \in V(i)$, and consider the following commutative diagram:

$$\begin{array}{ccc} P_i(a) & \xrightarrow{T_e} & P_i(b) \\ \downarrow \varphi_a & & \downarrow \varphi_b \\ V(a) & \xrightarrow{S_e} & V(b) \end{array}$$

Then define $\varphi^v : P_i \rightarrow V$, $\varphi^v(\gamma) = S_\gamma(v)$. Then

$$S_e(\varphi_a^v(\gamma)) = S_e S_\gamma(v) = S_{e\gamma}(v) = \varphi_b^v(e\gamma) = \varphi_b^v(T_e(\gamma))$$

proving commutativity of the diagram. If $v \neq w$, then since $\varphi_i^v(i) = v \neq w = \varphi_i^w(i)$, $\varphi^v \neq \varphi^w$. Next, if $\varphi \in \text{Hom}_Q(P_i, V)$, then if $(\gamma_1, \gamma_2, \dots, \gamma_n)$ is a basis for $V(i)$, consider

$$(\varphi_i(\gamma_1), \dots, \varphi_i(\gamma_n)) = (v_1, \dots, v_n)$$

and define $v = \sum_i v_i$. Then extending linearly we get that $\varphi_i(\sum_i c_i \gamma_i) = \sum_i c_i v_i$, and then extending φ_i to any other φ_j through a chain of commutative diagrams, we get that all of φ is determined by where v maps to in any path in V . Thus, $\varphi = \varphi^v$, and so $\text{Hom}(P_i, V)$ and $V(i)$ are in bijective correspondence. Finally, By linearity of each map in φ , and each map assigned to an edge in P_i and $V(i)$, it we see that:

$$\varphi^{v+cw} = \varphi^v + c\varphi^w \in \text{Hom}(P_i, V)$$

giving us that the map is in fact a linear map, completing the proof.

Definition 3.2.58: Path Algebra

Let $Q = (V, E)$ be a quiver. Then the *path algebra* $P(Q)$ is defined as follows:

1. As a vector-space $P(Q) = \text{span}\{\text{paths in } Q\}$, where each vertex $i \in V$ we include an element e_i representing the constant path $i \rightarrow i$
2. multiplication is defined via concatenation when legal, that is:

$$p \cdot q = \begin{cases} pq & \text{if } t(q) = s(p) \\ 0 & \text{otherwise} \end{cases}$$